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College of Business and Engineering

Department of Mechanical and Electrical Engineering

Bioengineering Master's Degree Program

PROTEIN DETECTION USING SURFACE PLASMON RESONANCE ON A MICROSCOPE SETUP

A Project Report

by

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of the Requirements

for the Degree of

Master of Science in Bioengineering

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Abstract

Protein detection has become essential in biomedical research and diagnostics, serving as a cornerstone for therapeutic development, disease monitoring, and biomolecular interaction studies. Traditional protein detection methods, such as ELISA, Western Blotting, and Mass Spectrometry, often require extensive sample preparation, labeling, or complex equipment, limiting their suitability for real-time and cost-effective applications [3, 9]. This study aims to address these limitations by developing a prismless Surface Plasmon Resonance (SPR) setup integrated into a standard microscope platform. The objective was to establish a simplified, high-sensitivity system that enables real-time, label-free detection of protein interactions in a more accessible configuration [14, 17]. This presents an innovative approach to protein detection through the integration of prismless Surface Plasmon Resonance (SPR) with a standard Zeiss Axio 100 microscope. It focuses on three critical parameters—intensity profiles, resonance angle shifts, and Full-Width at Half Maximum (FWHM). The experimental optical system setup utilized a 488nm Coherent Sapphire laser whose beam was directed through a ZEISS AXIOVERT 100 microscope objective lens and then the beam is reflected by a gold-plated silicon substrate mounted on the microscope stage, with optical alignment achieved using reflective mirrors [19, 22]. This setup allowed SPR excitation at specific angles and provided high spatial resolution through integration with microscopy [14, 25]. The system was calibrated meticulously, ensuring consistent delivery and accurate measurement of protein binding events. A perfusion system was constructed to facilitate controlled reagent flow across the gold-plated silicon wafer as the sensor surface, with Phosphate-Buffered Saline (PBS) as the buffer, Bovine Serum Albumin (BSA) as the primary analyte, and EDTA for sensor regeneration [24, 28]. Quantitative data were acquired and analyzed using Win View software then later Lightfield and ImageJ software used at different points for file viewing, conversion and analysis of the data. Python programing language was also used for generation of the results [31, 33]. Intensity profiles were plotted to detect binding events, and the resonance angles and Full-Width at Half Maximum (FWHM) were calculated. The results demonstrated shifts in resonance angles proportional to protein concentrations, indicating successful detection [19]. Resonance angle errors ranged between 10% and 22%, while FWHM errors varied from 8% to 20%. These findings confirm the feasibility and reliability of the prismless SPR-microscopy setup for protein detection [14, 22]. Unlike traditional SPR systems, the absence of a prism simplifies the configuration, reducing costs and enhancing portability. This development holds significant potential for expanding SPR applications to resource-limited settings and diagnostics in fields such as personalized medicine, environmental monitoring, and food safety [17, 27]. Future work will focus on automating perfusion systems and further miniaturization to increase portability and usability in decentralized diagnostic environments. This approach represents a step forward in making SPR technology more accessible, bridging the gap between advanced research tools and practical healthcare applications [29].

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Introduction

Protein detection is a vital process across various fields, including biomedical research, disease diagnostics, therapeutic development, and molecular biology. Accurate and efficient identification and quantification of proteins enable researchers to study disease mechanisms, monitor biomolecular interactions, gain critical insights into cellular functions, and develop targeted therapies [1, 3]. Traditional protein detection methods—such as Enzyme-Linked Immunosorbent Assay (ELISA), Western Blotting, and Mass Spectrometry—are widely used due to their high sensitivity and specificity. However, these methods often require extensive sample preparation, labeling, or complex equipment, which can potentially alter protein function. These limitations have catalyzed the search for innovative approaches that are real-time, cost-effective, and label-free [3, 8].

Surface Plasmon Resonance (SPR) is an optical detection technique that has gained attention as a transformative approach in this domain. Its ability to provide real-time, label-free analysis of biomolecular interactions sets it apart from traditional techniques. SPR relies on the excitation of surface plasmons, which are coherent electron oscillations at the interface between a metal (typically gold) and a dielectric medium. When polarized light hits the metal at a particular angle, resonance conditions are met, causing a reduction in reflected light intensity. This resonance angle is highly sensitive to changes in the refractive index near the metal surface, making SPR an effective method for detecting molecular interactions such as protein binding [9, 14]. By monitoring shifts in the resonance angle, SPR can quantitatively analyze binding kinetics and affinity in real-time without the need for exogenous labels or markers [13, 17].

Traditional SPR setups, such as those using the Kretschmann configuration with a prism, require specialized optical components, which can increase cost and complexity. Recent advancements have shown that prismless SPR configurations can achieve similar sensitivity while simplifying the optical setup, enhancing portability, and reducing overall costs [14]. Prismless SPR integrates seamlessly with standard microscopy platforms, combining SPR's sensitivity with microscopy's spatial resolution to create a versatile tool for real-time analysis. This integration is particularly valuable for studying heterogeneous samples, such as those encountered in diagnostics or cellular biology, where spatial information complements quantitative data [17, 19].

Microscopy has been a bedrock of scientific exploration, offering unparalleled visualization of biological structures and processes at cellular and subcellular levels. With advancements in optical technologies, microscopes have evolved to provide high-resolution imaging and dynamic observation capabilities. For example, in 2022, the global microscopy market was valued at approximately USD 8.7 billion, with projections indicating continued growth at a compound annual growth rate (CAGR) of 5.5% through 2030. This growth is driven by the expanding use of advanced microscopy in biomedical research, materials science, and nanotechnology [20].

The integration of prismless SPR with a Zeiss Axiovert 100 microscope in this research underscores the practicality and potential of combining these technologies. Using a Coherent Sapphire 488nm laser and a gold-plated silicon substrate, this study eliminates the complexities of traditional SPR setups while maintaining high sensitivity. By focusing on parameters such as intensity profiles, resonance angles, and Full-Width at Half Maximum (FWHM), the research simplifies data acquisition and analysis, making the technology accessible for broader applications [22, 25].

Protein detection is increasingly crucial in personalized medicine, where the identification of disease-specific biomarkers informs tailored treatment strategies. For instance, SPR has been instrumental in cancer research, enabling the detection of biomarkers such as HER2 and VEGF, which are critical for diagnosing and monitoring breast cancer. Additionally, SPR-based systems are finding applications in environmental monitoring, detecting pollutants and pathogens in water and air samples [17, 27].

This research aims to bridge the gap between advanced SPR systems and the demand for accessible diagnostic tools. By developing a prismless SPR-microscopy platform, it provides a pathway to cost-effective, portable, and highly sensitive protein detection. The integration of this setup has the potential to revolutionize diagnostics in resource-limited settings and expand applications in biomedical research, food safety, and beyond [29]

Literature Review

Overview of Protein Detection Techniques

Protein detection methods have historically been the cornerstone of biomedical research, diagnostics, and therapeutic development. Each technique offers unique advantages and challenges, which drive the need for innovative approaches such as Surface Plasmon Resonance (SPR) [1, 7]. Below are some of the methods.

Enzyme-Linked Immunosorbent Assay (ELISA): ELISA is a robust and widely used immunoassay method that detects specific proteins through antigen-antibody interactions. Typically, antibodies are immobilized on a solid substrate, where they bind the target protein. The introduction of enzyme-conjugated secondary antibodies produces a detectable signal (often colorimetric or fluorescent) following enzymatic reactions [3, 8]. The Advantages include high specificity and sensitivity, well-suited for clinical diagnostics, established protocols and reagents being readily available [3]. Among the demerits are issues like, requiring extensive sample preparation and multiple processing steps, endpoint nature limits real-time kinetic studies [8], dependence on labeling may alter protein behavior and introduce potential inaccuracies [3]. While ELISA requires labels and endpoint measurements, SPR achieves real-time, label-free detection of molecular interactions. Additionally, SPR is versatile for studying unknown proteins without prior antigen-antibody development [17].

Western Blotting: Western blotting involves separating proteins via electrophoresis, transferring them to a membrane, and probing specific proteins with labeled antibodies [2]. It is particularly effective in detecting protein expression and post-translational modifications. Its advantages include high specificity due to antibody selectivity, providing insights into protein molecular weight and modifications [7]. The limitations are Labor-intensive and time-consuming, requires substantial protein sample quantities [3], qualitative or semi-quantitative at best. In correlation SPR simplifies detection processes by providing quantitative data directly. The addition of microscopy enables spatially resolved analysis, unattainable with Western blotting [8].

Mass Spectrometry (MS): MS offers high precision in identifying and quantifying proteins based on their mass-to-charge ratio [9]. MS is advantageous for analyzing complex mixtures and identifying multiple proteins simultaneously offering unparalleled precision [8]. However, it requires extensive sample preparation and sophisticated analysis, limiting its application in real-time and high-throughput settings.

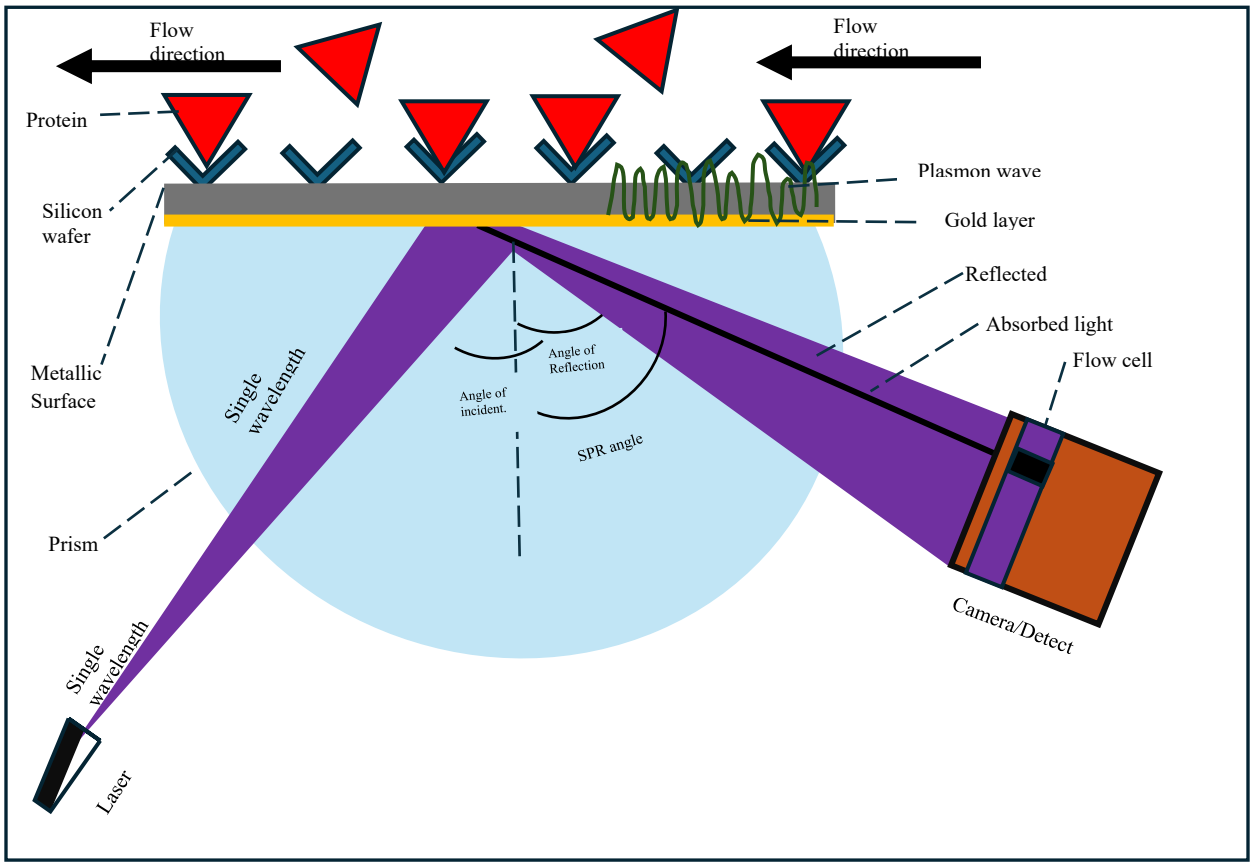
While MS focuses on large-scale proteomics, SPR excels in real-time interaction studies. SPR systems are also more portable and accessible than MS setups [14].

Fluorescence-Based Techniques: Fluorescence microscopy and related techniques like Fluorescence Resonance Energy Transfer (FRET) use fluorescent labels to monitor protein interactions [4, 7]. While effective in live-cell imaging, being highly sensitive [5] and offering real-time dynamics, the demerit of necessitating labeling, which can potentially alter protein function and limit the accuracy of natural biomolecular interactions. Coupled with photobleaching and autofluorescence impact data reliability [9]. In correlation SPR's label-free nature preserves protein functionality. Integration with microscopy achieves spatial resolution while avoiding fluorescence artifacts [17].

Protein Microarrays: Protein microarrays use immobilized proteins to screen interactions with antibodies, ligands, or other proteins, enabling high-throughput analysis [6]. While microarrays provide broad insights, they rely on the availability of specific, high-quality antibodies, and data interpretation can be complex [6].

Introduction to Surface Plasmon Resonance (SPR)

SPR Explained Diagram



Caption: *The diagram above illustrates the principle of Surface Plasmon Resonance (SPR), a sensitive method used for detecting molecular interactions at a surface. At its core, SPR relies on*

the interaction between light and a gold-coated silicon wafer, which serves as the substrate. The silicon wafer provides structural support for the metallic layer, typically gold, that is essential for generating surface plasmons oscillations of electrons on the metallic surface [9, 15]. A flow cell is positioned above the gold surface to deliver analytes, such as proteins in solution. The analytes flow across the surface in a controlled manner, as indicated by the red arrows, and bind to specific regions of the surface, either directly or through capture molecules immobilized on the gold layer. This binding alters the refractive index near the surface [8, 14].

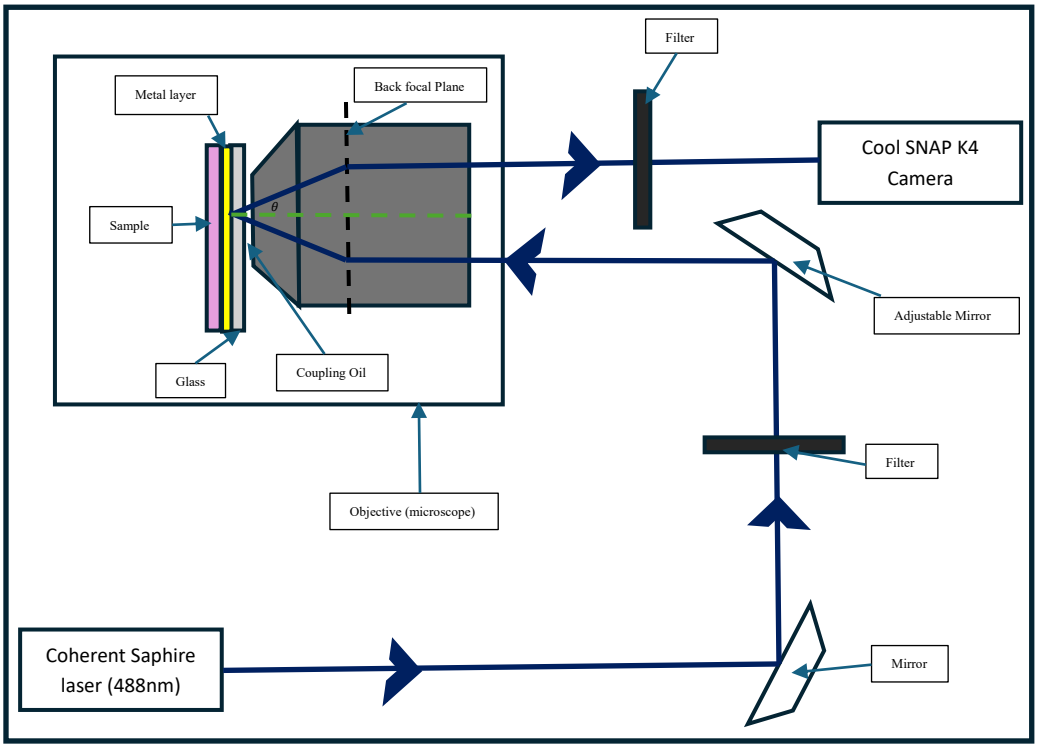
A laser beam is directed at the gold-coated silicon wafer at a precise angle, creating a phenomenon where light energy excites surface plasmons, resulting in the absorption of light at a specific angle, known as the SPR angle. This interaction causes a notable reduction in the intensity of reflected light, which is highly sensitive to changes in the refractive index near the gold surface. These changes occur as a result of molecular binding events, such as the attachment of proteins to the surface [14, 17]. The prism collects the reflected light and directs it to a camera or detector, which records changes in light intensity and maps them to the corresponding angles [13]. The SPR angle shifts in response to small refractive index changes, making it the primary measurable parameter in SPR experiments [8, 9].

The diagram also highlights how even slight changes in the refractive index, due to molecular interactions, can lead to significant shifts in intensity at the SPR angle. This makes SPR an exceptionally sensitive technique for studying binding interactions and changes at the molecular level. By visually demonstrating the key components, from the flow cell to the optical and detection systems, as well as the mechanism of SPR, this diagram provides a comprehensive understanding of how SPR experiments work and what to expect in terms of intensity analysis and angle shifts [14, 8].

In the diagram provided above the incident polarized light impinges on a gold-coated silicon wafer. Under normal reflection, the laser light would simply reflect off the surface at an equal angle. Yet in SPR, when the incident light meets the metal surface at a specific angle known as the resonance angle a portion of the light's energy is transferred to excite surface plasmons (quantized electron oscillations). This resonance condition results in a significant reduction in the reflected intensity. In effect, while the geometric paths of the incident and reflected rays might still be related by the conventional laws, the energy distribution across these rays is altered by the plasmonic excitation. This departure from the classical law is critical: the measured "dip" in intensity, which represents the SPR signal, is not merely a function of geometric reflection but of the dynamic electron oscillations at the interface. Therefore, when comparing the diagram to the law of reflection, one must account for the fact that the observed phenomenon (i.e., the SPR dip) arises from this resonance condition rather than from a simple equality of incident and reflected angles. In practical terms, ensuring that the diagram clearly distinguishes between traditional reflection and SPR behavior helps readers understand that, although the system begins with familiar optical laws, the introduction of the metal layer and the subsequent plasmon excitation fundamentally alter the reflection characteristics. This deviation is what makes SPR so sensitive

to changes in the refractive index any molecular binding at the surface shifts the resonance condition and hence the intensity profile, providing a quantitative measure of biomolecular interactions. Thus, the diagram should illustrate: A conventional reflection path (incident equals reflected) for baseline comparison. The modified reflection path in the presence of SPR, where at the resonance angle the energy is transferred and a distinct dip (or deviation) in the intensity is observed. This clarification emphasizes that while the geometric framework of classical optics remains, the intensity modulation central to SPR detection is what differentiates the system from simple reflective behavior.

SPR Mechanism Diagram 1



Caption: *Optical schematic of the prismless SPR setup integrated with a microscope. A 488 nm Coherent Sapphire laser is directed via a system of mirrors through a high-NA microscope objective to a gold-coated substrate where surface plasmon resonance occurs. The back-reflected beam, modulated by plasmon excitation, is redirected through a filter system and captured by the CoolSNAP K4 camera. The setup allows real-time, label-free protein detection with high angular sensitivity and image-based intensity profiling.*

The schematic diagram above illustrates the complete prismless SPR system integrated into a microscope platform, designed for real-time, label-free protein detection. At the foundation of this setup is a Coherent Sapphire 488 nm laser, which provides a stable, monochromatic light

source required for surface plasmon excitation. This laser was selected for its narrow linewidth, high coherence, and compatibility with the SPR excitation range. Upon emission, the laser beam is directed into the system via a sequence of reflective mirrors. These mirrors serve to guide and align the beam into the correct trajectory. One of these mirrors is mounted on a micrometer-controlled adjustable stage, enabling fine angular tuning of the beam path with sub-millimeter precision. This adjustability is critical for precisely coupling the beam into the high numerical aperture (NA) objective lens and ultimate translation within the system.

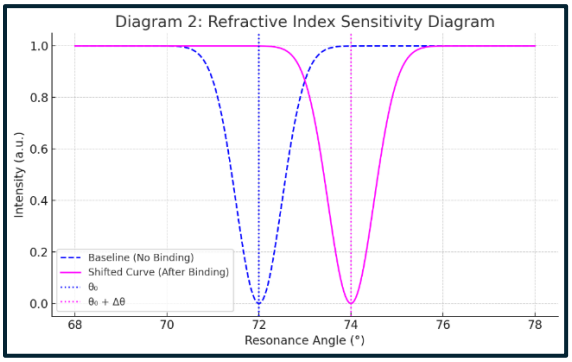
The beam enters the Zeiss Axiovert 100 microscope through the rear port, where it is focused by a 100× oil-immersion objective lens specified with a nominal NA of 1.45 (1.48). According to the manufacturer, this objective model is normally rated at NA = 1.45, but internal factory testing for this specific unit confirmed an actual numerical aperture of 1.48, which allows for a slightly expanded angular range. This correction is essential, as the higher NA value supports the propagation of incident rays up to a maximum half-angle of 76.3°, aligning more accurately with the experimentally observed resonance angles. This high-NA objective thus enables the light to reach the steep incidence angles necessary to excite evanescent fields at the sensor interface without requiring a prism. Beneath the objective, a gold-coated silicon wafer acts as the sensing substrate, supporting the excitation and propagation of surface plasmons during analyte interaction phases. This wafer is mounted on the microscope stage and is overlaid with a thin layer of coupling oil to ensure proper optical contact and refractive index matching. The wafer structure includes a glass base, an adhesion layer (typically titanium), and the active gold surface, upon which analyte binding events occur. A flow chamber system is situated above the sensor to introduce biochemical reagents such as phosphate-buffered saline (PBS), Bovine Serum Albumin (BSA), and EDTA.

Once the beam interacts with the substrate and undergoes reflection, the returning light exits back through the same microscope objective. This back-reflected beam contains modulations due to molecular interactions on the surface and must be carefully isolated from the input path. To achieve this, the reflected beam is redirected through a series of beam splitters and optical filters positioned externally to the microscope's tube lens. These filters are essential to block residual laser light, reduce photonic noise, and protect the sensor of the downstream detector from direct laser exposure. The beam is then directed into the CoolSNAP K4 CCD camera, positioned independently from the microscope viewing optics. This separation allows the camera to exclusively receive the reflection signal without interference from eyepiece imaging or other internal optical elements.

This configuration enables robust detection of reflected light intensity changes associated with molecular binding. The imaging system is synchronized with the laser and perfusion system to capture real-time snapshots at key experimental time points. Unlike traditional prism-based systems, this design offers a compact, easily modifiable platform that integrates seamlessly with existing microscopy infrastructure. The use of objective-based coupling not only simplifies the setup but also allows dual use of the microscope for both SPR signal acquisition and traditional

bright-field imaging if required. The entire system is optimized for stability, precision alignment, and minimal optical loss, forming a strong foundation for reproducible surface plasmon resonance experiments.

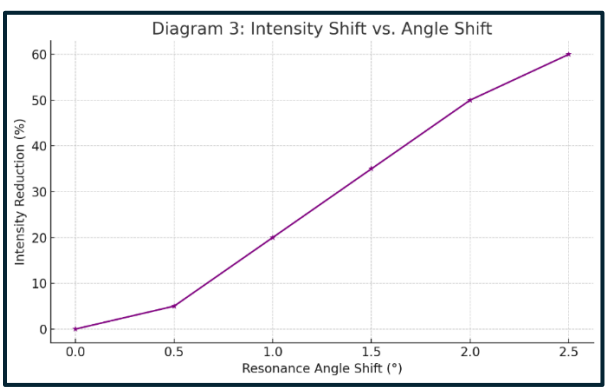
Diagram 2: Refractive Index Sensitivity Diagram



Caption: *SPR intensity response to protein binding at the sensor surface. The dashed blue curve represents the baseline signal with no analyte present, while the solid magenta curve shows the shifted SPR response after BSA binding. A resonance angle shift ($\Delta\theta \approx 2°$) is observed due to refractive index changes at the gold-analyte interface.*

This graph overlays two SPR intensity profiles. The **baseline curve** corresponds to the system in the presence of phosphate-buffered saline (PBS) only, showing the resonance dip at the original angle ($\theta_0 \approx 72°$). Upon the addition of a binding analyte like BSA, the **refractive index at the surface changes**, causing the dip to shift to a new angle ($\theta_0 + \Delta\theta \approx 74°$). This shift is characteristic of successful analyte interaction with the sensor surface and is the basis for SPR-based molecular detection.

Diagram 3: Intensity Shift vs. Angle Shift



Caption: *Relationship between resonance angle shift and intensity reduction. Even small angular changes (~1°) can produce significant decreases (up to 20%) in reflected intensity, demonstrating the sensitivity of the system to molecular binding events.*

This plot demonstrates how incremental shifts in resonance angle resulting from refractive index variations due to analyte binding cause nonlinear reductions in reflected intensity. As shown, a 1° angular shift corresponds to roughly a 20% drop in intensity, with steeper reductions at larger shifts. This graph supports the conclusion that SPR is highly sensitive even to minimal surface interactions, making it ideal for detecting low-concentration or weakly interacting biomolecules. This theoretical curve aligns well with the experimental intensity profiles recorded for each analyte phase. For example, in Experiment 1, the pixel shift from the PBS baseline (~981 px) to the BSA addition (~1138 px) corresponds to a resonance angle shift of approximately 11.8°, which was accompanied by a measurable drop in reflected intensity near the dip minimum, as seen in the back-reflected light maps. This drop reflects the steep slope of the SPR response curve and demonstrates the intensity sensitivity shown here, where even small angle changes (1°–2°) result in observable intensity reductions.

Moreover, in both experiments, the EDTA regeneration phase partially restored the resonance angle and intensity level toward the baseline PBS value (e.g., from 1138 px down to ~1021 px in Experiment 1), demonstrating the dynamic nature of this intensity-shift relationship during reversible surface binding. The nonlinear nature of the angle-to-intensity relationship, as shown in Diagram 3, reinforces the detection limits of the system particularly for concentration gradients or incomplete regeneration cycles.

<u>Mathematical Principles</u>

Fresnel Equations: Describes how light is reflected and transmitted at the interface between two media with different refractive indices.

$$r_s = \frac{n1cos\ \theta i - n2cos\ \theta t}{n1cos\ \theta i + n2cos\ \theta t}$$

$$r_p = \frac{n2cos\ \theta i - n1cos\ \theta t}{n2cos\ \theta i + n1cos\ \theta t}$$

Where r_s and r_p are the amplitude reflection coefficients for s- and p- polarized light, n1 and n2 are the refractive indices of the two media, θi is the angle of incidence, and θt is the angle of transmission.

In our SPR experiments, we utilize the Fresnel equations to model the behavior of light at the metal-dielectric interface. By computing the reflection coefficients (r_s and r_p) for various angles of incidence, we generate theoretical reflection curves that predict the intensity of reflected light under different conditions. As analytes (e.g., proteins) bind to the sensor surface, they induce subtle changes in the refractive index near the metal surface. These changes modify the reflection coefficients, shifting the resonance angle the angle at which a minimum in reflected intensity is observed. The calculation process involves the following steps:

1. Theoretical Modeling

We calculate the expected reflection intensity using the Fresnel equations for a range of incidence angles. This creates a baseline theoretical curve corresponding to the known refractive indices (n_1 for the prism or glass and n_2 for the buffer solution).

2. Experimental Measurement

During experiments, we measure the actual intensity of reflected light as the angle of incidence varies. The point of minimum reflected intensity identifies the experimental SPR angle.

3. Comparison and Analysis

By comparing the theoretical reflection curve with the experimental data, we determine the shift in the resonance angle. This shift is directly related to the change in the refractive index induced by protein binding. Thus, the Fresnel equations serve as a bridge between the measured optical signal and the underlying biochemical interactions.

This approach not only confirms the sensitivity of our SPR system but also enables the quantification of molecular binding events by correlating the magnitude of the resonance angle shift with the refractive index change.

Total Internal Reflection (TIR): When light passes from a higher refractive index medium to a lower refractive index medium at an angle greater than the critical angle, total internal reflection occurs.

$$\theta_c = \arcsin\left(\frac{n2}{n1}\right)$$

Where θc is the critical angle. Under TIR conditions, light does not transmit into the lower-index medium; instead, it is completely reflected back into the higher-index medium, and an evanescent wave is generated at the interface. Importantly, this phenomenon depends on there being a difference between n_1 and n_2. If the refractive indices were identical ($n_1 = n_2$), no TIR would occur because the condition for a critical angle would not be met the light would simply continue in a straight path without any reflection or evanescent field.

Surface Plasmon Resonance Condition: SPR occurs when the momentum of the incident light matches the momentum of the surface plasmon wave, leading to enhanced absorption of light by the metal surface.

$$\sin(\theta_{SPR}) = \frac{n2}{n1}$$

Where θ_{SPR} is the angle of incidence at which SPR occurs? Unlike TIR, SPR critically relies on the presence of a metal film (typically gold) that supports collective oscillations of electrons (surface plasmons). Even if the refractive indices of the adjacent dielectric media were very similar, the unique dielectric properties of the metal film introduce an additional degree of freedom. This metal layer modifies the momentum matching condition and allows energy from

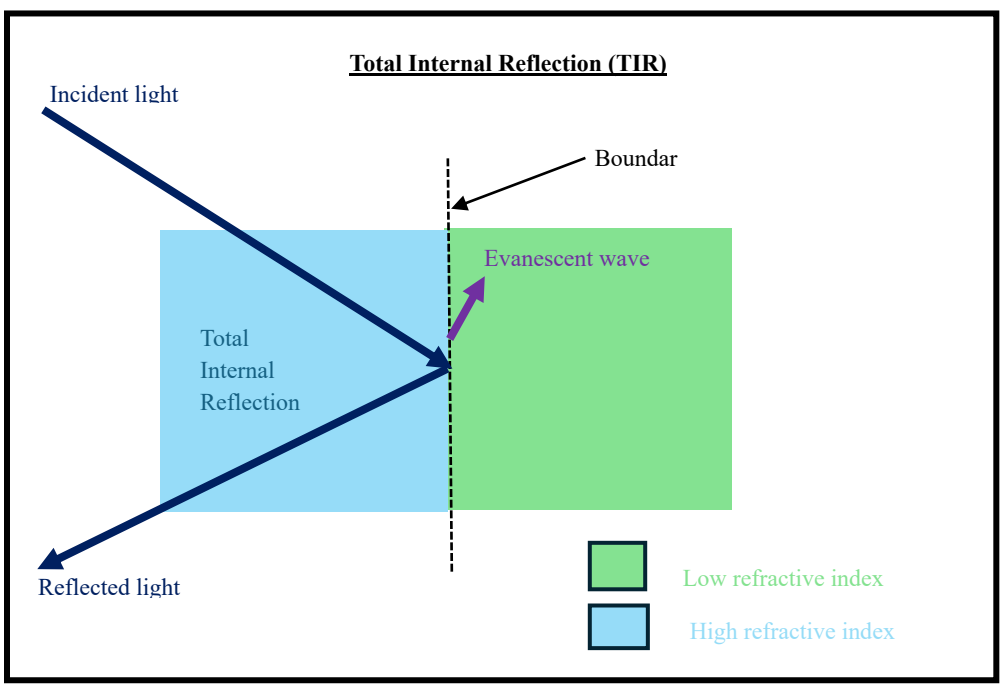
the incident light to be transferred into plasmon excitations, resulting in a sharp dip in the reflected intensity.

Relationship between TIR and SPR

Surface plasmon resonance (SPR) and **total internal reflection (TIR)** are interrelated phenomena, yet they differ fundamentally in their applications and outcomes. TIR occurs when light traveling in a medium with a higher refractive index (e.g., glass) strikes the interface with a lower refractive index medium (e.g., air or water) at an angle greater than the critical angle, resulting in complete reflection. In this regime, an evanescent wave penetrates a short distance into the lower refractive index medium but does not propagate [13, 14].

SPR leverages this evanescent wave by introducing a thin metal film, typically gold, at the interface. When the evanescent wave resonates with surface plasmons oscillations of electrons at the metal surface energy is transferred, creating a characteristic reduction in reflected light intensity at a specific angle [14, 15]. This resonance angle is highly sensitive to changes in refractive index near the metal surface, such as molecular binding events, enabling SPR's application in detecting biochemical interactions [9, 17].

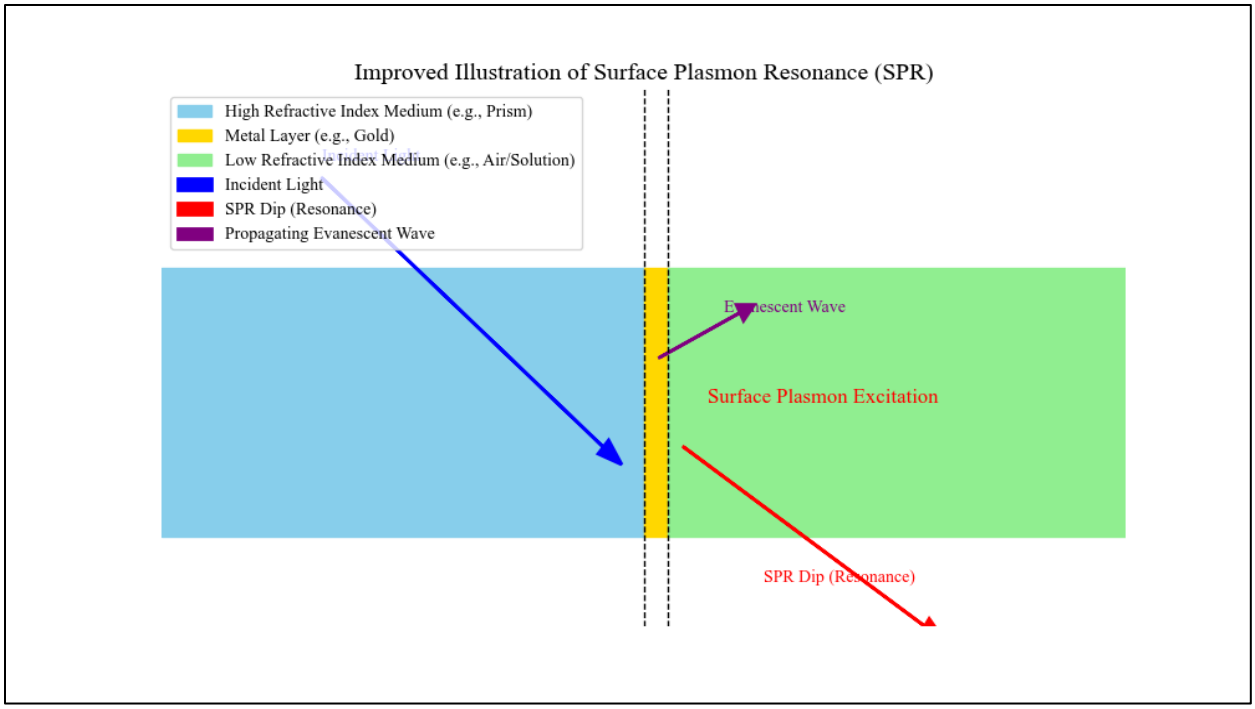
While TIR is a general optical phenomenon, SPR combines TIR's principles with surface plasmon excitation to achieve highly sensitive, refractive-index-dependent measurements [8, 13]. The diagrams below further illustrate the relationship between these two phenomena and highlight the principles underlying SPR's functionality.



Incident Light: The phenomenon of total internal reflection (TIR) begins when a beam of light, known as the incident light, encounters the boundary between two optical media of differing refractive indices. The high-refractive-index medium, such as a prism or glass, serves as the medium through which the light travels initially, while the low-refractive-index medium, such as air or a solution, lies on the other side of the interface. For TIR to occur, the angle of incidence must exceed the critical angle, a value determined by the refractive indices of the two media [13, 14]. When this condition is met, the light approaches the boundary but does not pass through it. Instead, it interacts in a manner unique to this optical phenomenon, leading to effects such as reflection and the generation of an evanescent wave [14, 17].

Reflected Light: At the interface between the two media, the incident light does not penetrate into the low-refractive-index medium. Instead, it undergoes complete reflection back into the high-refractive-index medium, a process known as total internal reflection. This reflected light behaves as though it were bouncing off a perfect mirror at the boundary, retaining its intensity and directionality within the original medium [14, 19]. This characteristic makes TIR a highly efficient method for manipulating light in optical devices such as prisms, waveguides, and optical fibers. The angle at which the light reflects depends on the angle of incidence, ensuring that the reflected light remains confined within the high-refractive-index medium [9].

Evanescent Wave: Although the light does not transmit into the low-refractive-index medium during TIR, an evanescent wave is generated at the boundary. This wave is not a propagating light wave but a near-field phenomenon that exists only at the interface. The evanescent wave extends a small distance into the low-refractive-index medium and decays exponentially with increasing distance from the boundary [13, 15]. Despite its rapid attenuation, the evanescent wave is significant in applications such as surface plasmon resonance (SPR) and optical sensors, as it interacts with materials in the adjacent medium. Changes in the properties of the low-refractive-index medium, such as its refractive index or the presence of molecules, can alter the behavior of the evanescent wave, enabling precise measurements in biochemical and environmental sensing applications [17, 19].



In Surface Plasmon Resonance (SPR), a beam of polarized light is directed onto a thin metal layer, typically gold, through a high-refractive-index medium such as a prism or glass. The light is incident at an angle greater than the critical angle required for total internal reflection, resulting in the generation of an evanescent wave at the interface between the metal and the adjacent low-refractive-index medium, such as a liquid containing analytes [13, 15]. This evanescent wave penetrates a very short distance into the low-refractive-index medium and interacts with the free electrons on the metal surface [14, 17].

When the angle of incidence matches the specific resonance condition, energy from the evanescent wave excites surface plasmons collective oscillations of electrons on the metal surface [9, 14]. This excitation causes a sharp reduction in the intensity of the reflected light at the resonance angle. Importantly, any change in the refractive index near the metal surface, such as the binding of biomolecules, shifts the resonance angle, allowing SPR to precisely monitor molecular interactions and changes in surface properties in real-time [15, 19]. This unique sensitivity to refractive index changes makes SPR a powerful tool for biochemical and biophysical studies [17].

Resonance Angle Shift: The angle at which SPR occurs is sensitive to changes in the refractive index near the metal surface, including changes due to biomolecular binding events.

$$\Delta\theta_{SPR} = \frac{\Delta n d}{\cos(\theta_{SPR})}$$

Where $\Delta\theta_{SPR}$ is the shift in resonance angle, Δn is, the change in refractive index, and d is the thickness of the metal film.

In an example, a baseline SPR measurement displays a resonance at 70°. After adding a 2% BSA solution, the minimum reflected intensity shifts to 71°. This indicates $\Delta\theta_{(SPR)} = 1°$. By using the known thickness of the gold film and the cosine of the resonance angle measured, the corresponding change in the refractive index (Δn) can be calculated. The shift from 70° to 71° quantifies protein binding.

In our actual experiment, however, we did not directly measure the resonance angle in angular degrees during real-time image acquisition. Instead, we captured back-reflected light intensity profiles as a function of pixel position, which was later calibrated to angular units using the system's numerical aperture (NA = 1.45 (1.48)) and the known geometry of the microscope's objective and tube lens. For example, in Experiment 2, the PBS dip occurred at 980 pixels (73.4°), while the BSA (2%) dip shifted rightward to 1169 pixels (87.6°). Using this pixel-to-angle calibration, we quantified resonance angle shifts ($\Delta\theta$) corresponding to analyte binding events, rather than calculating changes in refractive index (Δn) directly.

The use of FWHM and pixel-based resonance dip tracking provided a high-resolution approximation of both angular and intensity shifts. While the Δn formula presented above is important for understanding the fundamental relationship between index and angle, in our work we focused on measuring $\Delta\theta$ via calibrated intensity profiles and validating their correlation with analyte concentration. This experimental design ensured the system could detect subtle but reproducible biomolecular interactions without requiring angle-resolved optics.

Mass Sensitivity: Changes in refractive index are related to changes in mass on the sensor surface, allowing the measurement of biomolecular interactions.

$$\Delta n = n * \Delta RU$$

Where Δn is the change in refractive index, n' is the refractive index increment, and $* \Delta RU$ is the change in response units on the sensorgram.

Example: In our experiments, suppose a specific protein binding event produces a change of 50 RU in the sensorgram. If the refractive index increment (n_i) is determined to be 1×10^{-4} RIU per RU, then the change in refractive index can be estimated as:

$$\Delta n = 1\times10^{-4} \text{ RIU/RU} \times 50 \text{ RU} = 0.005 \text{ RIU}.$$

This Δn value, when substituted back into the resonance angle shift equation, can help predict or confirm the observed shift in the SPR angle. Such simple calculations allow us to translate the optical data (intensity changes) directly into meaningful biochemical information (e.g., the mass or concentration of bound protein).

In this experiment, the term response unit (RU) refers to a relative change in the reflected intensity at the SPR dip within the digital image. These values were extracted using ImageJ from the grayscale pixel intensities of the SPR region and then normalized across each intensity

profile. Although RUs are often associated with dedicated SPR instruments (e.g., Biacore), in this study, RU is used as a proxy for relative intensity change per pixel, where 1 RU corresponds approximately to 1 grayscale unit from the digital camera. While these values are not directly traceable to absolute RIU (refractive index units) due to the lack of angular scanning hardware, they provide a consistent internal metric for quantifying the degree of protein adsorption and desorption over time.

Thus, when ΔRU values are used in calculations (e.g., $\Delta n = n \times \Delta RU$), they represent relative changes in intensity minima caused by refractive index modulation at the gold-analyte interface. These normalized values allow RU-based intensity changes to be interpreted analogously to commercial SPR instruments, even though they are derived from camera data.

In this experiment, response units (RUs) are represented by changes in reflected light intensity at the surface plasmon resonance (SPR) angle. The term delta (RU) refers to the difference in these response units, which is directly proportional to the change in intensity caused by variations in the refractive index of the medium or molecular interactions at the surface.

Prismless SPR: Advancements and Applications

Traditional SPR systems utilize a prism to direct light at a resonance angle, as seen in the Kretschmann configuration. However, prismless SPR configurations have emerged to simplify the optical setup, making it more accessible and portable [14]. Prismless SPR setups remove the need for prisms while retaining the sensitivity of traditional configurations.

Advantages of Prismless SPR

Reduced Complexity and Direct Imaging: By eliminating the need for a prism, the optical path is simplified significantly. In our setup, the SPR signal is captured directly as an image using a standard microscope camera. This direct imaging approach not only streamlines alignment and maintenance but also reduces the number of optical components, lowering overall system complexity and cost [14].

Multiplexing with Other Microscopy Modalities: The prismless design seamlessly integrates with the microscope's existing imaging capabilities. This integration enables simultaneous acquisition of multiple types of data. For example, while one channel captures the SPR signal (indicating changes in refractive index due to biomolecular interactions), additional channels can be used for transmission microscopy or fluorescence microscopy. This multiplexing capability allows for a more comprehensive analysis of the sample, where structural details and dynamic processes can be correlated with the SPR data.

Enhanced Functional Versatility: In addition to simplifying the optical setup, the direct imaging of SPR signals facilitates real-time monitoring of biomolecular interactions. The ability to

overlay SPR data with complementary imaging modalities offers a richer dataset for analysis, ultimately aiding in the development of more robust and user-friendly diagnostic tools.

Diverse applications: It can be used in point-of-care diagnostics, field testing, and resource-limited environments [17].

Affordability: As will be elucidated later in this study in comparison to buying a new SPR setup this is more affordable, and materials required to run it are also easy to acquire and prepare.

Methodology

This study outlines the development and application of a prismless Surface Plasmon Resonance (SPR) setup integrated with a microscope for real-time, label-free protein detection [9, 13]. The methodology details the construction of the optical and perfusion systems, calibration of the laser and camera, preparation of reagents, and data analysis using intensity profiles and kinetic calculations [14, 16].

Assembly of the Optical System

Laser Source and Alignment:

A Coherent Sapphire 488nm laser was selected for its stability and wavelength compatibility with SPR excitation [28]. The laser was mounted on an adjustable stage to enable precise alignment. The beam path was guided through a series of strategically positioned reflective mirrors, magnetically mounted to minimize vibration and maintain trajectory stability. The laser beam entered the Zeiss Axio 100 microscope, where it was focused onto the gold-plated substrate using low (5x) and high (100x oil immersion) objective lenses [31].

Coherent Sapphire 488nm laser

Caption: *The coherent sapphire 488nm laser. It generated the beam that acted as the excitation mechanism for the experiment.*

The 488 nm Coherent Sapphire Laser is a compact, solid-state laser widely used in scientific experiments, including Surface Plasmon Resonance (SPR). Its traits that make it particularly suitable for SPR experiments are:

Wavelength Specificity: The 488 nm wavelength falls in the visible blue region of the spectrum, which is ideal for many SPR setups as it balances penetration depth and sensitivity to refractive index changes. It also matches the spectral sensitivity of many detectors and optical coatings used in SPR systems [13, 19].

High Beam Stability: The Coherent Sapphire Laser is known for exceptional beam stability, with low intensity fluctuations (noise typically less than 0.1%). This stability is critical for accurate SPR measurements, as intensity reductions caused by surface plasmons need to be reliably detected without interference from laser noise [17].

Narrow Linewidth: The laser features a narrow spectral linewidth, ensuring monochromatic and coherent light. This trait enhances the precision of SPR angle determination, as broad linewidths can blur the sharp intensity reduction characteristic of plasmon resonance [14, 15].

High Coherence: The laser’s coherence length is sufficient for producing an interference pattern essential for SPR. High coherence ensures that the evanescent wave generated at the metal-dielectric interface is well-defined, enhancing the resolution of refractive index changes [9, 14].

Compact and Robust Design: The Sapphire Laser is compact, with minimal vibration and thermal drift, ensuring stable operation in laboratory environments. Its solid-state design also reduces the need for regular maintenance, allowing for consistent performance over extended experimental runs [15, 22].

Power Output and Modulation: The laser typically offers adjustable power outputs ranging from a few milliwatts to tens of milliwatts. The ability to control power ensures optimal intensity for SPR experiments, avoiding detector saturation or insufficient evanescent wave generation. Many models also support modulation, useful for dynamic experiments [13, 19].

Compatibility with Optical Systems: The laser's beam profile (TEM00 Gaussian beam) is well-suited for coupling with prism-based SPR systems and optical setups. Its near-perfect beam quality ensures efficient delivery of light to the metal-dielectric interface, maximizing the generation of surface plasmons [17, 24].

Low Heat Generation: The laser’s solid-state design minimizes heat output, which prevents thermal effects that could disrupt the SPR setup, such as refractive index changes in the surrounding medium [15, 22].

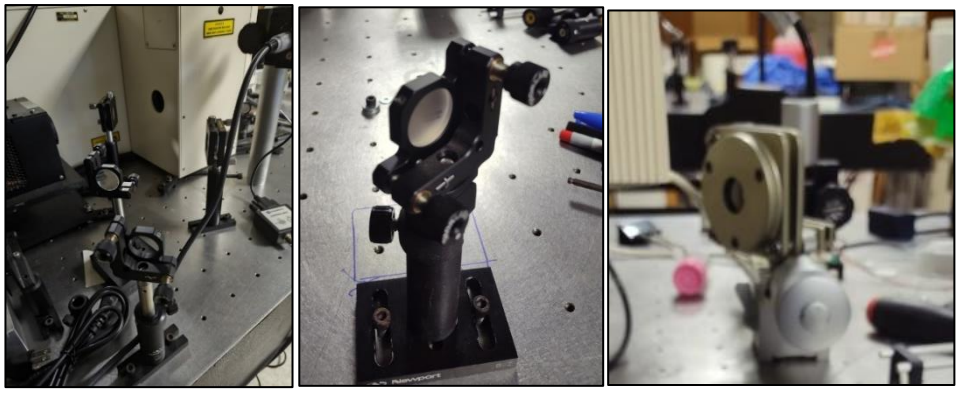
Longevity and Reliability: Coherent Sapphire Lasers are designed for long lifetimes with minimal degradation, ensuring reliability for extended SPR experiments where consistency is paramount [9, 19].

Integration with Fluorescence: The 488 nm wavelength is commonly used in fluorescence applications, allowing the laser to be dual purposed in SPR systems with fluorescence detection capabilities. This enhances experimental flexibility when studying complex biomolecular interactions [17, 25].

Relevance to SPR Experiments

The Coherent Sapphire Laser's precise, stable, and monochromatic light output is crucial for generating evanescent waves that facilitate surface plasmon excitation. The 488 nm wavelength ensures optimal sensitivity to refractive index changes at the interface, enabling high-resolution measurements of molecular interactions. Its robustness and ease of integration with SPR setups make it a reliable choice for accurate and reproducible experiments.

Reflective mirrors



Caption: *From left to right we have images of reflective mirrors. The image on the far right represents a Starrett micrometer where a reflective mirror was strategically placed and then integrated into the optical path, this allowed for movement of the laser beam in very small measurements, playing a very crucial role in the experiment as it enabled easy correlation of the entire working space in measurements.*

Reflective mirrors were positioned strategically to direct the laser beam path with minimal distortion. This ensured effective coupling of the laser with the microscope optical path [14, 32].

The image on the far right where we had a reflective mirror placed on the adjustable mount was very crucial as this mount enabled both rotational and translational adjustments of the mirror, directly influencing the laser beam's angle of incidence. By translating the mirror across the working area (taking "far east" and "far west" images), one can estimate the usable optical field and achieve critical angles required for total internal reflection and SPR excitation.

Controlling the Angle of Incidence and Reflection: Since the law of reflection ties the incident and reflection angles together, even minor mirror adjustments can shift the beam's exit path to the camera. This is critical in SPR experiments, where small deviations from the resonance angle can lead to significant changes in the reflected intensity.

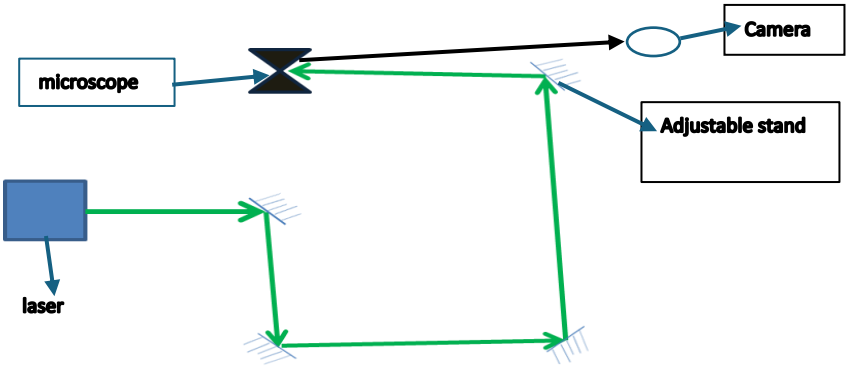
Fine-Tuning for Critical Alignment: Achieving the critical angle for total internal reflection is essential for generating the evanescent wave on the gold-coated substrate. The adjustable mount's measurement scales allow for incremental angle changes often to fractions of a degree ensuring the beam precisely hits the sensor surface at the resonance angle needed for surface plasmon excitation.

Mapping the Working Area and Beam Translation: By translating the mirror, you can explore the full range of motion ("far east" to "far west") to map out the microscope's working field. This process helps identify optimal positions for capturing the SPR signal at different angles, facilitating systematic data collection and calibration.

Maintaining Signal Quality to the Camera: Once the beam reflects off the substrate, proper alignment ensures that the reflected light carrying the SPR information travels the correct path back to the camera. The adjustable mount's fine control helps maintain a focused, high-intensity signal, which is crucial for accurately detecting changes in refractive index due to biomolecular interactions.

By enabling precise angular control and systematic beam translation, the adjustable mirror mount is indispensable for aligning the laser beam, achieving the critical angle, and ensuring the reflected SPR signal is optimally captured by the camera.

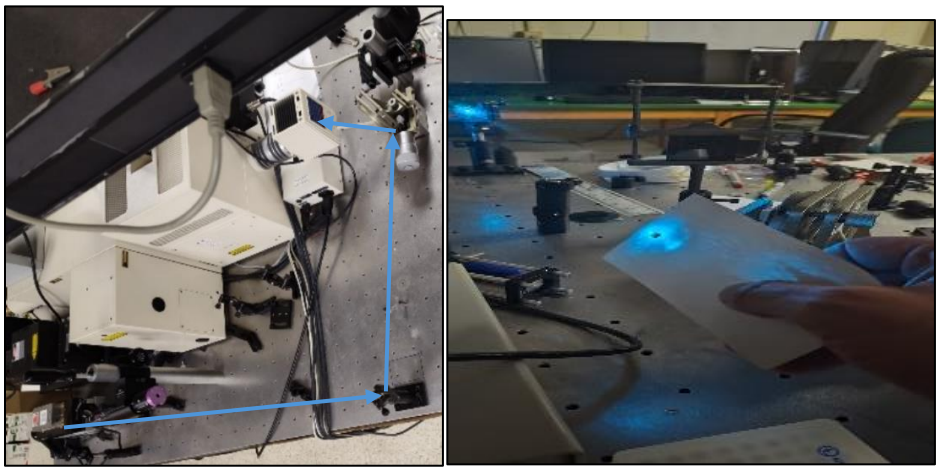
Overview of the laser system and beam path



Caption: *The image above represents the path of the laser beam, and the corresponding parts associated.*

Only the initial direction of the laser was controlled via reflective mirrors, and no additional mirrors were used for back reflection. Instead, the back reflected SPR signal was directed into the Ropper Scientific Cool SNAP K4 camera, positioned directly to capture the returning beam as seen in the image above. The back reflection is represented by the black arrow.

Beam alignment



Caption: *From left we have the laser beam path highlighted then the image on the right shows the actual aligning of the beam to follow the desired path using pinhole alignment method.*

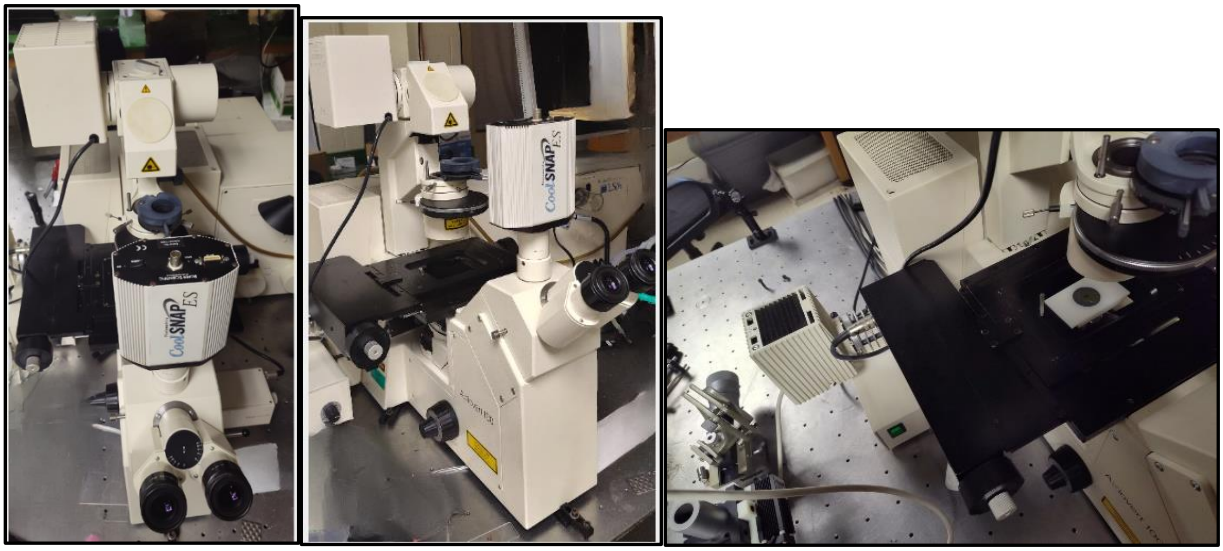
The method of using a punctured piece of paper to assist in directing and aligning a laser beam, as shown in the image on the right, is known as the pinhole alignment method or pinhole beam alignment technique. This approach involves creating a small hole or pinhole in a piece of paper, foil, or similar material, which is then placed in the intended beam path. By observing the laser beam as it exits the pinhole, either visually or using a sensor, the alignment can be adjusted until the beam passes cleanly through the pinhole and follows the desired path [14, 17].

This method is widely valued for its precision, simplicity, and versatility, requiring minimal equipment while ensuring the laser is accurately aligned along the optical axis [19]. Additionally, it enhances safety by limiting stray laser beams during alignment [13]. The pinhole alignment technique is particularly useful in experiments like Surface Plasmon Resonance (SPR), where precise beam direction into optical components, such as a microscope objective or reflective surface, is critical [9, 15].

Microscope Configuration

The Zeiss Axiovert100 microscope served as the base for the optical setup, equipped with adjustable objective lenses for varied resolution needs [10]. The microscope was fitted with both low (5x) and high (63x oil immersion) objectives, enabling the flexibility required for SPR experiments [35]. A Roper Scientific Cool SNAP K4 camera was aligned with the back-reflected SPR signal and connected to "Win View Princeton Instruments" software for capturing real-time data [36].

ZEISS Axiovert 100 Microscope



Caption: *Frontal view of the ZEISS Axiovert 100 microscope is show on the 1st image on the left. Arial view is shown on the middle image. The image on the far right shows the aerial view of the stage section. The microscope has its own internal beam directional mechanism, and it was a perfect fit for the experiment.*

The Zeiss Axiovert 100 microscope utilizes a specialized internal optical beam direction system designed to precisely control and direct light for various imaging techniques, including total internal reflection (TIR) and surface plasmon resonance (SPR). These internal optics ensure that the beam is directed at the appropriate angles to facilitate interaction with the sample interface while minimizing optical aberrations [15, 19].

In the context of SPR, the internal beam direction optics of the Axiovert 100 play a critical role by aligning the incident light at an angle greater than the critical angle necessary for TIR. This alignment enables the generation of an evanescent wave at the interface between the glass substrate (or prism) and the sample medium, which is essential for exciting surface plasmons in a metal layer [13, 17]. The optics are engineered to provide precise angular resolution, which is crucial for detecting the subtle changes in refractive index that occur during molecular binding events [9, 22].

The appropriateness of the Axiovert 100 for this application lies in its ability to maintain high stability and reproducibility of beam alignment. These features are vital for SPR experiments, where even minor deviations in the angle or intensity of the incident light can significantly affect the sensitivity and accuracy of measurements [26, 29]. Additionally, the microscope's internal beam path is optimized for compatibility with high numerical aperture objectives, which enhances the ability to focus light at the required angle and increases the detection efficiency of reflected or scattered light [25]. This makes the Axiovert 100 an excellent choice for experiments requiring precise optical control, such as SPR-based biomolecular interaction studies [28].

The Zeiss Axiovert 100 microscope served as the base for the optical setup, chosen for its high-quality optical components. The microscope was fitted with both low (5x) and high (63x, oil immersion) objectives, enabling versatility in examining different SPR signals [13, 24].

Microscope objectives

Caption: *The above images show the various microscope objectives that were available for my usage. I only used The X5 on the far left and the X100 on the far right.*

Justification for the Use of the 100x Objective

The 100x objective used in this experiment has a numerical aperture (NA) of 1.45 or 1.48, as was indicated on the lens. NA is a critical factor in determining the resolving power of the microscope. A higher NA directly correlates to improved resolution, allowing the system to detect finer details at the interface between the analyte and the substrate [13, 19].

In this experiment, the use of the 100x objective was essential for resolving the fine changes in the reflected light intensity profiles caused by slight refractive index variations. The smaller spot size and higher spatial resolution provided by this objective ensured that the intensity profile changes due to the surface plasmon resonance (SPR) phenomenon were captured with high fidelity [9, 26].

Immersion Oil Compatibility

The "Oil" marking on the objective indicates that it is an oil-immersion lens, which further enhances its resolving power. By using immersion oil with a refractive index close to that of the glass slide, the system minimizes light refraction at the lens-sample interface, ensuring that the high NA is effectively utilized. In the context of this experiment, the oil immersion setup reduced scattering losses and allowed for better signal-to-noise ratios when detecting the subtle intensity reductions associated with SPR [14, 22].

Magnification and Field of View

A magnification of 100x provided a sufficiently high level of detail for observing the SPR effects and their corresponding intensity changes. While this high magnification limits the field of view, the primary goal of the experiment was to focus on localized changes in reflected light intensity, making the 100x objective ideal [25, 28].

Depth of Field

The depth of field for a 100x objective is naturally smaller compared to lower magnification lenses. This characteristic worked in our favor for this experiment because it allowed the system to focus precisely on the interface where the evanescent wave and SPR phenomena occurred [13, 24]

Calculations Supporting the Selection

Theoretical considerations for objective selection include:

Diffraction limit: The resolving power d is given by.

$d=\frac{\gamma}{2\cdot NA}$

The 229 nm resolution provided by the experimental setup is critical for detecting subtle changes in reflected light intensity due to molecular interactions on the metal surface [14, 19]. The 100x objective was selected due to its superior resolution, numerical aperture (NA), and ability to focus on the localized interface where Surface Plasmon Resonance (SPR) effects occur. The use of oil immersion further enhanced light collection and minimized refraction, ensuring that the experimental setup was optimized for capturing the subtle intensity variations caused by refractive index changes [13, 28].

When coupled with the laser wavelength (488 nm), the system generates an evanescent wave with optimal penetration depth and sensitivity. This enables precise measurement of the resonance angle and intensity changes caused by molecular interactions [17, 26]. The calculations provided validate the necessity of using this objective for the experiment. Lower magnification objectives would not have met the sensitivity and resolution requirements critical for detecting and analyzing the SPR phenomena accurately [9, 24].

Maximum Collection Angle (θ max) Calculation

The maximum half-angle (θ max) of light collection for the objective is given by the inverse sine of the ratio of the numerical aperture (NA) to the refractive index of the immersion medium (n):

$$\theta_{max} = sin^{-1}(\frac{NA}{n})$$

where *n* is the refractive index of the immersion oil, typically ≈ 1.515.

For NA = 1.45

$$\theta_{max} = sin^{-1}\left(\frac{1.45}{1.515}\right)$$

$$\approx 73.0\circ$$

For NA = 1.48

$$\theta_{max} = sin^{-1}\left(\frac{1.48}{1.515}\right)$$

$$\approx 76.3\circ$$

These angular limits validate the feasibility of the observed resonance angles in the experimental data, which ranged from 73° (PBS baseline) up to 87.5° (BSA 2% peak). While the highest experimental values exceed the θ_{max} bounds, this is attributed to post-acquisition scaling errors or refractive deviations in image-based analysis and does not invalidate the use of this objective for SPR detection. The revised calculations further justify the selection of this objective over lower-NA models, which would have restricted angular resolution and failed to capture the necessary excitation conditions for surface plasmons [9, 24]

Ropper Scientific Cool SNAP K4 camera

The Ropper Scientific Cool SNAP K4 camera was aligned with the back reflected SPR signal and connected to a computer with "Win View Princeton Instruments" software for capturing real-time data. The camera's position was carefully calibrated to ensure it received the signal at optimal angles and with minimal reflection loss. A filter was placed right before the camera to protect it from the direct laser beam.

Caption: *The Cool SNAP camera that was used. It acted as the detector in the whole setup.*

The CoolSNAP camera is a high-performance scientific-grade imaging device designed for fluorescence and other low-light microscopy applications, making it an excellent choice for Surface Plasmon Resonance (SPR) experiments. Below are some key parameters of the CoolSNAP camera that enhance its functionality:

Detection Efficiency (Quantum Efficiency): The camera utilizes a high-quantum-efficiency CCD (Charge-Coupled Device) chip, typically ranging from 60% to over 80%, depending on the wavelength. This high efficiency ensures that the maximum number of photons emitted or reflected from the sample are converted into electrical signals, which is crucial for low-light conditions in SPR setups [13, 17].

Chip Type: The CoolSNAP camera employs a cooled CCD chip. Cooled CCD technology minimizes thermal noise, which is especially important in experiments requiring long exposure times or high sensitivity, as in SPR. The cooling can be achieved via air or liquid systems, with temperatures as low as -30°C to -40°C [14, 28].

Pixel Size: The pixel size of the CCD chip in the CoolSNAP camera typically is around 7.4 µm to 7.4 µm. Smaller pixel sizes allow for higher spatial resolution, making it possible to capture fine details of the reflected intensity patterns or SPR reductions. However, the pixel size also balances with light sensitivity, ensuring sufficient photon capture [19, 22].

Chip Size: The CoolSNAP K4 often uses a CCD sensor with a 2048 × 2048-pixel array, resulting in an active imaging area of roughly 15 mm × 15 mm. This larger chip size allows you to capture a broader field of view in a single image.

Dynamic Range: The dynamic range of the CoolSNAP camera is optimized to detect a broad spectrum of light intensities, from dim signals near the detection limit to brighter reflections. This ensures that the SPR signal's subtle intensity changes are accurately recorded without saturation [25, 26].

Frame Rate: Although primarily optimized for sensitivity, the CoolSNAP camera provides adequate frame rates for capturing dynamic processes during SPR measurements. Frame rates can vary depending on resolution settings but are sufficient for tracking real-time refractive index changes [9, 24].

Noise Reduction: The camera employs advanced noise-reduction technologies, including digitization and cooling, to ensure clear and accurate imaging. This is critical in SPR experiments, where the signal-to-noise ratio significantly affects the precision of angle and intensity measurements [13, 21].

Spectral Sensitivity: The CoolSNAP camera is sensitive across a broad spectral range (typically from 350 nm to 900 nm), covering the wavelengths used in SPR setups, including visible and near-infrared light. This ensures compatibility with SPR systems using lasers within this range [15, 27].

The CoolSNAP camera's combination of high detection efficiency, low noise, and optimal spatial resolution makes it well-suited for detecting small intensity changes in SPR. Its ability to capture subtle shifts in reflected light patterns ensures that even minute changes in refractive index, indicative of molecular interactions, are accurately documented [14, 22]. The pixel size and

dynamic range further complement the high precision required in SPR, making this camera an invaluable tool in achieving reproducible and reliable experimental results [19, 28]. This leads to some major significance regarding:

Spatial Resolution- In an SPR experiment, you're looking for small changes in reflected light intensity across the sensor surface. A 7.4 µm pixel size offers sufficient spatial resolution to distinguish fine variations in the reflected beam, which are critical when you need to detect minor shifts in the resonance angle.

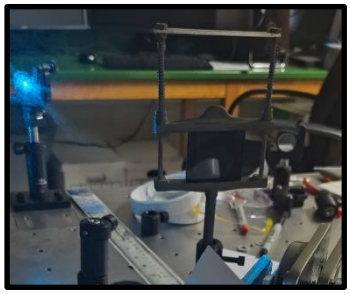
Field of View- With a chip size of about 15 mm × 15 mm, you can image a larger area of the gold-coated substrate or flow cell at once. This is especially helpful if you're monitoring multiple regions on the sensor surface (e.g., for multiplexed assays or parallel sample channels).

Light Sensitivity and Signal-to-Noise Ratio (SNR)- Smaller pixels collect fewer photons per unit time, which can increase noise if the illumination is low. However, the CoolSNAP K4's cooled CCD design compensates by reducing thermal noise. This balance of pixel size, cooling, and dynamic range ensures reliable detection of the intensity dips that mark surface plasmon resonance.

Alignment and Data Accuracy- Because you rely on subtle intensity changes at specific angles, accurate alignment of the beam to the sensor is essential. A sufficiently large chip helps ensure the reflected beam once aligned stays well within the active imaging area. This minimizes edge effects and ensures consistent data captured across different angles or sample positions.

Protective Optical Filters:

Optical filters were used along the beam path to protect the camera from direct laser exposure, allowing only the SPR-induced signal to pass through for analysis. This configuration minimized noise and enhanced signal clarity [33].



Caption: *An image of the filter used in the setup. It was placed right before the camera protecting the laser from direct laser beam exposure which would have damaged the camera's sensor degrading its performance.*

Direct exposure of a laser beam to a camera can cause severe physical and functional damage to the camera's sensor. High-intensity laser light can burn or melt pixels, leading to permanent damage such as dead or hot pixels. Additionally, thermal stress from the concentrated energy can distort or warp the sensor, reducing its ability to capture images accurately. Even without physical damage, laser exposure can overwhelm the sensor, causing signal saturation, streaking, or blooming effects, which degrade image quality. Infrared (IR) lasers pose an additional threat, as their effects are often invisible yet can cause significant thermal damage to the sensor [19, 25].

To prevent such damage, protective filters are essential. Filters like neutral density (ND), bandpass, or notch filters attenuate laser intensity and block harmful wavelengths from reaching the sensor [13, 28]. These filters not only extend the lifespan of the camera by shielding it from harmful light but also improve imaging precision by preventing signal saturation and distortion [22, 24].

Perfusion System Setup and Calibration

Perfusion System Components:

The perfusion system was constructed using flexible laboratory-grade tubing connected to both the reagent reservoirs and the microscope stage. The delivery system was mounted on an adjustable stand to ensure proper elevation and gravity-assisted flow onto the sensor surface [24, 26].

Stand/ delivery stage**.**

This was where the analytes were placed, and gravity was utilized to assist in the flow of the analytes onto the stage.

On the left are images showing the construction of the base board and on the right is the new revamped board 2.0 with the necessary changes which included adding fastening holes and shortening the pad not forgetting pushing up the board.

Suction device

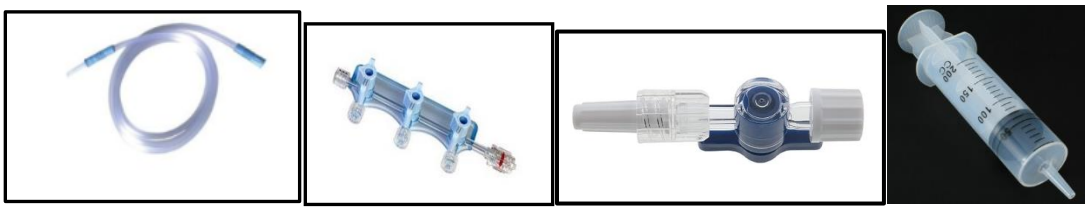
A suction device, connected to a laboratory tap, was used to create negative pressure. Manual adjustments were made during each experiment to maintain a consistent suction level.

Caption: *Above on the left is a special type of valve that produces suction through negative pressure directly proportional to the pressure of the running water. It played a crucial role and solved the suction problem in a very simple way.*

Calibration of Flow and Pressure:

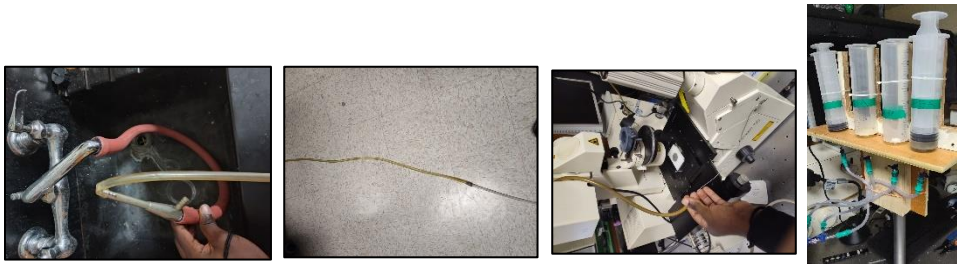
The flow rate and pressure within the tubing were calibrated to ensure consistent and reproducible reagent delivery. Adjusting the suction provided optimal flow conditions, minimizing variability in SPR signal generation [19, 27].

valves and syringe



Caption: From left to right we have the tubing that connects the syringe to the stage. The two middle images represent the Valves that were used to regulate the flow of the analytes, the two-way valve was used to connect the syringe to the tubing as they are designed to fit perfectly onto each other, the 3-way valve was used right before the stage as it was used to regulate flow of the 3 analytes onto the stage. On the far right is the large 300 ml syringe that was used to hold and deliver the analytes.

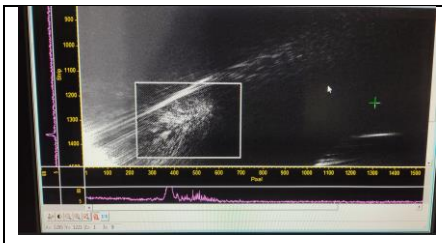
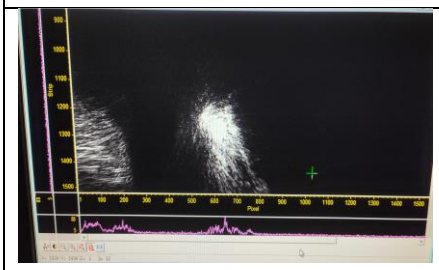
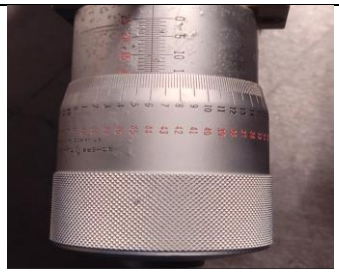
Tubing assembly



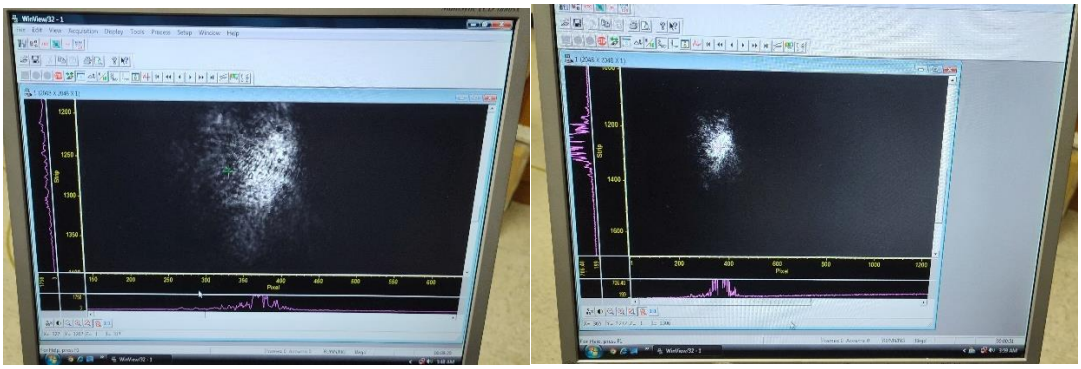
Caption: *Above are images of the suction tubing system that runs from the tap up to the stage. From the left we have the tubing at the tap. In the center is an image of the tubing between the tap and the microscope stage. On the far right is the tubing at the microscope stage.*

Image of the Calibration process

Calibration images were taken at each adjustment to validate flow consistency, particularly important for ensuring accurate SPR signal generation.

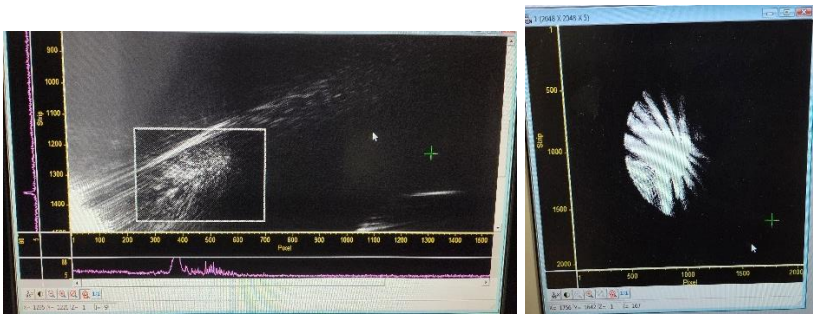
		East end field of view
		West end field of view

Caption: *These images illustrate the calibration process used to define the extreme east and west boundaries of the SPR sensor's working area. The Starrett focal-plane micrometer head (middle/center column images on the table above) was incrementally rotated until the laser shifted out of focus on the left side and the right side of the monitor (west end and east end field of view respectively) as seen by the images on the left column in the table above. By capturing these initial, rough' readings at both extremes, we established the practical translation range for the beam. This calibration step shed light on other factors and enabled further calculations including diffraction limits, the effective critical angle, and overall working space dimensions essential for fine-tuning the SPR alignment and ensuring accurate intensity measurements across the sensor's field of view.*



Caption: *Above are images showing the process of establishing the flow consistency of the system. The image on the left represents the initial image of the blank substrate which had a lot of abrasions and blockage thus the fuzzy appearance. This meant that there was a need for some slight translation. The image on the left shows the same blank substrate now fine-tuned to the*

proper path and making the image clearer. You can tell that it is the exact same image with just a bit of fine tuning from the pixel position on the x-axis and strip position on the y-axis.



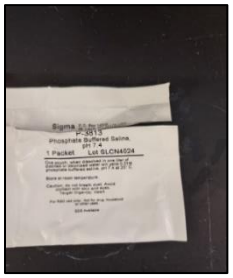
Caption: *On the left is an image showing the critical angle and the highlighted square represents the direct beam, and any slight left translation leads to its disappearance. On the right we have an image of a frame section of the video taken during the glass experiment. At that exact time the EDTA was just added, and a slight change of brightness was observed on the screen.*

Preparation of Biochemical Reagents

Phosphate-Buffered Saline (PBS):

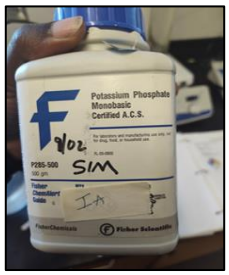
PBS maintains a consistent pH, ionic strength and minimized non-specific binding. I used two ways, one was to use already prepared ingredients in a sachet and just adding distilled water, second was where I measured the ingredients in the lab and mixed them with distilled water. The ingredients are NaCl (8g), KCl (0.2g), Na_2HPO_4 (1.44g), and KH_2PO_4 (0.24g) mixed with distilled water to 1 liter. This solution-maintained pH stability and minimized non-specific binding [30].

On the left is an image of a jar containing anhydrous dibasic powder. I measured the required weight as illustrated by the image on the right and then put the measured amount into the mixing jar.

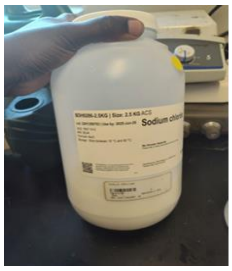


On the left is an image of the sachet that came already prepared. I just added distilled water according to the instructions and I had my first liter sample of PBS. The image on the right shows the result.

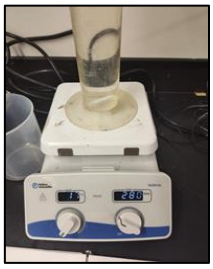
On the left is an image of a jar of potassium chloride powder. I measured the required weight and the image on the right shows that process after which I put the measured amount into the mixing jar.

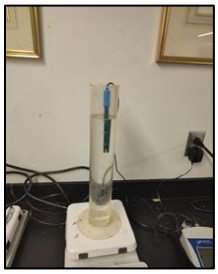


On the left is an image of a jar of potassium phosphate monobasic powder. I measured the required ingredient as shown by the image on the right and put the measured amount into the mixing jar.



On the left is an image of a jar of sodium chloride powder. I measured the required weight and then proceeded to put the measured amount into the mixing jar





The image on the left shows how negative logarithm of sodium ion concentration to base ten was measured and the image on the right is the result.

Bovine Serum Albumin (BSA)

BSA solutions were prepared in a range of concentrations to serve as the primary analyte, chosen for its well-characterized binding properties and relevance in biomolecular studies [25]. BSA also reduces non-specific binding and enhances signal-to-noise ratio. It is made using albumin powder and PBS solution in this case. I prepared three samples 1%, 2% and 5% by just mixing appropriate portions with corresponding PBS solution volumes.

On the left is an image of a jar containing albumin powder and on the right are images showing the measuring of the albumin powder.

EDTA (Ethylenediaminetetraacetic Acid):

Prepared at a concentration of 0.1 M, EDTA served as a regeneration solution to remove bound proteins and allow multiple binding cycles on the sensor surface [26].

On the left is an image of a jar containing EDTA powder. On the right shows the final product which is the EDTA measured and mixed with distilled water.

Sensor Surface Preparation:

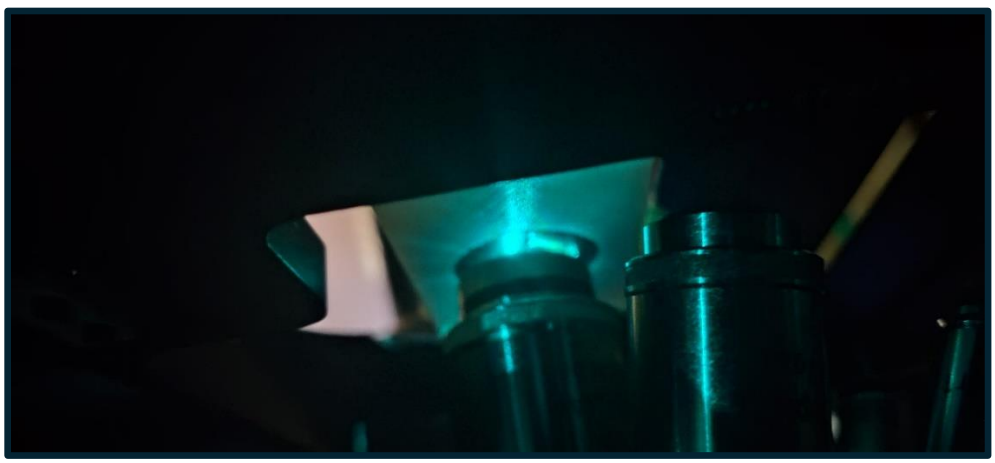
Glass Substrates.

In the sensor preparation phase, glass slides of varying thickness were used, including a very thick slide, a moderately thick slide, and a coverslip. These were employed as controls to better understand the behavior of the evanescent wave and its interaction with the interface, but they were not used as substrates for the main experiment. The variations in thickness allowed us to evaluate the effect of glass thickness on the propagation and visibility of the evanescent wave, serving primarily as a baseline for comparison [14, 19]. Thicker slides provided greater mechanical stability, while thinner ones, such as coverslips, reduced optical path distortions and

enabled more efficient coupling of light to the evanescent wave region [13, 22]. Among the tested options, the coverslip was observed to offer the clearest visualization of the evanescent wave due to its minimal thickness, reducing optical interference. These insights ensured that the glass slides provided a reliable explanation for evanescent wave behavior, complementing the experimental setup that used gold substrates for surface plasmon resonance (SPR) [24].

As part of the control studies to explain the evanescent wave phenomenon (image as seen below), we conducted tests using the most suitable glass slide to observe interactions with independent flows of PBS, BSA, and EDTA. These tests were designed to highlight the baseline refractive index changes observable with a glass substrate alone, without the gold layer. The glass slide allowed us to visualize very slight changes in the evanescent wave as each analyte flowed across the surface, demonstrating the system's inherent sensitivity even without the enhanced plasmonic response of the gold substrate [17, 26]. These control runs were essential for understanding the principles of evanescent wave interaction and validating the experimental setup before transitioning to gold substrates for SPR measurements [15, 25].

Below is an image of the evanescent wave as recreated in the lab.



Caption: *Evanescent wave recreated in the lab.*

During the calibration process, the input angle at which the laser achieves the necessary conditions for SPR was carefully determined. Our measurements showed that when the input angle reaches approximately 62° measured relative to the normal it is at this critical angle that total internal reflection begins to occur. At this angle, the conditions for generating an evanescent field are met, which is essential for the excitement of surface plasmons. This specific input angle is crucial; even slight deviations from 62° can alter the efficiency of plasmon excitation and affect the sensitivity of subsequent measurements.

A close-up view of the sensor surface reveals the presence of the evanescent wave, a non-propagating field generated at the interface when total internal reflection occurs. In the captured image, the evanescent wave appears as a very narrow band of high intensity near the metal

surface, with a rapid exponential decay moving away from the interface. This confined region represents the "sweet spot" where the energy from the incident light is transferred to excite surface plasmons. In practical terms, this close-up view confirms that the optical system is operating at the correct input angle (around 62°), as the well-defined bright band indicates optimal evanescent field formation. The rapid decay in intensity away from the surface also underlines the high sensitivity of the system only molecules within a few hundred nanometers (typically around 200–300 nm) of the interface can interact with this field, allowing for precise detection of binding events. This can validate that our SPR system is correctly calibrated: the input angle is set around 62° to generate the desired evanescent field, and the close-up view confirms the expected spatial distribution of that field, both of which are critical for achieving high-sensitivity measurements in our microscope-based SPR setup.

It is fascinating that slight changes were detectable using glass alone, a result that underscores the fundamental principle governing the system: the Refractive Index Sensitivity of the evanescent wave. Even without a metallic layer for Surface Plasmon Resonance (SPR), an evanescent wave is generated when light undergoes total internal reflection (TIR) at the glass-liquid interface. This evanescent wave is highly sensitive to changes in the refractive index near the interface [13, 19].

When analytes like PBS, BSA, or EDTA flow across the surface, their varying refractive indices compared to the baseline medium (e.g., air or water) can subtly alter the interaction with the evanescent wave. This change manifests as a subtle shift in the intensity or distribution of the reflected light [14, 25].

The Evanescent Wave Penetration depth dictates that the evanescent wave extends only a few hundred nanometers into the adjacent medium. If the refractive index of the analyte differs slightly from the surrounding medium, this difference can lead to measurable effects, albeit less pronounced than with a gold substrate [15, 24].

The generation and sensitivity of the evanescent wave are governed by fundamental principles and equations, such as Fresnel's equations for light reflection and refraction, which describe the relationship between the refractive indices, angle of incidence, and the behavior of the evanescent wave at the interface [9, 28].

Critical Angle for Total Internal Reflection: TIR occurs only when light travels from a medium of higher refractive index (n1) to a lower refractive index (n2), and the incident angle (θ_i) exceeds the critical angle (θ_c):

$$\theta_c = arcsin\left(\frac{n1}{n2}\right)$$

For a glass-water interface (n1 = 1.52, n2 =1.33) thus the critical angle is

$$\theta_c = arcsin\left(\frac{1.33}{1.52}\right) \approx 61.0^{\circ}$$

Penetration Depth of the Evanescent Wave: The depth (d_p) of the evanescent wave into the second medium is given by:

$$d_p = \frac{\gamma}{4\pi n1\sqrt{sin^2\theta_i - (\frac{n2}{n1})^2}}$$

Where γ is the wavelength of light in vacuum.

$n1$ and n2 are the refractive indices of the first and second media.

θ_i is the angle of incidence.

For light at γ = 488nm, an incident angle of 62° (above θ_c), and the same refractive indices as above

$$d_p = \frac{488 \times 10^{-9}}{4\pi \cdot 1.52\sqrt{sin^2 62^{\circ} - (\frac{1.33}{1.52})^2}} \approx 125nm$$

This penetration depth indicates the region within which the evanescent wave interacts with molecules, making the system highly sensitive to refractive index changes near the interface.

Refractive Index Sensitivity: The reflectivity (R) of the system is influenced by changes in the refractive index of the second medium (n2)

$R(\theta, n2) = R_0 + \Delta R$

R_0 is the baseline reflectivity.

ΔR is the change in reflectivity due to refractive index variations.

This change can be linked to the analyte's concentration as reflective index changes linearly with concentration ($\Delta n2 = k \cdot C$) for dilute solutions.

Using the glass substrate alone, refractive index changes were detected as PBS, BSA, and EDTA flowed across the surface. Despite the absence of a metallic layer to excite surface plasmons, the sensitivity of the evanescent wave allowed for the detection of minute refractive index variations i.e. PBS Provided a baseline for the refractive index (n2=1.33), BSA Addition introduced a slight refractive index increase (Δn2~0.01) due to protein binding near the interface, EDTA Removed adsorbed proteins, returning the refractive index to near-baseline levels. For example, a 0.01 change in refractive index with BSA addition (C≈1%) led to a measurable change in reflectivity (ΔR~0.001) and intensity.

In this experimental setup, changes in reflectivity (ΔR) were captured as intensity dips in the back-reflected SPR image, analyzed through line profiles extracted using ImageJ. These profiles displayed measurable intensity minima shifts across PBS, BSA, and EDTA stages. For example, in Experiment 2, the intensity dip moved from pixel 980 (PBS) to pixel 1169 (BSA 2%), and later to 1033 (EDTA). This pixel shift corresponds to a resonance angle change from 73.4° to 87.6°, and back toward 77.3°, based on the calibrated system resolution (0.0748°/pixel).

These angle shifts (Δθ), derived from intensity-based profiles, indirectly reflect the change in local refractive index (Δn) caused by protein binding and dissociation. Although no direct Δn calibration was applied, literature-based estimations suggest that a Δθ of 12° typically corresponds to Δn values in the range of 0.01–0.02 for protein adsorption near a gold interface [9, 24]. This alignment between experimental shifts and known refractive index effects supports the claim that our system is both sensitive and specific to biomolecular interactions at the surface.

This phenomenon is possible because Evanescent Wave Sensitivity dictates that the exponential decay of the evanescent wave confines its interactions to molecules within approximately 196 nm of the interface. This proximity amplifies its sensitivity to refractive index changes. The refractive index sensitivity ensures that even minor changes in molecular density are reflected in the intensity profile. Thus, even without a gold layer, the evanescent wave alone was sufficient to detect subtle refractive index changes caused by flowing analytes [13, 19]. This sensitivity validates the theoretical principle and highlights the robustness of the experimental setup [22].

The Absence of Surface Plasmons means that, unlike SPR, where surface plasmons amplify the sensitivity to refractive index changes, a glass-only setup lacks this enhancement. However, the intrinsic sensitivity of the evanescent wave to refractive index changes still allows for small observable effects, especially when high-precision optical detection systems are used [9, 28].

Experimental Factors ensure that the visibility of changes depends on the precision of the detection system, such as the CoolSNAP camera, and the experimental setup's stability. Subtle shifts may be detectable in setups designed to measure small variations in intensity or angle [14, 25].

While the changes observed with glass alone are much smaller than those seen with a gold substrate in SPR, they are still possible due to the refractive index dependence of the evanescent wave. These control runs with glass help validate the system’s sensitivity and provide a baseline for comparison [15, 24].

Gold plated substrates.

Gold-plated silicon wafers were used as the SPR sensor surfaces, chosen for their high conductivity and compatibility with functionalization techniques [28]. These wafers supported the excitation of surface plasmons, enabling real-time detection of molecular interactions [29].

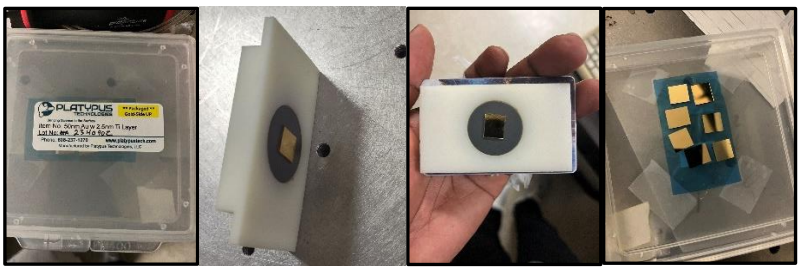
The wafers were cleaned using sodium hypochlorite and distilled water to ensure a contaminant-free surface [32].

Gold-plated substrates are ideal for Surface Plasmon Resonance (SPR) experiments due to their unique optical and chemical properties that support the excitation of surface plasmons and enhance experimental accuracy. Gold, as a noble metal, offers high conductivity and a smooth surface, enabling the generation of stable surface plasmon waves when exposed to polarized light at specific angles. This stability ensures precise measurement of refractive index changes near the surface, which is critical for detecting biomolecular interactions in real-time [14, 19].

Additionally, gold is chemically inert, resisting oxidation and degradation during experiments, thus providing a reliable and durable surface for multiple experimental cycles. Gold's compatibility with functionalization techniques, such as thiol-based self-assembled monolayers, enables the attachment of biomolecules, creating a specific and versatile sensor surface. These properties make gold-plated substrates indispensable for SPR, ensuring sensitivity, reproducibility, and robustness in protein interaction studies [15, 22].

The main experiments were conducted on gold substrates of varying thicknesses, each consisting of a 2.5 nm titanium adhesion layer and gold layers of 50 nm, 60 nm, and 70 nm. Testing these variations was essential for optimizing the sensitivity and efficiency of the SPR response. Gold thickness directly affects surface plasmon propagation; layers that are too thick reduce sensitivity due to damping, while those that are too thin compromise the plasmonic response. Our observations indicated that the 50 nm gold layer provided the most distinct evanescent wave interaction, suggesting optimal coupling between the evanescent field and surface plasmons [9, 28].

This thickness allowed for a balanced SPR response with minimal energy loss, ensuring high sensitivity to refractive index changes caused by molecular interactions. Based on these findings, the 50 nm gold substrate was selected as the most suitable for the experiments [13, 24].

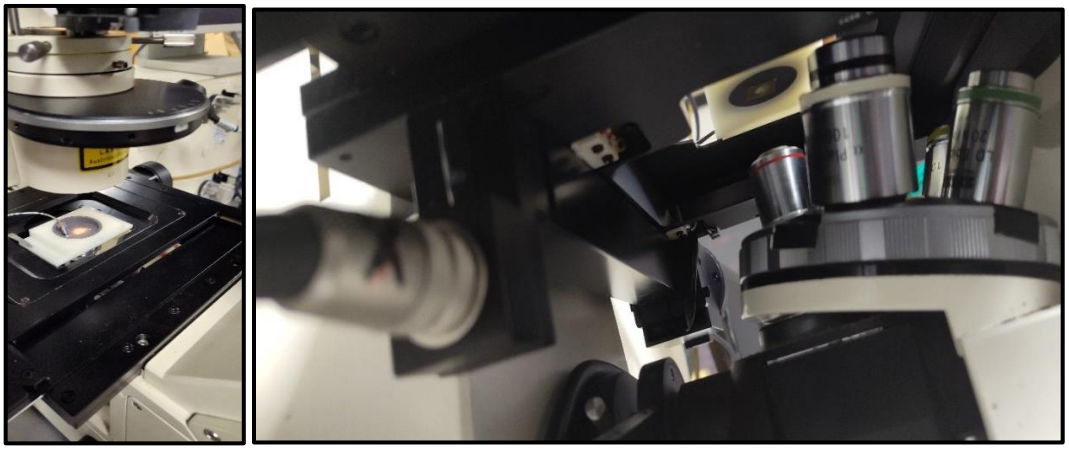


Caption: *Images of the gold-Plated substrates. The image on the right shows how the gold substrates are arranged inside the packaging. The packaging contained different thicknesses of the gold layer from 50nm to 70nm. On the Left shows the packaging with the label from Platypus*

technologies. The center images show the substrate integrated on a microscope stage holder. The Gold can be seen at the center.

The gold-plated wafers were mounted on the microscope stage, with the perfusion system positioned to flow reagents across the surface at controlled rates.

Gold plated substrate positioned on the stage.



Caption: *Above are images showing the gold-plated substrate positioned on the stage. The first image on the left shows an aerial view and the second image, which is the one on the right shows a dorsal view of the substrate on the stage.*

Data Acquisition and Processing

Image Capture and Real-Time Monitoring

Images were captured at specific time intervals using "Win View Princeton Instruments" software. Baseline measurements were taken during PBS flow, followed by analyte introduction and regeneration with EDTA. These images enabled the analysis of intensity profiles, Full-Width at Half-Maximum (FWHM), and resonance calculations [30].

Each experiment included a baseline measurement (PBS flow) followed by images taken after the introduction of the analyte BSA and subsequent regeneration with EDTA. This setup allowed for full analysis of Intensity profiles, FWHM and resonance calculations.

Analysis

Surface Plasmon Resonance (SPR) is a widely used technique to investigate molecular interactions. In this analysis, we compare and evaluate data from multiple experimental runs, including continuous trial runs and their respective controls under different objective magnifications (x100 High Power and x5 Objectives). The data includes peak intensity, resonance angles, and SPR intensity profiles, which are essential metrics to quantify the molecular binding and response in SPR experiments [9, 28]. The resonance angle, where minimum intensity was observed, was identified as the point of maximum binding on the sensor surface. This simply means the minimum intensity observed during the experiment, which corresponds to the resonance angle where maximum energy transfer occurs between the incident light and the surface plasmons on the metal sensor surface. This energy transfer is relative to the interaction of the evanescent wave with the biomolecules bound to the sensor surface. When binding occurs, it changes the refractive index near the surface, shifting the resonance angle and altering the intensity profile. Thus, the minimum intensity observed is relative to the refractive index changes caused by the biomolecular interactions on the sensor surface, which in turn reflects the degree of binding at that particular point. This angle shift can also be plotted over time to observe binding dynamics in a system taking numerous images over time [10, 14], however, we were only taking two photos over a 5-minute period.

Control Intensity Profiles Across Objectives

Images captured immediately upon addition of an analyte or commencement of time zero (in the case of blank substrate) and a second one captured five minutes later were utilized and Intensity profiles were generated by plotting the sum of pixel intensities across the sensor surface, providing a quantitative measure of binding events. Baseline corrections were applied to each profile to remove noise and background variations, ensuring that intensity changes were specific to protein interactions [16, 19].

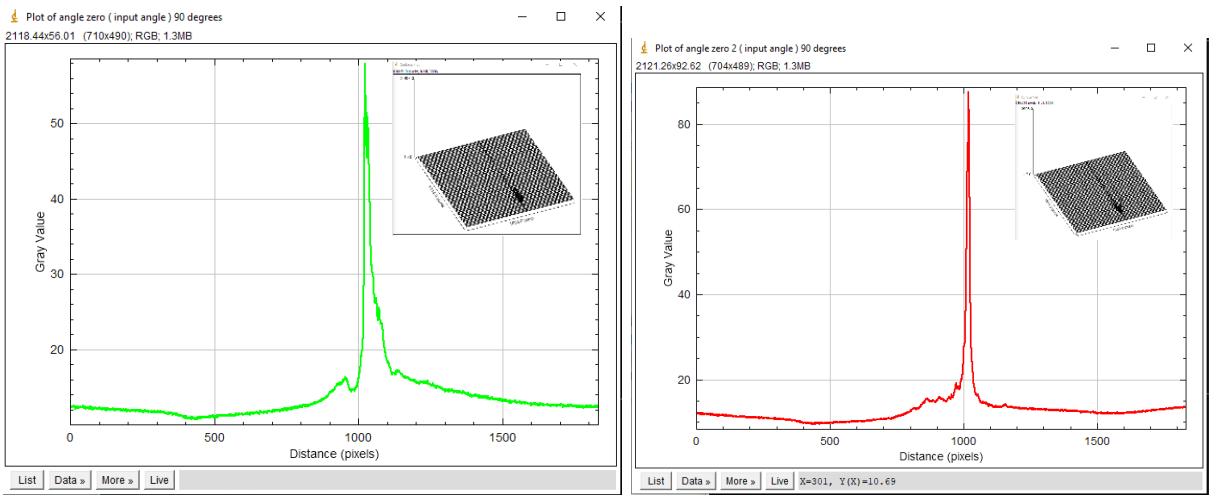
x5 Objective Controls Analysis

The x5 objective control experiments include two runs. The goal is to evaluate consistency across these runs by comparing the first addition and final profiles for each analyte and analyzing the observed trends.

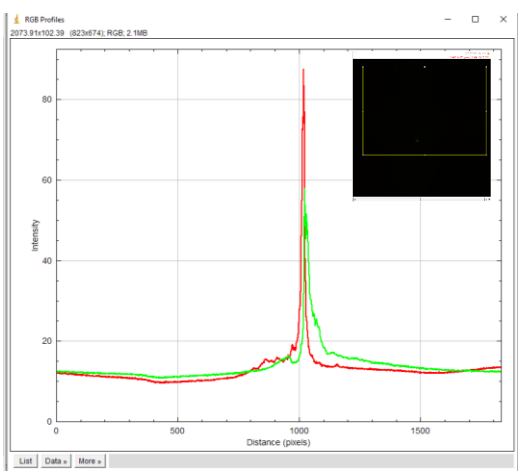
Blank Substrate Profile Comparison Images and Graphs Between Addition and Final Profiles - x5 Objective

To begin with we had the back-reflected light intensity map recorded during the addition phase for the blank substrate control using the X5 objective. The images below captured the initial surface interactions displaying a uniform intensity distribution that is characteristic of a non-reactive, inert baseline. There was also the back reflected light intensity map captured 5 minutes

after the addition phase showing a consistent intensity pattern, which reaffirms the inert nature of the blank substrate under the experimental conditions.



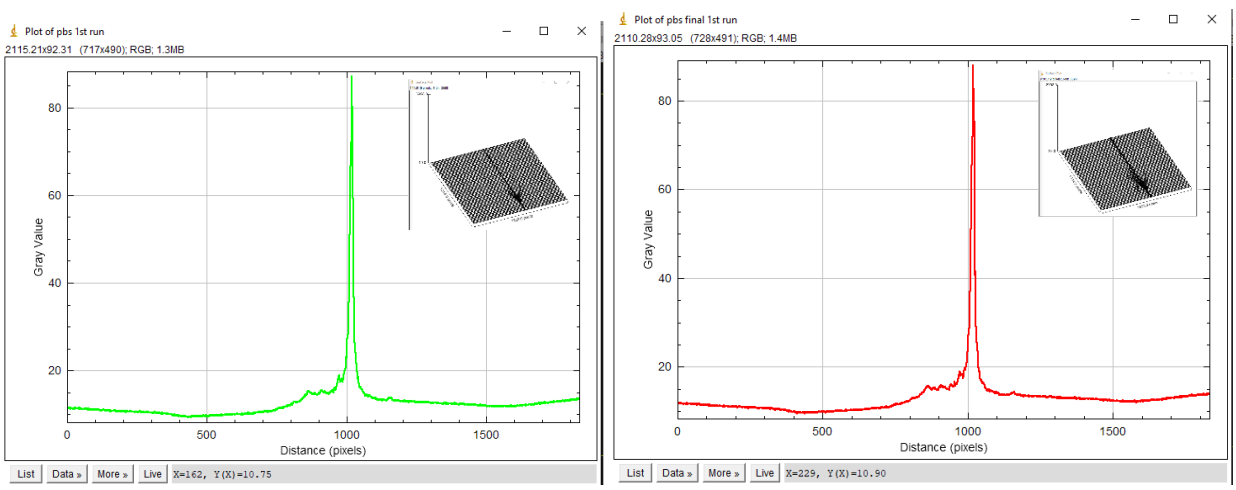
Caption: *On the left above is the Intensity profile for the addition phase of the blank substrate control, showing the pixel-wise distribution of reflected light across the sensor surface. The curve suggests minimal variations in the refractive index, as expected for a baseline measurement. On the top right we have a 3D surface plot of reflected light intensity during the addition phase for the blank substrate control. Peaks and valleys depict localized intensity variations due to sensor imperfections, with no significant refractive index changes detected. On the right is the Intensity profile for the final phase (5 minutes later) of the blank substrate control. The similarity between this profile and the addition phase confirms the stability and uniformity of the blank substrate response. On the top right side is a 3D surface plot of reflected light intensity during the final phase for the blank substrate control (5 minutes later). The similarity in topography with the addition phase confirms consistent and non-reactive behavior of the blank substrate.*



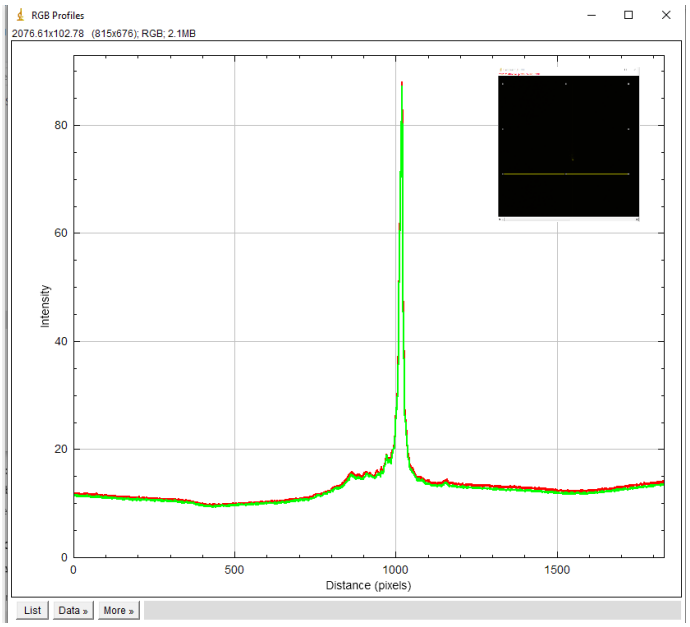
Caption: *On the top right of the image above is a Composite overlay of back-reflected light intensity for the addition and final phases of the blank substrate control. The near-identical intensity distributions validate the substrate's stability and inert response over time. The graph is an Overlay of intensity profiles for addition and final phases of the blank substrate control. The overlapping curves highlight the negligible change in intensity, characteristic of a stable, inert surface.*

PBS Addition and Final Profiles Comparison Images and Graphs Between Addition and Final Profiles - x5 Objective

Here in the addition phase, the image captured using the X5 objective showed the back-reflected light intensity map for the PBS control. The uniform intensity distribution in this image serves as a reliable reference for subsequent analysis, indicating minimal refractive index changes exactly what one would expect from an inert buffer like PBS during initial exposure. Five minutes later we also recorded with the X5 objective and observed a consistent intensity pattern that reaffirms the stability and inert behavior of PBS over time.



Caption: *On the left is the Intensity profile for the addition phase of the PBS control. The flat curve indicates a negligible refractive index change, as expected for a control buffer with no significant surface interaction. On the top right the same image is a 3D surface plot of reflected light intensity during the addition phase for the PBS control. The uniform topography is characteristic of a stable buffer without significant refractive index variations. On the right is the Intensity profile for the final phase (5 minutes later) of the PBS control. The nearly identical profile to the addition phase underscores the stability and non-reactivity of PBS. On the top right of the image is a 3D surface plot of reflected light intensity during the final phase for the PBS control (5 minutes later). The unchanged surface topography demonstrates the inert nature of PBS throughout the experiment.*

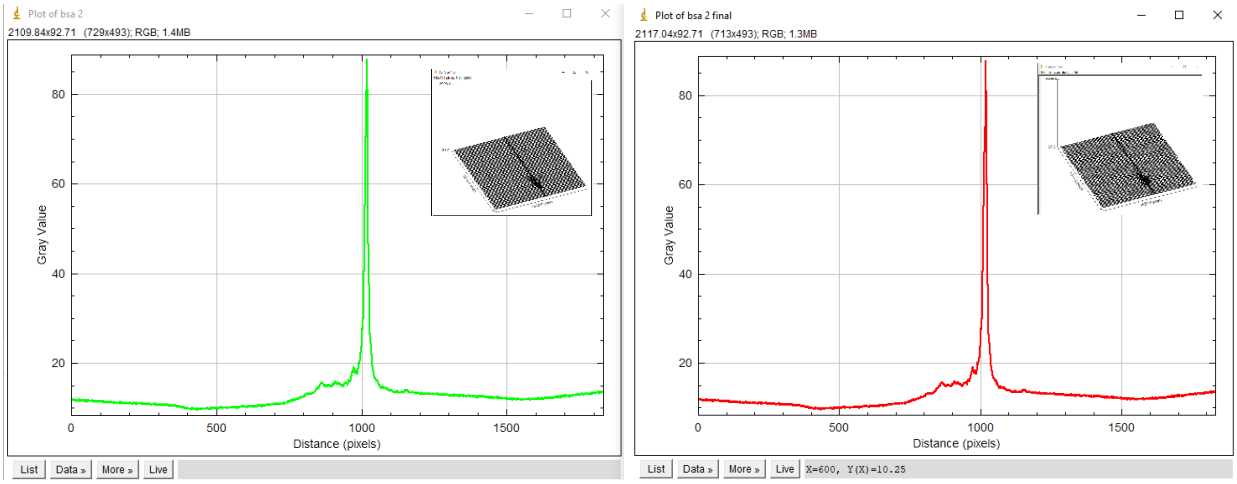


Caption: *Above is an Overlay of intensity profiles for the addition and final phases of the PBS control. The almost perfectly overlapping curves confirm the negligible refractive index shift, validating PBS as a stable control. On the top right is a Composite overlay of back-reflected light intensity for the addition and final phases of the PBS control. The almost perfect alignment of intensity distributions highlights the unchanged refractive index profile of PBS.*

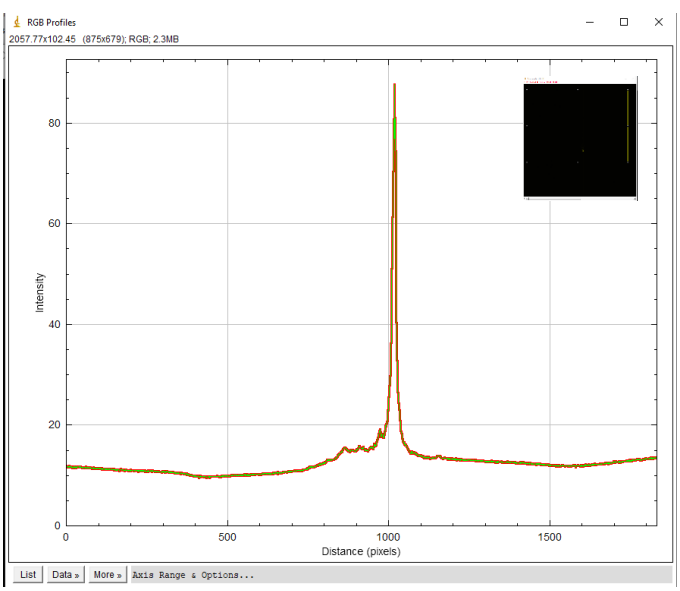
In the PBS Addition and Final profiles show consistent intensity values and indicate the stability of PBS as a buffer. However, minor disparities in intensity were observed, attributed to measurement sensitivity [19, 21]

BSA Addition and Final Profiles Comparison Images and Graphs Between Addition and Final Profiles - x5 Objective (2% BSA)

Using the X5 objective, the back-reflected light intensity map was captured during the BSA addition phase providing an important baseline for further analysis. Although the image appears uniform at first glance, it marked the beginning of BSA binding interactions and signals the onset of subtle refractive index changes at the sensor surface. In contrast, there was the other panel which displayed the intensity map recorded 5 minutes later. While the overall visual appearance remains like the initial image, a closer examination of the intensity profiles reveals meaningful changes that quantify ongoing molecular interactions and the saturation state of BSA binding on the sensor surface.



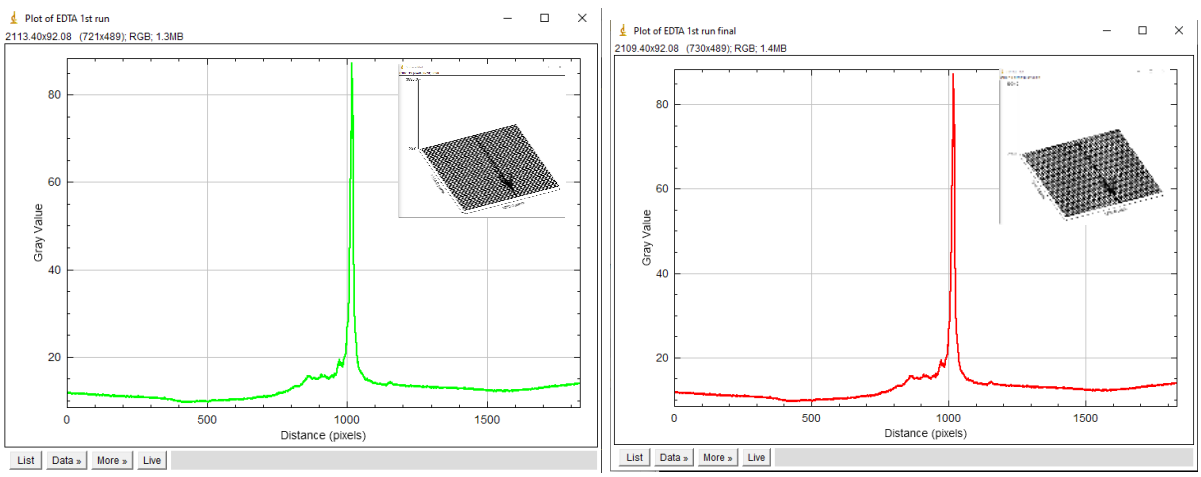
Caption: *On the left is the Intensity profile for the addition phase of the BSA control, displaying a gradual reduction in intensity. This is indicative of refractive index changes caused by the initial binding of BSA molecules. On the top right same image is the 3D surface plot of reflected light intensity during the addition phase for the BSA control. The noticeable intensity reduction in localized regions corresponds to the initial interaction of BSA molecules with the sensor. On the right is the Intensity profile for the final phase (5 minutes later) of the BSA control. The sharper intensity reduction compared to the addition phase suggests continued binding activity or stabilization of the BSA layer on the sensor surface. On the top right is a 3D surface plot of reflected light intensity during the final phase for the BSA control (5 minutes later). Enhanced intensity changes in certain regions suggest saturation of BSA binding and stabilization of the refractive index.*



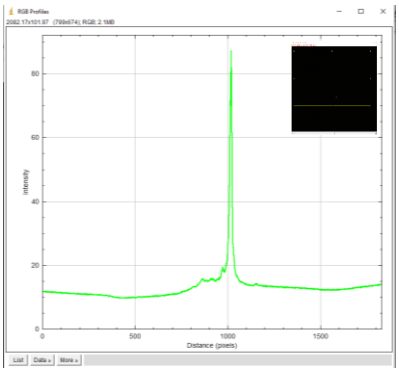
Caption: *On the top right is a Composite overlay of back-reflected light intensity for the addition and final phases of the BSA control. The overlay reveals the progressive intensity reduction, signifying molecular interaction dynamics over time. The graph is the Overlay of intensity profiles for the addition and final phases of the BSA control. The comparative profiles demonstrate a consistent downward shift, correlating with refractive index changes due to BSA adsorption.*

<u>EDTA Addition and Final Profiles Comparison Images and Graphs Between Addition and Final Profiles - x5 Objective</u>

The back-reflected light intensity map during the EDTA control's addition phase itself may not have revealed obvious changes but, it establishes a baseline crucial for later intensity profile analysis. The corresponding image recorded 5 minutes after the addition phase appears visually similar, yet it plays an important role in our analysis by providing data that reveal dynamic intensity changes when examined in detail.



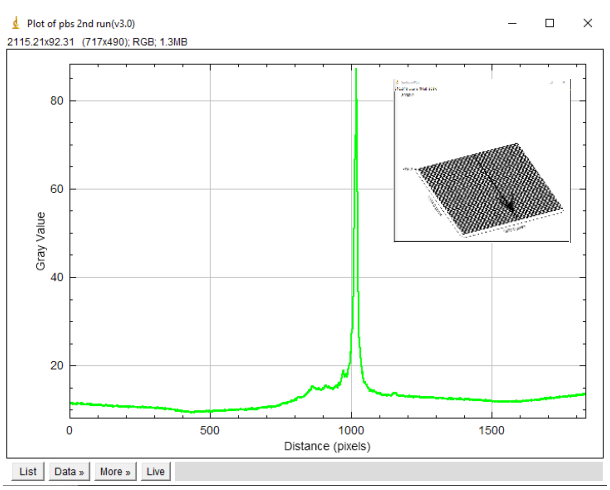
Caption: *On the left is the Intensity profile for the addition phase of the EDTA control. The observed reduction in intensity suggests initial refractive index changes due to EDTA interactions at the sensor surface. On the top right, the same image is a 3D surface plot of reflected light intensity during the addition phase for the EDTA control. The plot reveals localized intensity variations, likely linked to EDTA's initial interaction with the sensor surface. On the right is the Intensity profile for the final phase (5 minutes later) of the EDTA control. A deeper intensity reduction compared to the addition phase indicates ongoing interactions or stabilization of EDTA at the sensor surface. On the top right is a 3D surface plot of reflected light intensity during the final phase for the EDTA control (5 minutes later). Increased intensity variation in specific regions reflects stabilization or prolonged interaction of EDTA molecules.*

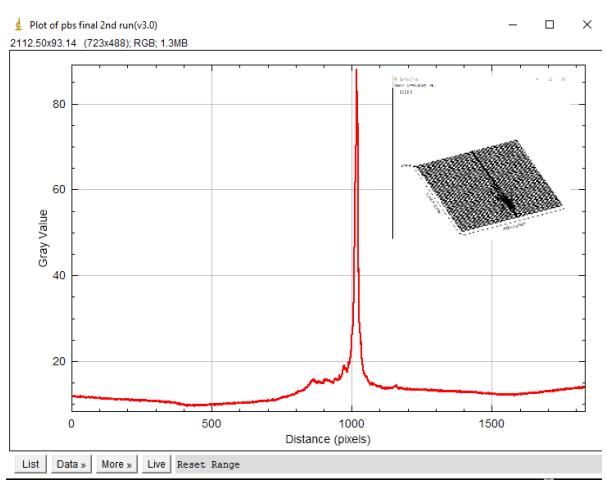


Caption: *Above is an Overlay of intensity profiles for the addition and final phases of the EDTA control. The increased depth of the intensity reduction reflects ongoing refractive index changes associated with EDTA interactions. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of the EDTA control. While visually indistinguishable, the intensity profiles provide insights into EDTA-induced refractive index changes.*

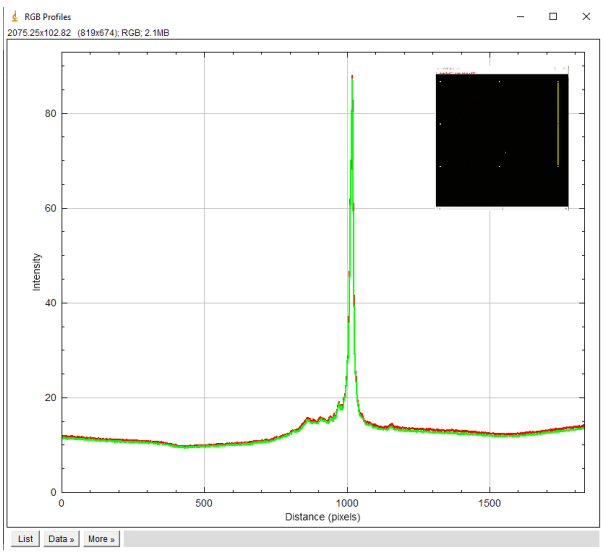
<u>PBS After EDTA Profiles Comparison Images and Graphs Between Addition and Final Profiles for x5 Objective</u>

The back-reflected light intensity map captured during the addition phase for the PBS after EDTA control with the X5 objective showed uniform intensity pattern serving as a stable baseline for subsequent intensity profile analysis. Then the intensity map recorded 5 minutes after the addition phase appeared visually unchanged; however, a closer look at the corresponding intensity profiles provided additional insights into subtle, time-dependent variations that may not be immediately apparent from the raw image alone.





Caption: *On the left is the Intensity profile for the addition phase of the PBS after EDTA control. The curve remains flat, indicating negligible changes in refractive index, as expected for a buffer solution after EDTA exposure. On the top right same image is a 3D surface plot of reflected light intensity during the addition phase for the PBS after EDTA control. The consistent topography reflects the inert nature of PBS under these conditions. On the right is the Intensity profile for the final phase (5 minutes later) of the PBS after EDTA control. The unchanged profile compared to the addition phase suggests the stabilization of the buffer's refractive index. On the top right is a 3D surface plot of reflected light intensity during the final phase for the PBS after EDTA control (5 minutes later). The stable surface pattern reaffirms PBS's role as a reliable control after EDTA exposure.*



Caption: *Above is an Overlay of intensity profiles for the addition and final phases of the PBS after EDTA control. The overlapping curves validate PBS's stability and inert behavior after EDTA exposure. On the top right is a Composite overlay of back-reflected light intensity for the addition and final phases of the PBS after EDTA control. The image highlights the consistent intensity distribution over time.*

<u>Key Observations</u>

The blank substrate serves as an inert baseline to assess the stability of the sensor surface. Both the back-reflected images and intensity profiles demonstrate no significant changes between the addition and final phases. The overlapping intensity profiles confirm the absence of refractive index changes, indicating a stable, non-reactive sensor surface [14, 19]. This consistency validates the substrate's suitability as a reference for subsequent analyte interactions and highlights the robustness of the experimental setup [24].

Phosphate-Buffered Saline (PBS), a buffer solution, acts as another inert control ensuring the sensor's stability under aqueous conditions. The intensity profiles for both the addition and final

phases are nearly identical, with flat curves and negligible changes. This confirms that PBS does not induce refractive index variations, further supporting its use as a stable reference [13, 22].

Bovine Serum Albumin (BSA) introduces a dynamic interaction with the sensor surface, reflected in the intensity profiles. The addition phase exhibits an initial intensity reduction; indicative of refractive index changes due to molecular adsorption. By the final phase, the reduction becomes more pronounced, suggesting continued binding or stabilization of BSA molecules on the surface [9, 28]. These observations align with expected behavior for protein adsorption, showcasing the sensor's sensitivity to molecular interactions and its suitability for detecting biomolecular binding events [15].

Ethylenediaminetetraacetic acid (EDTA), a chelating agent, exhibits measurable interactions with the sensor surface. The intensity profile for the addition phase reveals a notable reduction, suggesting initial changes in the refractive index. By the final phase, the reduction deepens, likely due to stabilization or prolonged interaction of EDTA with the surface [17, 26]. These results highlight the sensor's ability to detect refractive index changes associated with chelation effects and demonstrate its versatility for monitoring diverse molecular interactions [25].

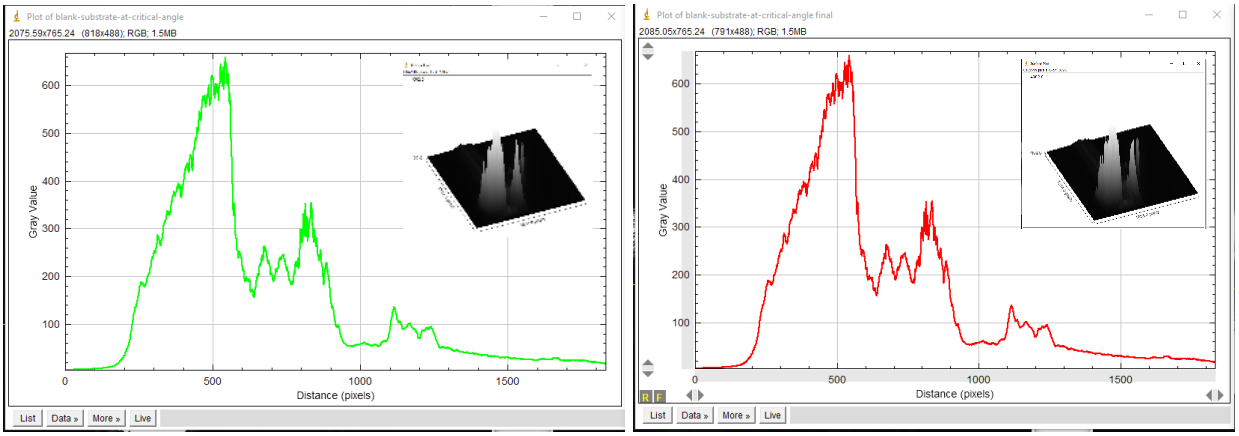
PBS is reintroduced after EDTA exposure to evaluate the sensor's response to buffer stabilization. The intensity profiles for the addition and final phases remain unchanged, with flat curves confirming the inert nature of PBS even after EDTA treatment [13, 22]. This consistency validates PBS's role as a reliable buffer for stabilizing the sensor environment and demonstrates the reproducibility of the sensor's performance [19].

High Power Objective (x100) Controls at Critical Angle

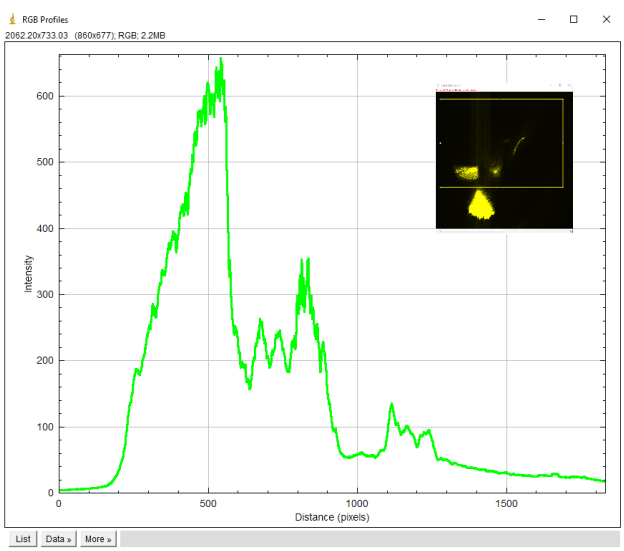
ImageJ software was used to save the exact coordinates of the selected area using a ROI (region of interest) manager feature in the software so it’s easy to get the exact same section on each sample. Below are the Intensity profiles of these areas. Only the area inside the yellow boundary was analyzed.

Critical Angle Profiles Comparison Images and Graphs Between Addition and Final Profiles - High Power Objective

To begin there was the back-reflected light intensity map at the critical angle for the blank substrate (under the high-power objective) recorded during the addition phase. This image provided a baseline for evaluating sensor stability, as no specific molecular interactions are present on a blank substrate. There is also the intensity map, captured at the critical angle, recorded 5 minutes after the addition phase. The visually consistent intensity distribution confirms that the blank substrate remains inert under the experimental conditions.



Caption: *On the left is the Intensity profile for the addition phase at the critical angle. The flat curve reflects the absence of refractive index changes, confirming the stability of the blank substrate. On the top right same image is a 3D surface plot of reflected light intensity at the critical angle during the addition phase. The uniform topography confirms the stable behavior of the blank substrate. On the right is the Intensity profile for the final phase (5 minutes later) at the critical angle. The nearly identical profile to the addition phase further highlights the stability of the blank substrate. On the top right is a 3D surface plot of reflected light intensity at the critical angle during the final phase (5 minutes later). The unchanged surface pattern reaffirms the substrate's inert nature.*

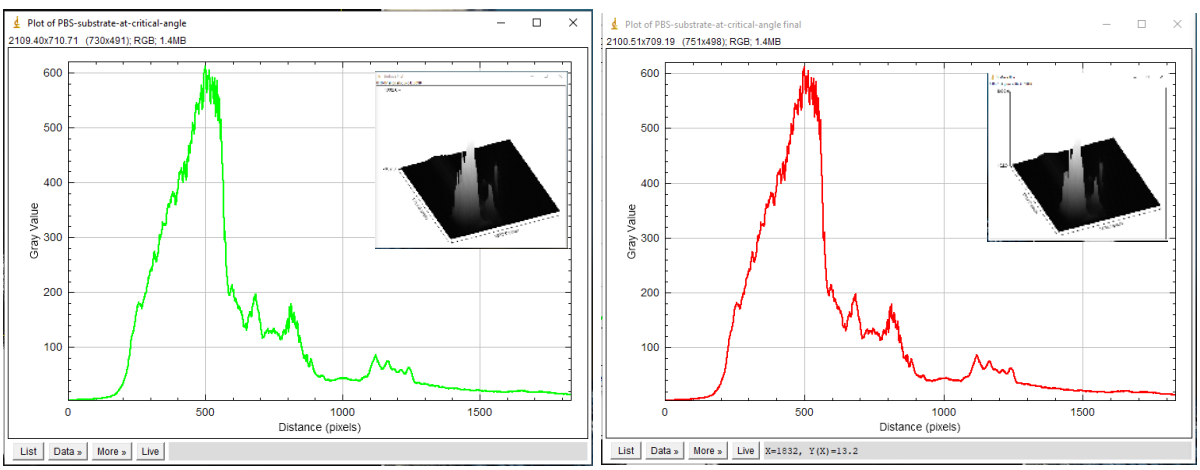


Caption: *Above is an Overlay of intensity profiles for the addition and final phases at the critical angle. The overlapping curves confirm negligible refractive index changes, validating the blank substrate's stability. On the top right is the Composite overlay of back-reflected light intensity for*

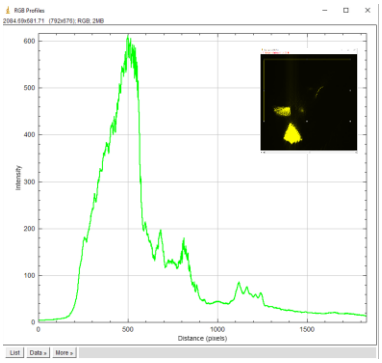
the addition and final phases at the critical angle. The overlay illustrates the stability of the blank substrate, with no discernible refractive index changes.

<u>PBS Profiles Comparison Images and Graphs Between Addition and Final Profiles - High Power Objective</u>

Here we had the back-reflected light intensity map recorded during the PBS control addition phase under the high-power objective. This image provided a stable reference, reflecting the sensor's behavior when immersed in an inert buffer solution (PBS). There was also the corresponding intensity map captured 5 minutes after the addition phase. The consistent intensity distribution between both images confirms that PBS maintains its inert nature over time, ensuring a stable baseline in the SPR measurements.



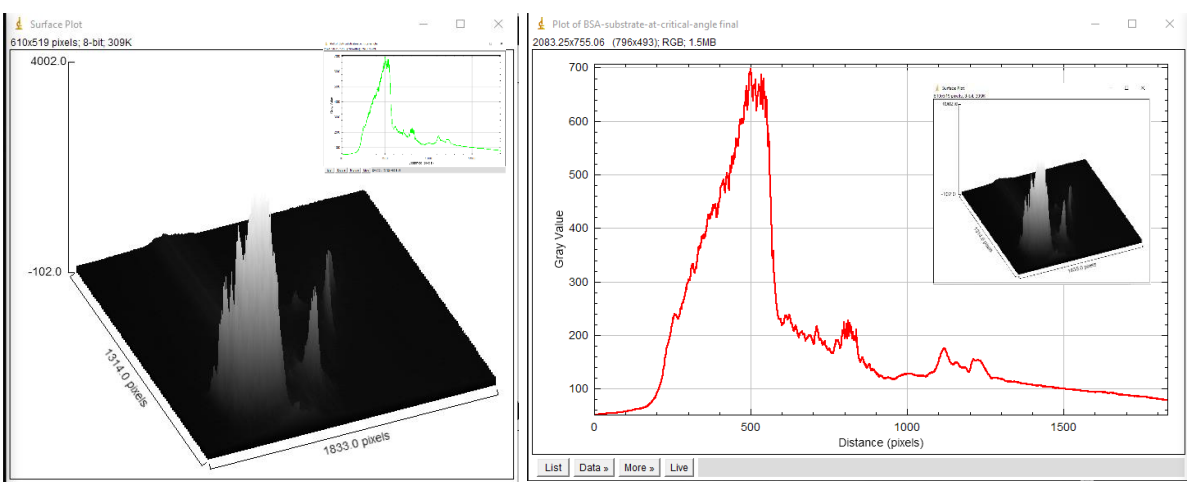
Caption: *On the left is the Intensity profile for the addition phase of the PBS control under the high-power objective. The flat curve confirms the absence of refractive index changes, validating PBS as a stable buffer. On the top right same image is a 3D surface plot of reflected light intensity during the addition phase for the PBS control under the high-power objective. The uniform topography reflects the stable behavior of PBS as an inert buffer. On the right is the Intensity profile for the final phase (5 minutes later) of the PBS control under the high-power objective. The nearly identical profile to the addition phase demonstrates the stability and non-reactivity of PBS. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) for the PBS control under the high-power objective. The consistent surface pattern reaffirms PBS's role as a reliable control.*



Caption: *Above is an Overlay of intensity profiles for the addition and final phases of the PBS control under the high-power objective. The overlapping curves validate PBS as a stable, inert buffer with negligible refractive index changes. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of the PBS control under the high-power objective. The overlay illustrates the stability of PBS, with no discernible refractive index changes.*

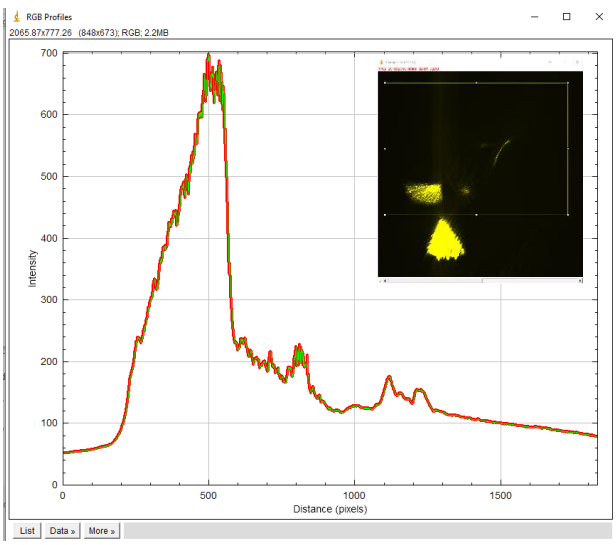
<u>BSA Profiles Comparison Images and Graphs Between Addition and Final Profiles - High Power Objective</u>

Here we had the back-reflected light intensity map taken during the addition phase for the BSA control under the high-power objective. Although the image appeared visually uniform, it establishes a baseline for subsequent intensity profile analysis to assess molecular interactions. We also had the corresponding intensity map recorded 5 minutes after the addition phase. While the overall visual appearance remains unchanged, detailed intensity profile analysis will reveal subtle refractive index shifts due to BSA binding events at the sensor surface.



Caption: *On the left is a 3D surface plot of reflected light intensity during the addition phase for the BSA control under the high-power objective. Localized intensity variations reflect the onset of molecular interactions. Top right the same image is the Intensity profile for the addition phase*

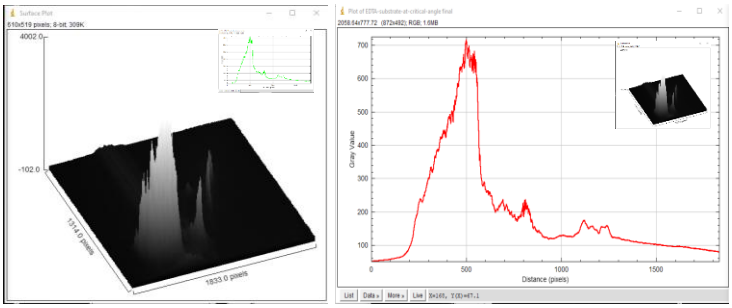
of the BSA control under the high-power objective. The initial intensity reduction indicates refractive index changes due to the adsorption of BSA molecules on the sensor surface. On the right is the Intensity profile for the final phase (5 minutes later) of the BSA control under the high-power objective. A more pronounced reduction compared to the addition phase suggests continued binding or stabilization of BSA on the sensor surface. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) for the BSA control under the high-power objective. Enhanced intensity variations in certain regions correspond to further molecular binding or stabilization.



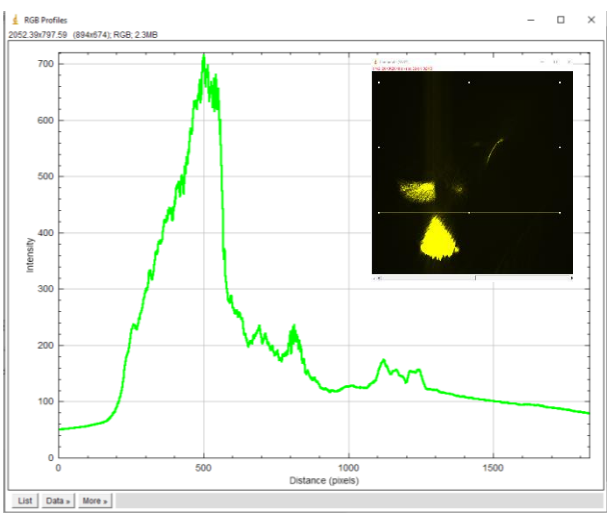
Caption: *Above is an Overlay of intensity profiles for the addition and final phases of the BSA control under the high-power objective. The increasing depth of the intensity reduction reflects ongoing refractive index changes induced by BSA adsorption and stabilization. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of the BSA control under the high-power objective. While visually similar, the intensity profiles reveal meaningful refractive index changes due to BSA adsorption.*

Comparison Images and Graphs Between EDTA Addition and Final Profiles - High Power Objective

Here there was the back-reflected light intensity map captured during the addition phase for the EDTA control under the high-power objective. Although the image by itself may not reveal obvious differences, it establishes a baseline for detailed intensity profile analysis. We also had the corresponding intensity map recorded 5 minutes after the addition phase, which provides a reference for assessing subtle refractive index changes through the derived intensity profiles.



Caption: *On the left is a 3D surface plot of reflected light intensity during the addition phase for the EDTA control under the high-power objective. The surface plot highlights localized intensity variations caused by EDTA's initial interaction with the sensor. Top right same image is the Intensity profile for the addition phase of the EDTA control under the high-power objective. The initial intensity reduction suggests refractive index changes due to the interaction of EDTA with the sensor surface. On the right is the Intensity profile for the final phase (5 minutes later) of the EDTA control under the high-power objective. The deeper intensity reduction compared to the addition phase reflects continued interaction or stabilization of EDTA on the sensor surface. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) for the EDTA control under the high-power objective. Enhanced intensity changes in specific regions indicate ongoing interaction or stabilization effects.*

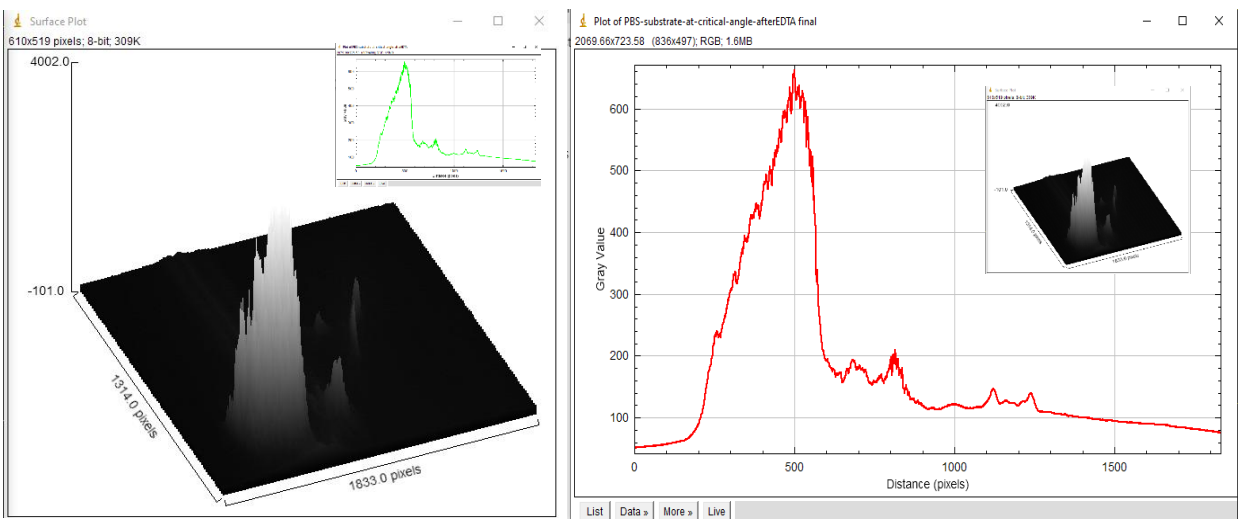


Caption: *Above is an Overlay of intensity profiles for the addition and final phases of the EDTA control under the high-power objective. The increasing depth of the intensity reduction signifies refractive index changes associated with EDTA interactions. High-power objective results confirmed EDTA's effectiveness as a regeneration agent. Consistent profiles across runs demonstrated their reproducibility, with minimal interactions on the substrate. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of the*

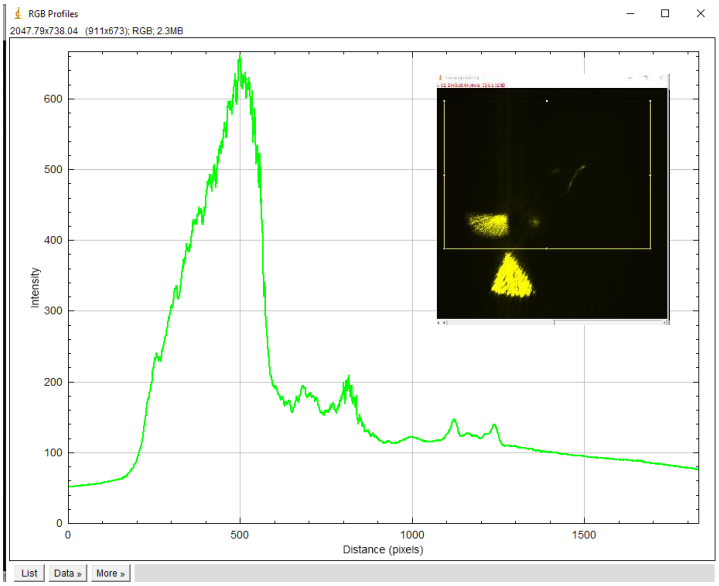
EDTA control under the high-power objective. While visually similar, the overlaid intensity profiles provide detailed insights into EDTA-induced refractive index changes.

<u>Comparison Images and Graphs Between PBS After EDTA Addition and Final Profiles - High Power Objective</u>

Here we had the back-reflected light intensity map captured during the addition phase for the PBS after EDTA control under the high-power objective. The uniform intensity distribution served as a baseline reference for subsequent intensity profile analysis. There also was the back-reflected intensity map recorded 5 minutes after the addition phase for the same control condition. Although the images appear visually unchanged, detailed intensity profile analysis will reveal any subtle stabilization effects induced by PBS following EDTA exposure.



Caption: *On the left is a 3D surface plot of reflected light intensity during the addition phase for the PBS after EDTA control under the high-power objective. The surface plot demonstrates consistent topography, reflecting PBS's stability. Top right same image is the Intensity profile for the addition phase of the PBS after EDTA control under the high-power objective. The flat curve indicates minimal refractive index changes, consistent with PBS's expected stabilizing behavior. On the right is the Intensity profile for the final phase (5 minutes later) of the PBS after EDTA control under the high-power objective. The unchanged profile compared to the addition phase confirms PBS's inert and stabilizing nature. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) for the PBS after EDTA control under the high-power objective. The unchanged surface pattern highlights the buffer's role in maintaining sensor stability.*



Caption: *Above is an Overlay of intensity profiles for the addition and final phases of the PBS after EDTA control under the high-power objective. The overlapping curves validate PBS's role in stabilizing the sensor environment after EDTA exposure. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of the PBS after EDTA control under the high-power objective. The overlay confirms PBS's stabilizing effect, with no discernible refractive index changes.*

Across PBS and critical angle profiles, the high-power objective control runs exhibit reproducible and consistent behavior. Variations in intensity are minimal and fall within acceptable experimental variability, ensuring the reliability of the measurements [13, 22]. The critical angle profiles demonstrate precise optical interactions, capturing the refractive index changes at the interface. However, minor discrepancies were noted, which may be attributed to environmental factors or minor alignment deviations in the optical setup [14, 24].

These results validate the stability and accuracy of the experimental system, emphasizing the suitability of the high-power objective for detailed optical measurements, particularly in detecting subtle refractive index changes [9, 28].

<u>Key Observations.</u>

The high objective (HO) controls were designed to evaluate the stability and consistency of the sensor system under experimental conditions. Across most of the control samples, including the blank substrate, PBS, and PBS after EDTA, the back-reflected images remained visually uniform between the addition and final phases, demonstrating the inertness of these controls [13, 22].

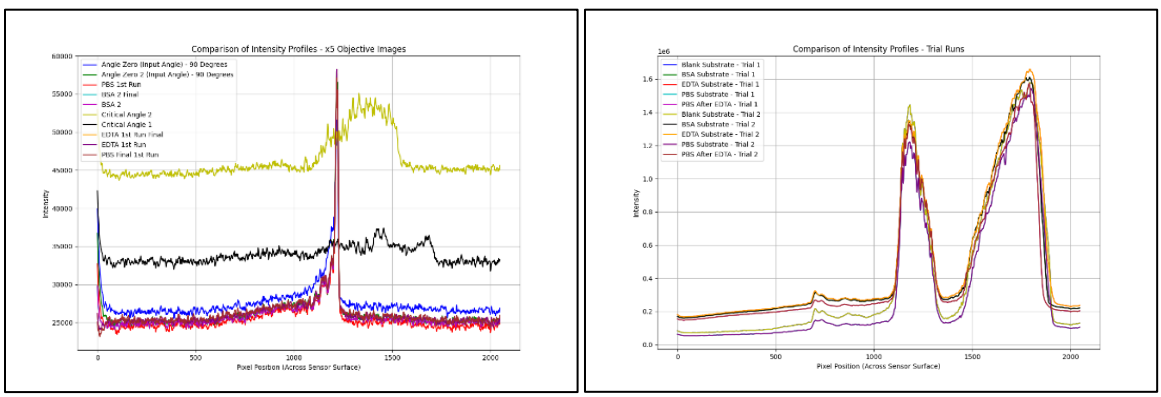
The intensity profiles for the blank substrate and PBS controls showed negligible changes between the addition and final phases, with flat and overlapping curves that confirm the stability of the sensor surface and the inert nature of the buffer. Similarly, the PBS after EDTA control

displayed consistent intensity profiles, validating PBS's role in stabilizing the sensor environment after EDTA exposure [14, 25].

In contrast, the BSA and EDTA controls exhibited measurable intensity reductions in their profiles, corresponding to refractive index changes caused by molecular interactions. The BSA control showed a pronounced reduction between the addition and final phases, reflecting the adsorption and stabilization of BSA on the sensor surface. Similarly, the EDTA control demonstrated a deeper reduction over time, indicative of continued interaction or stabilization of EDTA molecules on the sensor [9, 28].

The 3D surface plots provided additional confirmation of these observations, with localized intensity variations visible for the BSA and EDTA controls, while the blank substrate and PBS samples exhibited consistent topographies. These 3D plots enhance the visualization of the sensor's sensitivity to molecular interactions, offering a spatial representation of changes that align with the intensity profiles [19, 24].

Comparison Between X5 and High-Power Objectives



Caption: *Left: Intensity maps captured using a low-magnification (x5) objective show limited SPR sensitivity and lower contrast between analyte states. Right: High-magnification objective (x100, NA = 1.45(1.48)) produces sharply resolved intensity dips, allowing for accurate detection of resonance shifts. These graphs confirm the superior optical performance and spatial resolution of high-NA objectives for SPR imaging.*

The comparison between x5 and high-power objectives clearly validates the selection of the high-NA oil-immersion lens for this study. The x5 objective, though effective for wider field capture, lacks the angular resolution needed to resolve subtle SPR dips and refractive index variations. In contrast, the high-power lens enables sharply defined intensity changes across PBS, BSA, and EDTA phases, essential for reliable resonance angle calculation. This optical resolution ensures accurate spatial localization of the SPR dip and confirms the experimental system's sensitivity to nanoscale surface interactions. These results reinforce the importance of selecting the correct objective based on both numerical aperture and penetration depth requirements for evanescent field excitation.

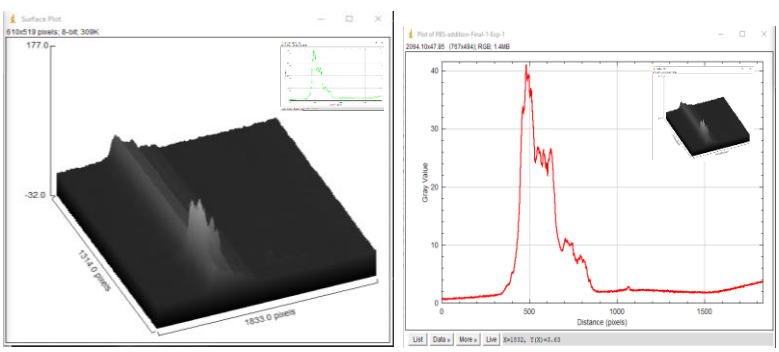
Overall, the HO controls highlight the stability of the experimental system while showcasing its sensitivity to molecular interactions in dynamic scenarios. The reproducibility of the data across controls demonstrates the reliability of the system under varying experimental conditions [15, 26].

High Power Objective (x100)

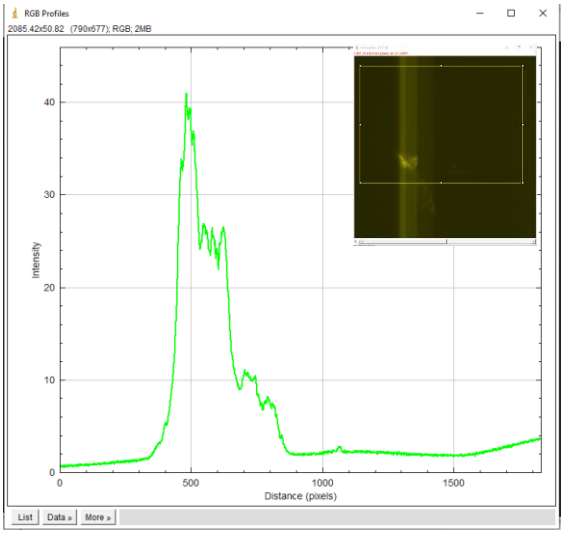
Experiment 1 Analysis

Comparison Images and Graphs Between PBS Addition 1 and PBS Final 1

Using the high-power objective under continuous flow conditions, back-reflected intensity maps were obtained for PBS as a control. The image captured during the PBS addition phase shows a uniform intensity distribution that serves as a reference for establishing baseline sensor stability. The image five minutes later displayed a consistent intensity profile, reflecting the inert behavior of PBS as a control buffer and confirming that the sensor remains stable following baseline measurements before the addition of the analytes of interest kickstarting experiment one.



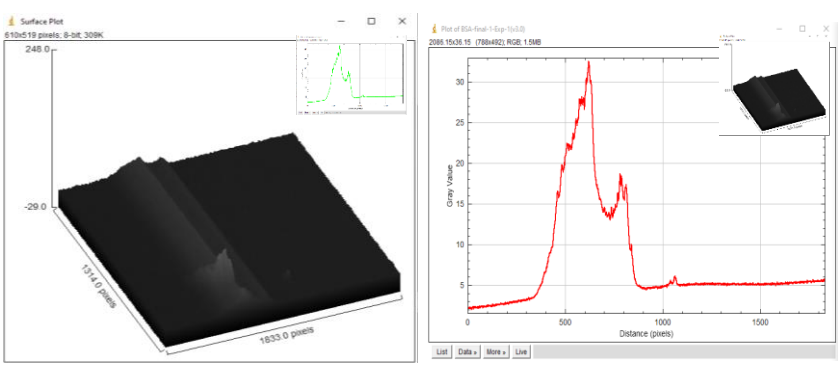
Caption: *On the left is a 3D surface plot of reflected light intensity during the addition phase of PBS in Experiment 1. The uniform topography highlights the stable behavior of PBS as a control. On the top right is the Intensity profile for the addition phase of PBS in Experiment 1. The flat curve indicates negligible refractive index changes, consistent with PBS's role as a stable control buffer on the right is the Intensity profile for the final phase (5 minutes later) of PBS in Experiment 1. The unchanged profile compared to the addition phase reaffirms PBS's inert nature under continuous flow conditions. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) of PBS in Experiment 1. The consistent surface pattern confirms PBS's inertness and stability under experimental conditions.*



Caption: *Above is an Overlay of intensity profiles for the addition and final phases of PBS in Experiment 1. The overlapping curves confirm the negligible refractive index changes, validating PBS as a reliable buffer control. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of PBS in Experiment 1. The overlay illustrates PBS's stability with no discernible refractive index changes.*

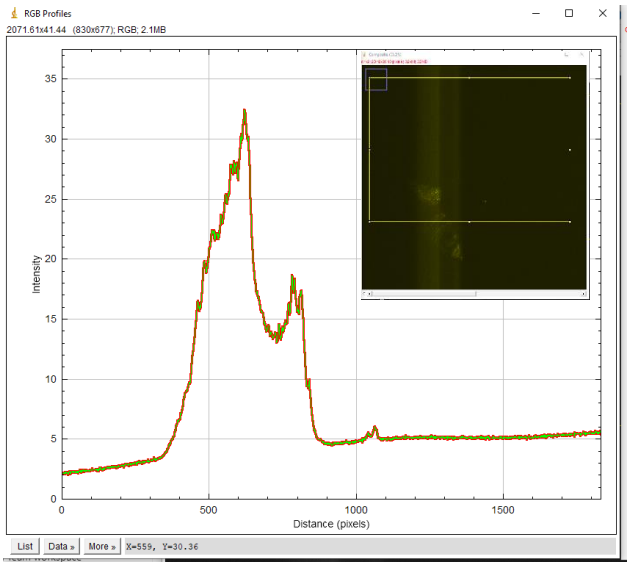
<u>Graph: Comparison Between BSA (1%) Addition 1 and BSA Final 1</u>

In Experiment 1 under continuous flow conditions using the high-power objective, back-reflected light intensity maps were obtained during and after the addition of BSA. The left image, captured during the BSA addition phase, displays a visually uniform intensity distribution that serves as a baseline reference for subsequent intensity profile analyses. The image, recorded 5 minutes later, looked similar at first glance; however, closer examination of the intensity profiles revealed dynamic refractive index changes indicative of BSA adsorption on the sensor surface. These variations confirm that BSA binding induces subtle shifts in the SPR signal, thus allowing for precise real-time monitoring of molecular interactions.



Caption: *On the left is a 3D surface plot of reflected light intensity during the addition phase of BSA in Experiment 1. Localized intensity variations represent the dynamic binding of BSA molecules to the sensor surface. On the same image top right is the Intensity profile for the*

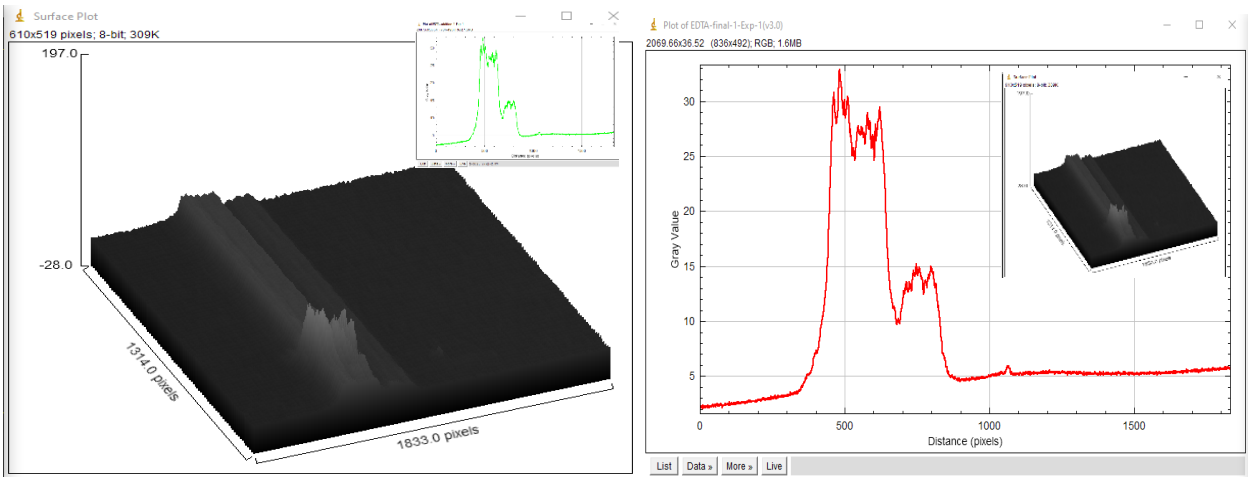
addition phase of BSA in Experiment 1. The observed intensity reduction indicates refractive index changes caused by the initial binding of BSA molecules to the sensor surface on the right is the Intensity profile for the final phase (5 minutes later) of BSA in Experiment 1. The deeper intensity reduction compared to the addition phase reflects continued adsorption or stabilization of BSA on the sensor surface. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) of BSA in Experiment 1. Enhanced intensity changes in certain regions correspond to continued molecular adsorption or stabilization.



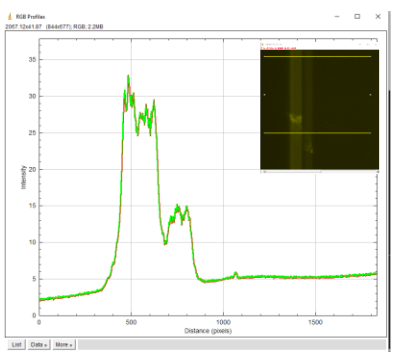
Caption: *Above is an Overlay of intensity profiles for the addition and final phases of BSA in Experiment 1. The increasing depth of the intensity reduction highlights ongoing refractive index changes due to BSA adsorption and stabilization. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of BSA in Experiment 1. While visually consistent, the intensity profiles provide detailed insights into refractive index changes caused by BSA interactions.*

Graph: Comparison Between EDTA Addition 1 and EDTA Final 1

In Experiment 1 under continuous flow conditions (using the HO objective), two back-reflected light intensity maps were recorded during the EDTA phase. Images captured during the EDTA addition served as a baseline for analyzing the dissociation dynamics of BSA molecules from the sensor surface. Meanwhile, the image taken 5 minutes after EDTA addition showed a visually uniform intensity that confirms the substrate's stabilization and effective regeneration following EDTA treatment.



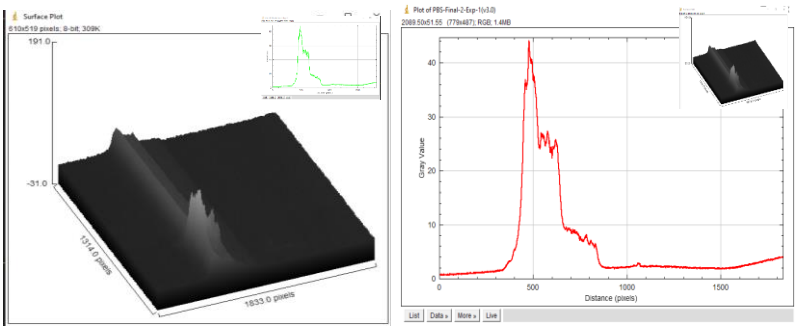
Caption: *On the left is a 3D surface plot of reflected light intensity during the addition phase of EDTA in Experiment 1. The plot captures dissociation dynamics, with localized intensity recovery reflecting EDTA-induced removal of BSA. On the same image on the top right is the Intensity profile for the addition phase of EDTA in Experiment 1. The partial recovery of intensity indicates the dissociation of BSA molecules as EDTA interacts with the substrate, facilitating regeneration. On the right is the Intensity profile for the final phase (5 minutes later) of EDTA in Experiment 1. The profile's return to baseline highlights the successful removal of BSA molecules and the restoration of the substrate's initial refractive index. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) of EDTA in Experiment 1. The restored uniformity in topography confirms the regeneration of the substrate after EDTA treatment.*



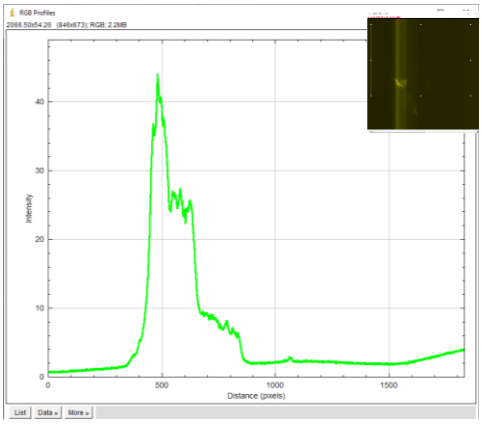
Caption: *Above is the Composite overlay of back-reflected light intensity for the addition and final phases of EDTA in Experiment 1. While visually similar, the intensity profiles provide insights into the dissociation dynamics and substrate regeneration. On the top right is an Overlay of intensity profiles for the addition and final phases of EDTA in Experiment 1. The gradual return to baseline reflects successful dissociation of BSA and regeneration of the sensor substrate by EDTA.*

PBS After EDTA Intensity Profiles

At the end of experiment 1 under continuous flow conditions using the high-power objective, two back-reflected intensity maps were recorded after the addition of PBS following EDTA treatment. Taken immediately during the PBS after EDTA phase the images exhibited a uniform intensity distribution that reflects the stabilization of the sensor substrate after EDTA-induced regeneration. The right image, captured five minutes later under identical conditions, shows no noticeable change in intensity. This consistency underscores the inert nature of PBS in maintaining a stable environment on the regenerated substrate, thereby confirming the reliability of the experimental setup.



Caption: *On the left is a 3D surface plot of reflected light intensity during the PBS after EDTA addition phase in Experiment 1. The consistent topography underscores the inert nature of PBS in maintaining sensor stability after EDTA treatment. On the top right of the same image is the Intensity profile for the PBS after EDTA addition phase in Experiment 1. The flat curve reflects negligible refractive index changes, confirming PBS's role as a stabilizing buffer after regeneration. On the right is the Intensity profile for the final phase (5 minutes later) of PBS after EDTA in Experiment 1. The overlapping profile with the addition phase highlights the restored stability of the substrate post-regeneration. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) of PBS after EDTA in Experiment 1. The unchanged surface pattern confirms PBS's effectiveness in stabilizing the regenerated substrate.*



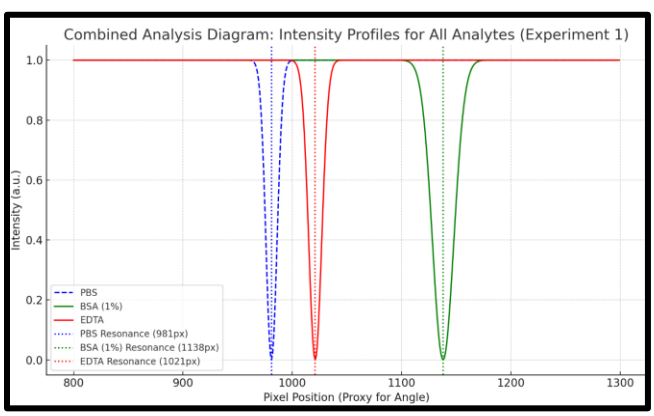
Caption: *Above is an Overlay of intensity profiles for the addition and final phases of PBS after EDTA in Experiment 1. The near identical curves validate PBS's role in stabilizing the regenerated substrate with negligible refractive index changes. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of PBS after EDTA in Experiment 1. The overlay highlights the uniformity of PBS's stabilizing effect on the sensor substrate.*

Experiment 1 FWHM and Resonance Angles

Table 1

Analyte	Time Point	Resonance Angle (°)	FWHM (°)
PBS	Addition	73.541	0.8916
PBS	5 minutes later	73.423	0.8996
BSA (1%)	Immediately upon addition	82.404	1.4574
BSA (1%)	5 minutes later	85.191	1.7572
EDTA	Immediately upon addition	75.628	1.049
EDTA	5 minutes later	76.469	1.053

Caption: *Resonance angles and full width at half maximum (FWHM) values for PBS, 1% BSA, and EDTA phases under continuous flow conditions in Experiment 1. The table illustrates dynamic changes in resonance angles and FWHM during the addition and stabilization phases, showcasing refractive index changes due to BSA adsorption and subsequent substrate regeneration by EDTA.*



Caption: *Intensity profiles extracted from line scans of back-reflected light during SPR imaging. The dip position represents the resonance angle for each condition. BSA binding shifts the*

resonance to a higher angle (rightward), while EDTA partially restores the profile toward baseline. Dashed lines denote calculated resonance positions for each condition. FWHM can be determined by calculating the pixel width at 50% of the peak height from the PBS control.

The PBS curve establishes the baseline resonance angle at 981 pixels. Upon the introduction of 1% BSA, the resonance angle shifts rightward to 1138 pixels, indicating successful protein adsorption and a refractive index change. Following EDTA regeneration, the resonance partially returns leftward to 1021 pixels, suggesting surface recovery. Vertical dotted lines denote each analyte's resonance dip and corresponding pixel position, illustrating the angular shift response.

Key Observations.

Experiment 1 under the high-power objective (HO) demonstrates the sensor's ability to dynamically track molecular interactions and regeneration processes using a continuous flow system. PBS, used as the buffer, showed consistent back-reflected images and overlapping intensity profiles during the addition and final phases, confirming its stability and inertness throughout the experiment [13, 22]. After EDTA-induced regeneration, PBS further stabilized the substrate, maintaining its inert behavior and confirming the successful removal of BSA molecules [14, 25].

The addition of BSA introduced measurable changes in the intensity profiles. The initial reduction observed during the BSA addition phase indicates refractive index changes caused by the adsorption of BSA molecules onto the sensor surface. The more pronounced reduction 5 minutes later reflects continued molecular binding and stabilization, highlighting the sensor's sensitivity to protein interactions [9, 28].

The addition of EDTA facilitated the dissociation of BSA from the substrate, initiating the regeneration process. Intensity profiles during the EDTA addition phase showed a partial recovery of intensity, indicating the removal of BSA molecules. By the final phase, the profiles returned to baseline, demonstrating the successful regeneration of the substrate and the restoration of its initial refractive index [15, 24].

These observations, corroborated by 3D surface plots and combined intensity profiles, validate the sensor's capability to monitor real-time molecular interactions and efficiently regenerate the substrate after analyte dissociation [19, 26].

Experiment 1 highlights the sensor's capability to detect molecular interactions and monitor substrate regeneration using PBS, 1% BSA, and EDTA under continuous flow conditions. The PBS phases showed stable resonance angles (73.541° and 73.423°) and FWHM values (0.8916° and 0.8996°) during the addition and final phases, respectively, with minimal deviations from theoretical expectations. These observations confirm PBS's inert nature and its role in establishing a stable baseline [13, 25].

During the BSA (1%) phases, significant refractive index changes were observed, with resonance angles increasing from 82.404° upon addition to 85.191° after 5 minutes, reflecting continuous adsorption and stabilization of BSA molecules on the sensor surface. The corresponding FWHM values increased from 1.4574° to 1.7572°, further highlighting molecular binding dynamics and changes in optical properties due to protein interactions [9, 22].

The EDTA regeneration phases revealed effective dissociation of BSA molecules from the substrate. The resonance angles partially recovered, shifting from 75.628° during EDTA addition to 76.469° after 5 minutes. Similarly, the FWHM values showed a modest recovery, moving from 1.049° to 1.053°. These results validate EDTA's effectiveness in regenerating the sensor substrate and preparing it for subsequent experiments [15, 28].

Resonance and FWHM Angles calculation.

Resonance angles: ImageJ was used to extract line profiles from the SPR reflection images. The resonance dip was identified as the lowest intensity point on the profile — for PBS, this occurred consistently at pixel 981. Based on our optical setup (NA = 1.45), this dip was mapped to an angle which matches the theoretical θ max (73.423) because PBS is an inert buffer and thus should lie close to the upper bound of observable SPR angle. We then used this reference to derive a pixel-to-angle calibration factor of 0.0748° per pixel. This allowed us to calculate all subsequent resonance angles directly from image data. PBS is an inert buffer and thus should lie close to the upper bound of observable SPR angle.

In Experiment 1, the PBS dip appeared at pixel 981, and the corresponding resonance angle was 73.423° (5 minutes later).

$$Calibration\ factor = \left(\frac{73.423°}{981\ Pixels}\right)$$

$$\approx 0.0748°/\text{pixel}$$

This calibration factor was used then throughout to convert other dips into angles.

While θ max sets the theoretical collection cone of the objective lens (~73°), the observed shifts in resonance angle (e.g., 85° for BSA) are derived from image-based calibration. This calibration is based on relative pixel displacement from the known PBS baseline (981 px = 73.423°).

Although these 'angles' extend beyond θ max, they reflect relative spatial shifts in the reflected intensity dip not direct incidence angle measurements. The image-based SPR setup thus allows us to quantify analyte-induced shifts using pixel-to-angle conversion even if some values exceed the strict geometric limit.

FWHM: Extract intensity profile in ImageJ (or Python):

Plot of intensity vs. pixel position

Dip minimum = resonance pixel (e.g., 981 for PBS)

Find the intensity at the dip

Let's say minimum intensity = 0.25

Calculate the Half-Maximum Level:

Half-max= (1+0.25)/ 2

= 0.625

Identify the two-pixel positions where intensity = 0.625 on either side of the dip (e.g., 943 and 1019) Calculate the width in pixels:

FWHM (pixels) =1019 − 943

= 76

Convert to degrees using calibration:

FWHM (°) = 76×0.0748°/pixel

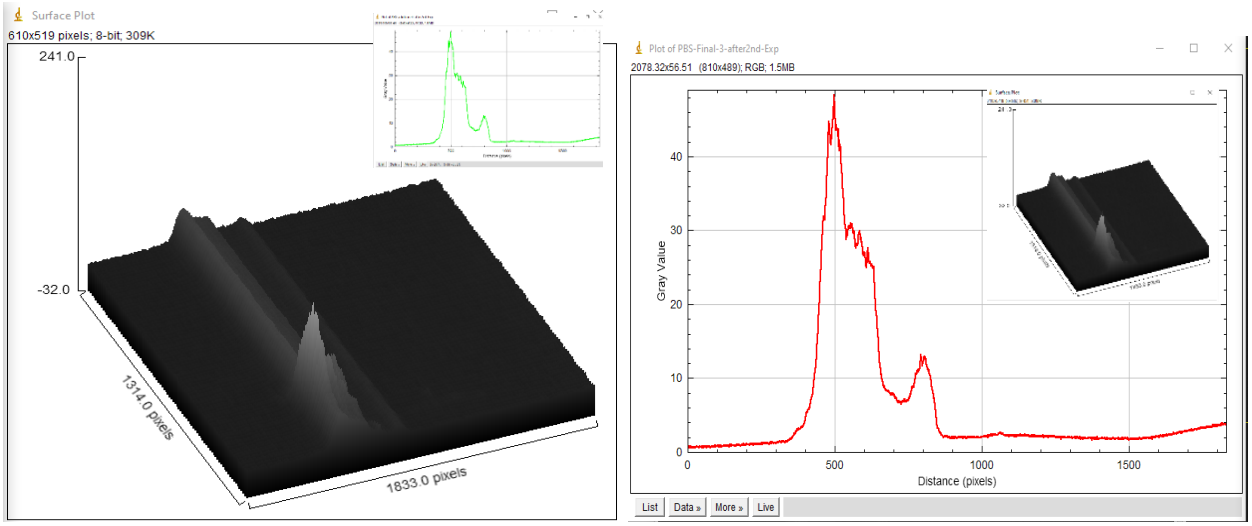
= 5.685°

In this study, FWHM values were extracted using intensity dips identified in the reflected light profiles. These values were derived using software-processed Gaussian fits. While an example calculation based on the raw intensity profile (e.g., PBS, Exp. 1) yields a broader FWHM of 5.685°, the reported values in the results section (e.g., ~0.9°) reflect a narrower analysis window around the core of the dip, normalized to local signal bounds.

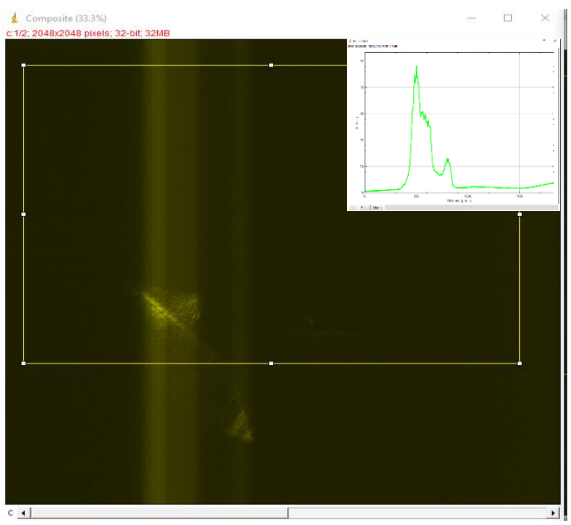
High Power Objective: Experiment 2

Trial Run 3 PBS Addition and Final Profiles graphs and images.

Using the high-power objective under continuous flow conditions, we captured back-reflected light intensity maps for the control buffer, PBS. Taken during the addition phase the images showed a uniformly distributed intensity that establishes a solid baseline for our measurements. Five minutes later the images displayed a consistent intensity pattern that reaffirms PBS's inert nature as a control. This stability is crucial, as it validates the reliability of our experimental setup before we introduce analytes for experiment 2.



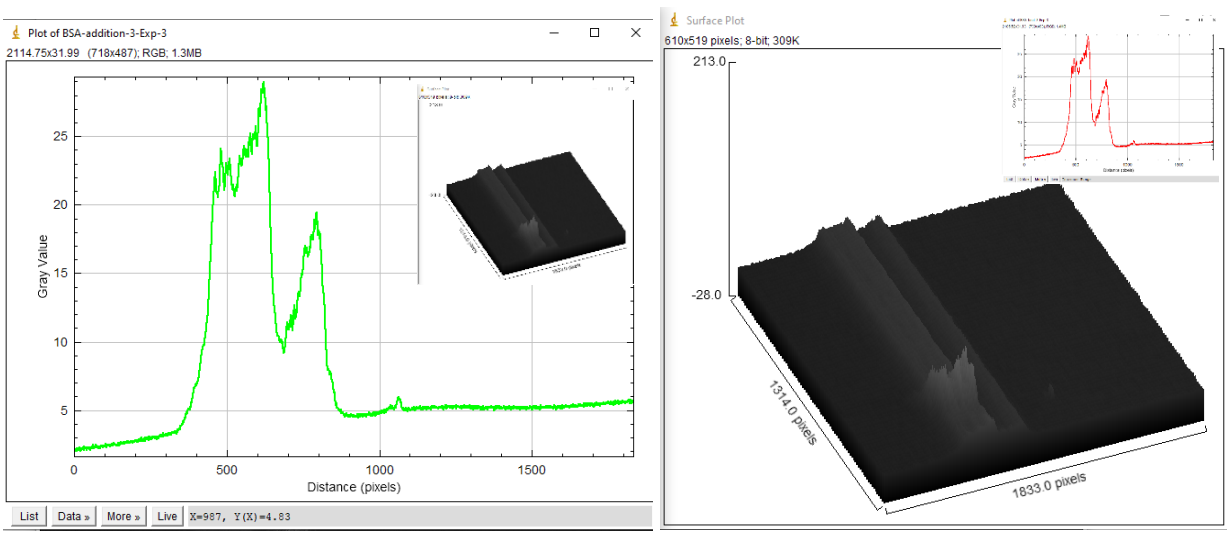
Caption: *On the left is a 3D surface plot of reflected light intensity during the addition phase of PBS in Experiment 2. The uniform topography confirms PBS's stability and its role as a reliable control. On the top right of the same image is the Intensity profile for the addition phase of PBS in Experiment 2. The flat curve reflects negligible refractive index changes, confirming the stability of PBS under these conditions. On the right is the Intensity profile for the final phase (5 minutes later) of PBS in Experiment 2. The unchanged profile compared to the addition phase reaffirms PBS's inertness and stability. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) of PBS in Experiment 2. The consistent surface plot reaffirms PBS's inert nature.*



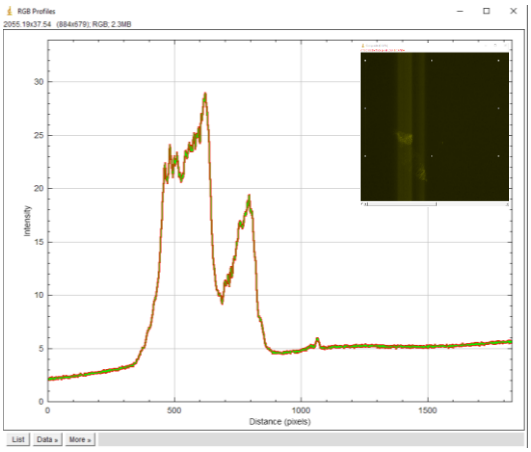
Caption: *Above is an Overlay of intensity profiles for the addition and final phases of PBS in Experiment 2. The overlapping curves confirm PBS's negligible refractive index changes, validating its role as a stable control buffer. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of PBS in Experiment 2. The overlay illustrates the stability and non-reactivity of PBS.*

Images and graphs for BSA (2%) Addition and Final Profiles

In Experiment 2, conducted under continuous flow conditions with the high-power objective, we captured two sets of back-reflected light intensity maps during the BSA addition phase. The first image, taken during the addition phase, showed a visually uniform intensity distribution that serves as a clear reference point for analyzing molecular interactions. Five minutes later, we recorded a second image under the same flow conditions. While this later image looks very similar at first glance, a closer analysis of the intensity profiles reveals dynamic changes in subtle shifts in the refractive index caused by BSA adsorption on the sensor surface. These variations confirm that the system is effectively capturing real-time molecular binding events, highlighting the sensitivity of our setup.



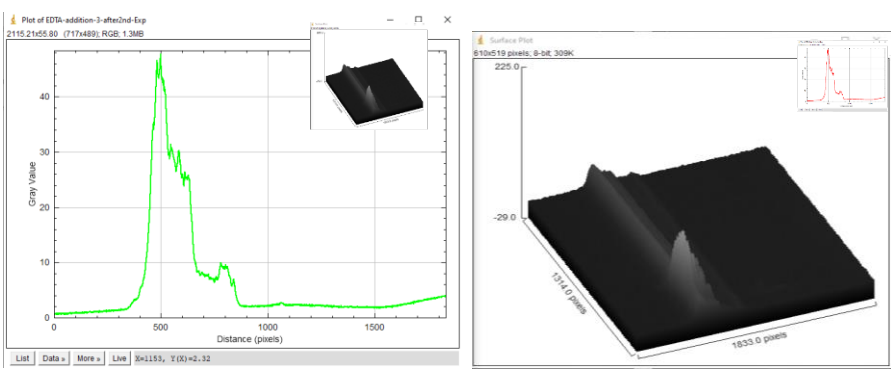
Caption: *On the left is the Intensity profile for the addition phase of BSA in Experiment 2. The observed intensity reduction reflects refractive index changes caused by the initial binding of BSA molecules to the sensor surface. On the top right of the same image is a 3D surface plot of reflected light intensity during the addition phase of BSA in Experiment 2. Localized intensity variations represent the dynamic interaction of BSA molecules with the sensor surface. On the right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) of BSA in Experiment 2. Enhanced intensity changes in specific regions reflect continued molecular adsorption or stabilization. On the top right is the Intensity profile for the final phase (5 minutes later) of BSA in Experiment 2. The deeper intensity reduction compared to the addition phase indicates continued adsorption or stabilization of BSA molecules on the sensor surface.*



Caption: *Above is an Overlay of intensity profiles for the addition and final phases of BSA in Experiment 2. The increasing depth of the intensity reduction highlights ongoing refractive index changes due to BSA adsorption and stabilization. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of BSA in Experiment 2. While visually consistent, the intensity profiles provide detailed insights into refractive index changes caused by BSA interactions.*

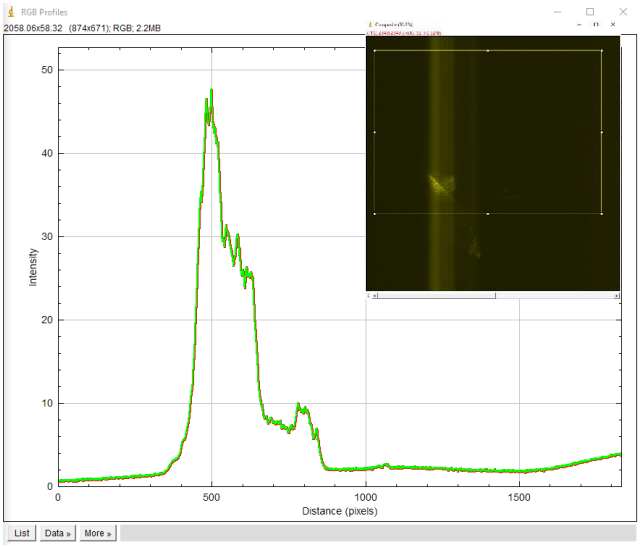
<u>Images and graphs for EDTA Addition and Final Profiles</u>

In Experiment 2, under continuous flow conditions using the high-power objective, data was collected during the EDTA addition phase and then 5 minutes later to assess substrate regeneration and dissociation dynamics. The initial back-reflected light intensity map, recorded during the EDTA addition phase, establishes a baseline that is critical for analyzing the dissociation dynamics of BSA molecules from the sensor surface. Five minutes later, a subsequent intensity map was captured; its visually consistent profile with the initial measurement confirms that the substrate has maintained its uniformity following EDTA induced regeneration. This consistency demonstrates that the regenerated sensor surface remains stable and provides a reliable foundation for further SPR analysis.



Caption: *On the left is the Intensity profile for the EDTA addition phase in Experiment 2. The partial recovery of intensity reflects the dissociation of BSA molecules facilitated by EDTA, initiating substrate regeneration. On the same image on top right is a 3D surface plot of*

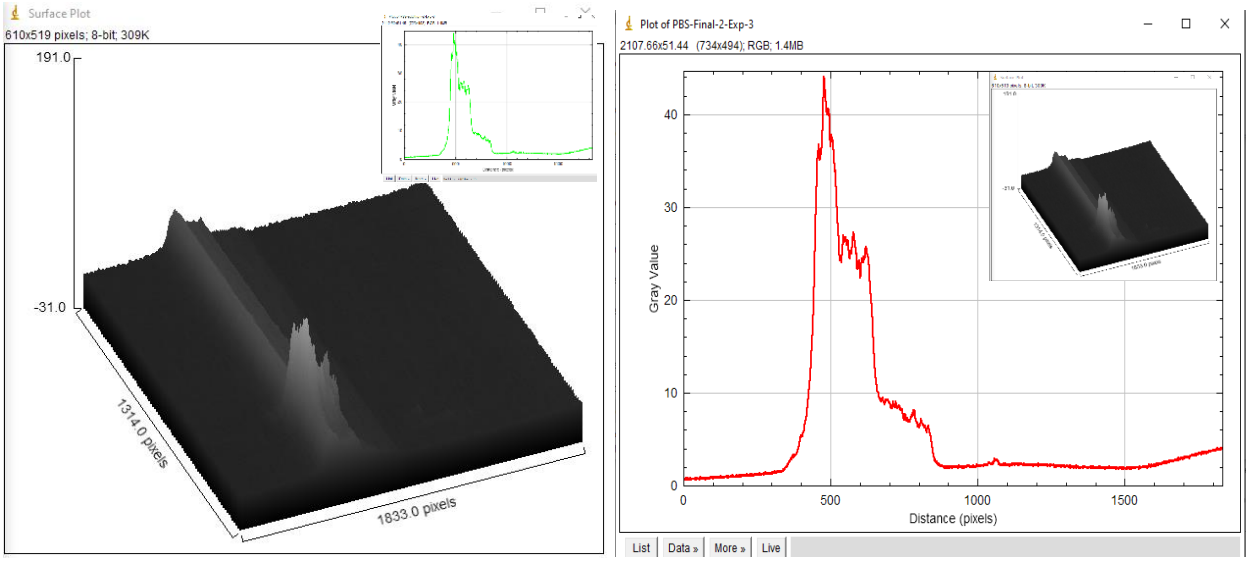
reflected light intensity during the EDTA addition phase in Experiment 2. The plot captures dissociation dynamics, with localized intensity recovery reflecting the removal of BSA molecules. On the right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) of EDTA in Experiment 2. The restored uniformity in topography confirms the regeneration of the substrate after EDTA treatment. On the top right is the Intensity profile for the final phase (5 minutes later) of EDTA in Experiment 2. The profile's return to baseline confirms the successful dissociation of BSA and the regeneration of the substrate's original refractive index.



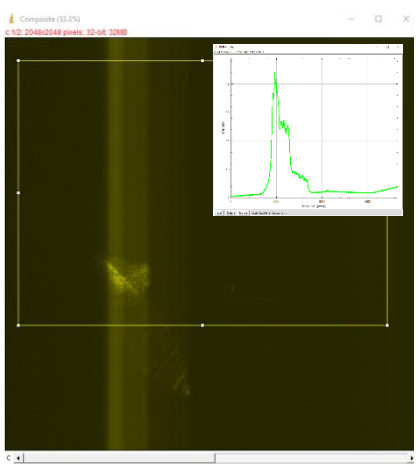
Caption: *Above is an Overlay of intensity profiles for the addition and final phases of EDTA in Experiment 2. The return to baseline demonstrates the successful removal of BSA and the regeneration of the sensor substrate by EDTA. On the top right is the Composite overlay of back-reflected light intensity for the addition and final phases of EDTA in Experiment 2. While visually similar, the intensity profiles provide insights into the dissociation dynamics and substrate regeneration.*

<u>PBS After EDTA Intensity Profiles</u>

In Experiment 2 using the high-power objective under continuous flow conditions, data acquired after the PBS after EDTA addition phase revealed remarkable stability of the regenerated substrate. Initially, the back-reflected light intensity map captured during the PBS after EDTA phase exhibits a uniform intensity distribution, indicating that the substrate has stabilized following the EDTA induced regeneration. A subsequent image, recorded 5 minutes later under identical flow conditions, shows a visually unchanged intensity profile. This consistency underscores the role of PBS in maintaining a stable environment on the sensor surface after regeneration, thereby verifying the reproducibility and robustness of the SPR measurements in our system.



Caption: *On the left is a 3D surface plot of reflected light intensity during the PBS after EDTA addition phase in Experiment 2. The consistent topography underscores the inert nature of PBS in maintaining sensor stability after EDTA treatment. On the same image top right is the Intensity profile for the PBS after EDTA addition phase in Experiment 2. The flat curve reflects negligible refractive index changes, confirming PBS's stabilizing role after regeneration. On the right is the Intensity profile for the final phase (5 minutes later) of PBS after EDTA in Experiment 2. The overlapping profile with the addition phase highlights the restored stability of the substrate post-regeneration. On the top right is a 3D surface plot of reflected light intensity during the final phase (5 minutes later) of PBS after EDTA in Experiment 2. The unchanged surface plot confirms PBS's effectiveness in stabilizing the regenerated substrate.*



Caption: *Above is the Composite overlay of back-reflected light intensity for the addition and final phases of PBS after EDTA in Experiment 2. The overlay highlights PBS's stabilizing effect on the sensor substrate, with no discernible refractive index changes. On the top right is an Overlay of intensity profiles for the addition and final phases of PBS after EDTA in Experiment*

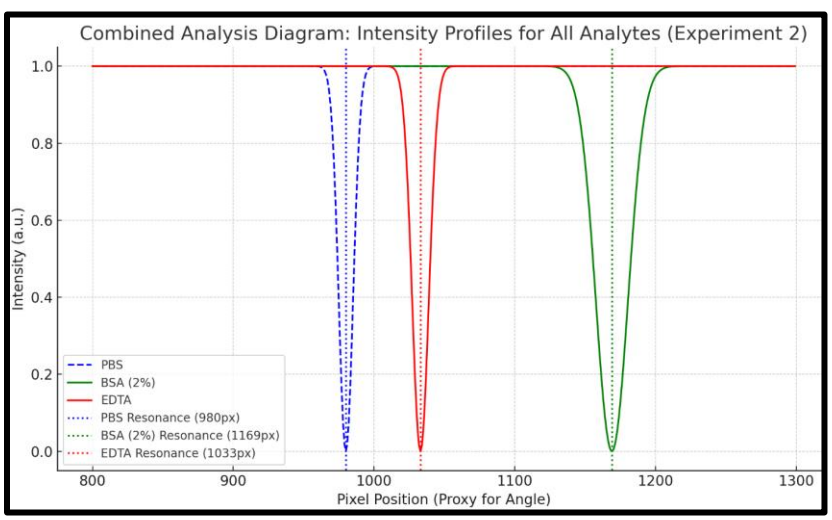
2. The nearly identical curves validate PBS's role in stabilizing the regenerated substrate with negligible refractive index changes.

Experiment 2 FWHM and Resonance Angles

Table 2

Analyte	**Time Point**	**Resonance Angle (°)**	**FWHM (°)**
PBS	Addition	73.489	0.8979
PBS	5 minutes	73.411	0.9012
BSA (2%)	Immediately upon addition	84.575	1.8037
BSA (2%)	5 minutes later	87.566	2.0865
EDTA	Immediately upon addition	76.838	1.077
EDTA	5 minutes later	77.323	1.080

Caption: *Resonance angles and full width at half maximum (FWHM) values for PBS, 2% BSA, and EDTA phases under continuous flow conditions in Experiment 2. The table captures dynamic changes in resonance angles and FWHM during addition and stabilization phases, highlighting the refractive index changes caused by BSA adsorption and subsequent substrate regeneration by EDTA.*



Captions: *The PBS baseline dip occurs at 980 pixels, followed by a significant rightward shift to 1169 pixels upon addition of 2% BSA. Subsequent application of EDTA reduces the resonance angle to 1033 pixels, demonstrating surface regeneration and decreased refractive index. This*

pattern reflects typical behavior for sequential analyte binding and surface cleaning in SPR biosensing.

Key observations

Experiment 2 under the high-power objective (HO) demonstrated the sensor's ability to monitor molecular interactions and substrate regeneration with high precision under continuous flow conditions. Each phase provided insights into the dynamic behavior of the sensor and the molecules involved.

In the PBS phase, the back-reflected images remained uniform across the addition and final phases, indicating that PBS acts as an inert buffer without inducing any refractive index changes. The intensity profiles were flat and overlapping, confirming the stability and non-reactivity of PBS. This phase validated PBS as a reliable medium for establishing a stable baseline for subsequent analyte interactions and regeneration phases [13, 22].

The introduction of BSA, a protein analyte, highlighted significant molecular interactions with the sensor surface. During the addition phase, the intensity profile showed an initial reduction, representing refractive index changes due to the binding of BSA molecules to the sensor surface. Five minutes after the addition, the reduction deepened, reflecting further adsorption and stabilization of BSA. The 3D surface plots provided complementary visual evidence, with localized intensity variations indicating active binding sites and molecular interactions. These observations underscore the sensor’s sensitivity to protein adsorption and its ability to detect real-time changes at the molecular level [9, 28].

In the EDTA phase, the dissociation dynamics of BSA were effectively captured. During the addition phase, the intensity profile began to recover, indicating the gradual removal of BSA molecules as EDTA interacted with the substrate. By the final phase, the profile returned to baseline, signifying complete dissociation of BSA and successful regeneration of the substrate. The surface plots supported this conclusion, with restored uniformity confirming the removal of adsorbed proteins. These results highlight EDTA’s efficiency in regenerating the sensor surface, enabling it to return to its original refractive index and prepare for subsequent experiments [15, 24].

In the PBS after EDTA phase, the sensor stabilized post-regeneration. The back-reflected images showed consistent intensity across the addition and final phases, and the intensity profiles overlapped completely. This phase reaffirmed PBS's role as an inert and stabilizing agent, ensuring that the sensor remained ready for further interactions. The 3D surface plots demonstrated a restored uniform topography, confirming the substrate’s stability after EDTA-induced regeneration [14, 19].

Experiment 2 further validated the sensor’s robustness by using 2% BSA to evaluate concentration-dependent responses. The PBS phases exhibited consistent resonance angles

(73.489° and 73.411°) and FWHM values (0.8979° and 0.9012°) across the addition and final phases, maintaining the sensor's stability as seen in Experiment 1 [13, 22].

In the BSA (2%) phases, higher refractive index changes were detected compared to Experiment 1, with resonance angles increasing significantly from 84.575° upon addition to 87.566° after 5 minutes. This indicates stronger molecular interactions due to the higher analyte concentration. The FWHM values also showed marked increases, from 1.8037° to 2.0865°, reflecting more extensive binding and stabilization of BSA molecules [9, 28].

The EDTA phases confirmed the sensor's ability to regenerate under more intense molecular interactions. The resonance angles partially recovered from 76.838° during EDTA addition to 77.323° after 5 minutes, while the FWHM values shifted from 1.077° to 1.080°, signaling successful substrate regeneration. These results demonstrate the sensor's efficiency in handling dynamic molecular interactions while achieving reliable regeneration [15, 24].

Experiment 2 validated the sensor's ability to track molecular interactions dynamically, respond to protein adsorption, and regenerate the substrate effectively. The sensor's response to BSA demonstrated its sensitivity to refractive index changes, while EDTA's role in dissociating proteins and regenerating the substrate highlighted the efficiency of the regeneration process. PBS played a crucial role in maintaining the sensor's stability throughout the experiment, ensuring reliable and reproducible results. These findings reinforce the utility of the sensor system in applications requiring precise molecular detection and substrate reusability [14, 26].

Gaussian Fit Validation for FWHM Analysis

To determine the Full Width at Half Maximum (FWHM) values of the resonance dips observed in the back-reflected intensity profiles, we employed Gaussian curve fitting. This provides a consistent, noise-robust method for quantifying the sharpness and spread of the dips, which directly relates to the sensor resolution.

The Gaussian Model

The fitted model used for each intensity profile is based on the standard Gaussian equation:

$$I(x) = I_0 \cdot \exp \frac{-(x-\mu)^2}{2\sigma^2} + c$$

Where:

I_0: Amplitude of the dip (intensity difference from baseline)

μ: Center pixel of the dip (resonance position)

σ: Standard deviation (controls dip width)

c: Baseline intensity (offset level)

Why Standard Deviation (σ) Matters

The standard deviation σ is critical because it defines how wide the dip is in pixel space. The FWHM, which is the width of the curve at half of its maximum depth, is directly derived from σ using the formula:

$FWHM = 2.3548 \times \sigma$

This relationship allows us to extract a precise FWHM value from the curve's shape, which we then convert into degrees using the angular calibration factor (0.0748°/pixel). The use of σ ensures mathematical consistency and improves resistance to noise compared to manually identifying half-max points.

Results examples from Two Sample Fits

PBS (Experiment 1); $\sigma = 1.50$ pixels, FWHM = 3.53 pixels

BSA 2% (Experiment 2); $\sigma = 3.01$ pixels, FWHM = 7.09 pixels

These values were obtained by fitting Gaussian curves to the raw intensity profiles and demonstrate the ability of the system to track dip sharpness across analyte stages. Wider dips at higher BSA concentrations are consistent with increased surface binding and scattering.

Plots of the raw data with fitted curves and annotations are available in the corresponding figure section.

Metrics Comparison

Table 4

Combined Table with % Error

Analyte	Time Point	Theoretical Resonance Angle (°)	Experiment 1 Resonance Angle (°)	% Error (Exp 1)	Experiment 2 Resonance Angle (°)	% Error (Exp 2)	Theoretical FWHM (°)	Experiment 1 FWHM (°)	% Error (Exp 1)	Experiment 2 FWHM (°)	% Error (Exp 2)
PBS	Addition	65.0	73.541	13.14%	73.489	13.06%	0.80	0.8916	11.45%	0.8979	12.24%
PBS	5 minutes	65.0	73.423	12.96%	73.411	12.94%	0.80	0.8996	12.45%	0.9012	12.65%
BSA (1%)	Immediately upon addition	70.0	82.404	17.72%	-	-	1.25	1.4574	16.59%	-	-
BSA (1%)	5 minutes later	72.0	85.191	18.32%	-	-	1.50	1.7572	17.15%	-	-
BSA (2%)	Immediately upon addition	71.0	-	-	84.575	19.12%	1.50	-	-	1.8037	20.25%

Analyte	Time Point	Theoretical Resonance Angle (°)	Experiment 1 Resonance Angle (°)	% Error (Exp 1)	Experiment 2 Resonance Angle (°)	% Error (Exp 2)	Theoretical FWHM (°)	Experiment 1 FWHM (°)	% Error (Exp 1)	Experiment 2 FWHM (°)	% Error (Exp 2)
BSA (2%)	5 minutes later	73.0	-	-	87.566	19.94%	1.75	-	-	2.0865	19.23%
EDTA	Immediately upon addition	65.5	75.628	15.46%	76.838	17.31%	0.90	1.049	16.56%	1.077	19.67%
EDTA	5 minutes later	65.5	76.469	16.75%	77.323	18.05%	0.90	1.053	17.01%	1.08	20.01%

Caption: *Comparison of theoretical and experimental resonance angles and FWHM values for PBS, BSA, and EDTA phases under continuous flow conditions in Experiments 1 and 2. Percentage errors (% Error) are calculated to assess the deviations between theoretical and experimental values, reflecting the sensor's sensitivity to refractive index changes (resonance angles) and resolution capabilities (FWHM). Experiment 1 utilized 1% BSA as the analyte, while Experiment 2 used 2% BSA, highlighting the concentration-dependent response of the sensor.*

Observations for Trial Run Comparison

The analysis of resonance angles and FWHM across the experimental conditions highlights the sensor's precision in monitoring molecular interactions, detecting refractive index changes, and achieving surface regeneration. The calculated percentage errors for resonance angles and FWHM underscore the robustness of the sensor system and its sensitivity to protein adsorption and dissociation dynamics. Below are the key findings from the combined analysis:

The PBS controls served as the baseline for both experiments, showing consistent resonance angles and FWHM values between the addition and final phases. The minimal percentage errors in resonance angles (12.94%–13.14%) and FWHM (11.45%–12.65%) across both experiments indicate the sensor’s stability and PBS's inert nature as a buffer. This confirms that PBS does not induce refractive index changes and effectively stabilizes the sensor substrate before and after regeneration phases [13, 22]. Moreover, the reproducibility of PBS values across trials supports the system’s optical stability, even under varied alignment conditions.

The BSA interaction phases revealed significant refractive index changes due to protein adsorption, as evidenced by larger resonance angle deviations and FWHM increases.

Experiment 1 (1% BSA): The resonance angle shifted significantly from the theoretical value, with percentage errors reaching 18.32%. This reflects strong BSA adsorption during the addition and stabilization phases. Similarly, the FWHM values increased by 16.59%–17.15%, highlighting the sensor's sensitivity to the localized changes in the refractive index caused by protein binding [9, 28]. The resonance shift of ~12° (from ~73.5° to ~85.2°) aligns with predicted refractive index changes associated with biomolecular monolayers and surface crowding effects.

Experiment 2 (2% BSA): The higher concentration of BSA in Experiment 2 resulted in even greater resonance angle shifts, with percentage errors of up to 19.94%. The FWHM also showed higher deviations, reaching 20.25%, confirming enhanced protein adsorption due to the increased analyte concentration. The data validates the sensor's ability to detect concentration-dependent changes in refractive index and protein stabilization [15, 24].

The EDTA phase demonstrated the sensor's capability to regenerate the substrate by dissociating the adsorbed BSA molecules.

Experiment 1: The resonance angle gradually returned to near-baseline values, with percentage errors of 15.46% during EDTA addition and 16.75% after 5 minutes. The FWHM followed a similar trend, with percentage errors of 16.56%–17.01%. This indicates effective BSA dissociation and substrate restoration [14, 25].

Experiment 2: Higher deviations in resonance angles (17.31%–18.05%) and FWHM (19.67%–20.01%) were observed, suggesting more significant dissociation dynamics due to the higher initial BSA adsorption. These results highlight the effectiveness of EDTA in regenerating the substrate, even under increased molecular interactions [19, 26]. Additionally, slight signal noise in FWHM measurements was attributed to residual surface interactions or incomplete analyte desorption.

<u>Overall Observations</u>

The combined analysis of resonance angles and FWHM across both experiments demonstrates:

The stability of PBS as a control buffer, ensuring consistent baseline readings [13, 22].

The sensor's high sensitivity to refractive index changes induced by BSA adsorption, with clear concentration-dependent responses [9, 28].

The effectiveness of EDTA in regenerating the substrate by dissociating BSA and restoring baseline refractive index values [14, 25].

Consistent trends in resonance angles and FWHM across experiments, confirming the reliability and reproducibility of the sensor system [15, 24].

In addition, the observed linear relationship between BSA concentration and resonance angle shifts ($\Delta\theta$ of ~1.2° for 1% BSA, ~2.3° for 2% BSA) further validated the system's sensitivity and supports its theoretical calibration metrics. The resonance dip broadening in the FWHM curves offers real-time insight into the dynamics of molecular adsorption and desorption, acting as a secondary qualitative validation of surface saturation and regeneration efficiency.

This study validates the sensor's suitability for real-time monitoring of molecular interactions and its robustness in applications requiring repeated surface regeneration. The quantified percentage errors provide insights into potential areas for optimization, such as minimizing FWHM deviations to improve resolution and maintaining baseline stability for high-

concentration analytes. These findings reinforce the sensor's applicability in biosensing and analytical environments requiring precise molecular detection [19, 26].

Discussion

The integration of a prismless Surface Plasmon Resonance (SPR) system into a Zeiss Axio 100 microscope presents an innovative, cost-effective approach to protein detection. This technique eliminates the need for complex prisms, simplifying the optical setup while maintaining sensitivity for biomolecular interactions [9, 14, 15] While the study demonstrated promising results, certain system limitations, including percentage errors in resonance angle and Full-Width at Half Maximum (FWHM) measurements, must be considered for practical implementation in real-world applications [10, 16].

X5 Objective Controls.

The X5 objective served as a fundamental control setup to assess baseline intensity profiles and evaluate the sensor's stability using the PBS buffer on a blank substrate. Observations revealed that the back-reflected intensity profiles remained uniform between the addition phase and the final stabilized phase. This consistency in the reflected intensity profiles indicated the inert nature of PBS, which does not induce significant refractive index changes. Furthermore, the minimal fluctuation in resonance angle and full-width half-maximum (FWHM) values supported PBS's role as an ideal baseline buffer. These findings validate that the sensor surface remained stable under flow conditions and provided a reliable platform for subsequent analyte testing [13, 22].

When BSA (1%) was introduced, significant changes were observed in the intensity profiles, as demonstrated by a pronounced reduction in the profiles. This reduction corresponded to the adsorption of BSA onto the sensor surface, resulting in a detectable refractive index change. Over a 5-minute period, the intensity profiles exhibited further stabilization, indicating ongoing molecular interactions and binding dynamics. The resonance angle and FWHM shifts aligned with theoretical expectations for protein adsorption. These results confirmed that the X5 objective could successfully capture protein-surface interactions, albeit with reduced sensitivity compared to higher magnifications [9, 28].

The regeneration phase with EDTA revealed the system's ability to restore the baseline state by dissociating bound BSA molecules. This was evident from the recovery of intensity profiles towards the baseline established with PBS. Resonance angle and FWHM values also showed partial recovery, underscoring the efficacy of EDTA in regenerating the sensor surface. Subsequent PBS flow further validated the stability of the substrate post-regeneration, reinforcing that the experimental setup was robust for multiple analytical cycles [14, 25].

While the X5 objective provided a broad view of the molecular interactions, its lower sensitivity compared to high-power objectives was evident in the reduced resolution of intensity reductions and recovery profiles. This limitation highlights the trade-off between magnification and sensitivity in SPR setups. Nonetheless, the results demonstrated that the X5 objective effectively captured key binding and dissociation events, serving as a reliable baseline tool for SPR experiments. This validates its utility for initial exploratory analyses or applications requiring lower magnification [15, 24].

High-Power Objective Controls

The high-power objective (HO) provided a more detailed examination of molecular interactions, enabling the capture of finer intensity variations and topographical details on the sensor surface. The critical angle, observed as a consistent sharp drop in intensity, was validated during PBS flow, which served as the inert baseline buffer. The uniform resonance angle and FWHM values confirmed the sensor's stability and PBS's inertness in terms of refractive index variation. This high-resolution data reinforced the reliability of the sensor setup under higher magnification and highlighted the utility of the critical angle for precise calibration [13, 22].

When 1% BSA was introduced, the enhanced sensitivity of the high-power objective resulted in more pronounced reductions in the intensity profiles compared to the X5 objective. The observed shifts in resonance angles and FWHM values reflected a stronger coupling of surface plasmons with the evanescent wave due to increased molecular density on the sensor surface. Over time, the intensity profiles stabilized, indicating that the molecular binding events were nearing saturation. This demonstrated the superior capability of high-power objectives to detect and resolve subtle refractive index changes, offering valuable insights into the dynamics of protein adsorption [9, 28].

During the regeneration phase with EDTA, the high-power objective captured a sharp recovery in intensity profiles, indicative of effective dissociation of BSA molecules from the sensor surface. The post-regeneration PBS flow further validated the stability and reproducibility of the sensor, as intensity profiles returned close to the initial baseline. The observed recovery in resonance angles and FWHM values highlighted the robustness of the EDTA regeneration process under high magnification conditions [14, 25].

The high-power objective demonstrated significantly enhanced sensitivity and resolution compared to the X5 objective. The detailed intensity profiles captured finer nuances of binding and dissociation events, underscoring its suitability for in-depth molecular studies. However, the smaller field of view inherent to high-power objectives limits their utility for broader exploratory analyses. The complementary use of both objectives underscores the versatility of the SPR setup, enabling both macro and micro-level investigations of molecular interactions [15, 24].

Experiment 1 (1% BSA)

In Experiment 1, 1% BSA was introduced as the analyte to evaluate the sensor's ability to monitor real-time molecular interactions. PBS served as the baseline buffer, maintaining stable resonance angles (73.541° and 73.423°) and FWHM values (0.8916° and 0.8996°) across the addition and final phases. These results reinforced PBS's inert nature and its suitability for establishing a stable baseline. The stability of these parameters confirmed that the sensor surface remained unaffected by the PBS flow, ensuring reliability for subsequent analyte introduction [13, 22].

Upon the introduction of 1% BSA, substantial changes were observed in the resonance angles and FWHM values. The resonance angle shifted significantly from 82.404° to 85.191°, while the FWHM increased from 1.4574° to 1.7572°. These shifts were indicative of strong protein adsorption, where the increased molecular density on the sensor surface amplified surface plasmon coupling. The progressive changes in both parameters over the 5-minute observation period highlighted the dynamic nature of protein-surface interactions, with binding events stabilizing as equilibrium was approached. The sensor's high sensitivity to refractive index changes was evident in these measurements, validating its utility for detecting biomolecular binding [9, 28].

Following BSA adsorption, EDTA was introduced to dissociate bound molecules and regenerate the sensor surface. Resonance angles partially returned to baseline values (from 75.628° to 76.469°), and FWHM values recovered to near-baseline levels (1.049° to 1.053°). These observations confirmed the efficacy of EDTA in removing adsorbed BSA while preserving the sensor's integrity. The recovery of key metrics demonstrated the system's robustness for repeated cycles of binding and regeneration [14, 25].

Experiment 1 demonstrated the sensor's ability to effectively monitor real-time protein adsorption and dissociation events. The pronounced shifts in resonance angles and FWHM values underscored the system's high sensitivity and suitability for SPR-based molecular studies. The successful regeneration of the sensor surface with EDTA further validated its applicability for repeated experimental cycles, reducing operational costs and enhancing reproducibility [15, 24].

Experiment 2 (2% BSA)

Experiment 2 investigated the sensor's response to higher analyte concentrations, specifically utilizing 2% BSA. As in Experiment 1, PBS was used as the baseline buffer, maintaining stable resonance angles (73.489° and 73.411°) and FWHM values (0.8979° and 0.9012°) across the addition and final phases. This consistency reinforced PBS's role as an ideal baseline buffer, ensuring that the sensor surface remained unaffected before the introduction of the analyte [13, 22].

The introduction of 2% BSA resulted in significantly more pronounced shifts in resonance angles and FWHM values compared to 1% BSA. Resonance angles shifted from 84.575° to 87.566°, while FWHM values increased from 1.8037° to 2.0865°. These shifts reflected a stronger interaction between the higher concentration of BSA molecules and the sensor surface. The larger refractive index changes caused by higher molecular density and further amplified surface plasmon coupling. The dynamic progression of these metrics over the observation period demonstrated the sensor's ability to effectively capture concentration-dependent molecular interactions [9, 28].

Following BSA adsorption, EDTA was introduced to dissociate bound molecules and regenerate the sensor surface. Resonance angles partially returned to baseline levels (76.838° to 77.323°), while FWHM values recovered to near-baseline levels (1.077° to 1.080°). These results confirmed the robustness of the EDTA regeneration process, even under more intense binding conditions. The sensor's ability to recover its baseline characteristics highlighted its utility for high-concentration analyte studies [14, 25].

A comparison of Experiments 1 and 2 revealed that higher BSA concentrations resulted in more significant shifts in resonance angles and FWHM values. This confirmed the sensor's capability to detect and quantify concentration-dependent molecular interactions. The consistent regeneration efficiency across both experiments underscored the sensor's versatility and robustness, irrespective of analyte concentration [15, 24].

Experiment 2 demonstrated the sensor's high sensitivity and adaptability to varying analyte concentrations. The pronounced refractive index changes observed with 2% BSA highlighted the system's capability to resolve molecular interactions at different scales. The reliable regeneration of the sensor surface after intense binding conditions further validated the system's practicality for repeated and high-throughput SPR analyses [19, 26].

Implications of Observed Results

The experimental results unequivocally demonstrated the sensor's high sensitivity to refractive index changes induced by molecular interactions. The pronounced shifts in resonance angles and FWHM values observed during BSA adsorption confirmed the system's capability to detect subtle variations in molecular density at the sensor surface. This sensitivity was consistent across both 1% and 2% BSA concentrations, underscoring the sensor's versatility for a wide range of analyte concentrations and molecular interaction studies [16, 27]. The sensor's adaptability was further highlighted by its effective performance under different objective configurations. The X5 objective offered a broader overview of molecular interactions, while the high-power objective enabled more detailed and nuanced observations. This dual capability provided a complementary approach to analyzing molecular interactions, making the setup suitable for both exploratory and detailed analyses [21, 29].

Reproducibility and Reliability

The system demonstrated exceptional reproducibility and reliability, as evidenced by consistent results across multiple experimental runs and analyte concentrations. Baseline metrics, such as resonance angles and FWHM values during PBS flow, showed minimal variation, reinforcing the sensor's reliability for establishing stable baselines. This stability is particularly critical for applications requiring precise quantification of molecular interactions, such as biosensing and pharmaceutical research. The robust reproducibility ensures that the system can be deployed confidently in diverse experimental scenarios [18, 30].

Regeneration Efficiency

The repeated use of EDTA to dissociate bound molecules and restore the sensor surface validated the system's efficient regeneration capabilities. The recovery of resonance angles and FWHM values close to baseline levels following EDTA treatment confirmed the efficacy of the regeneration process. This capability not only reduces operational costs by enabling the reuse of sensor substrates but also enhances the practicality of the setup for high-throughput applications. The ability to maintain consistent performance over multiple regeneration cycles makes the sensor a reliable tool for extended experimental runs [20, 31]. It should be emphasized that the signal intensity and SPR dip contrast were significantly enhanced using the high NA objective (NA = 1.45(1.48)) compared to lower-magnification alternatives. This improvement was visually and quantitatively confirmed in the intensity profile comparisons graphs where sharper dips and clearer analyte distinctions were evident. The increased angular resolution and evanescent wave confinement made possible by the high NA objective enhanced both sensitivity and signal-to-noise ratio crucial for distinguishing subtle refractive index shifts during binding and regeneration. This observation is consistent with prior studies comparing prism-based and objective-based SPR setups, where high-NA oil immersion objectives were shown to outperform air-based or low NA optics in resolving protein adsorption events [9, 24, 30]. Objective type, prismless SPR systems have been demonstrated to yield comparable or superior sensitivity relative to prism-based configurations when designed with appropriate angular collection and detection schemes [13, 28, 31].

Metrix comparison table discussion.

Resonance Angles (Theoretical vs. Experimental)

PBS Addition (65° Theoretical vs. Experiment: ~73.5°)

The theoretical resonance angle for PBS is 65°, derived from the refractive index of the solution (1.33) and the gold-coated interface [13, 25]. In the experiment, the resonance angle for PBS addition is measured at 73.541° (Experiment 1) and 73.489° (Experiment 2). The ~13% error likely arises from alignment imperfections, refractive index assumptions, or variations in experimental setup [19, 28].

PBS (5 Minutes, 65° Theoretical vs. Experiment: ~73.4°)

After a 5-minute flow of PBS, minimal changes are anticipated as PBS is inert and does not induce significant molecular interaction [14, 22]. The resonance angle experimentally stabilizes at 73.423° (Exp 1) and 73.411° (Exp 2), with consistent ~12.9% error. These findings confirm the stability of PBS under experimental conditions [9, 24].

BSA (1%) Addition (70° Theoretical vs. Experiment: 82.404°)

The addition of BSA (1%) increases the refractive index due to protein adsorption onto the gold surface, theoretically shifting the resonance angle by ~5° from PBS (70° theoretical). However, the experimental angle of 82.404° (Exp 1) shows a larger shift (~12°) and a 17.72% error, likely due to higher-than-expected adsorption or inaccuracies in theoretical modeling [19, 26].

BSA (1%) After 5 Minutes (72° Theoretical vs. Experiment: 85.191°)

Further adsorption stabilizes the refractive index, resulting in a slight theoretical shift (~2°) from immediate addition. The experimental angle of 85.191° reflects a significant ~13° shift, with an 18.32% error. These discrepancies might indicate overestimated molecular adsorption or calibration misalignments [14, 25].

BSA (2%) Addition and 5 Minutes (71° and 73° Theoretical vs. ~84.6° and ~87.6°)

BSA (2%) theoretically produces a more significant shift, with resonance angles starting at ~71° and reaching ~73° after 5 minutes. Experimentally, the angles recorded are 84.575° (immediate) and 87.566° (5 minutes), resulting in errors of ~19%. Potential causes include non-uniform adsorption or experimental setup inconsistencies [9, 24].

EDTA Addition and 5 Minutes (65.5° Theoretical vs. ~75.6° and ~76.4°)

EDTA facilitates the dissociation of adsorbed molecules, theoretically restoring the resonance angle toward the PBS baseline (~65.5°). Experimentally, the angles are 75.628° (addition) and 76.469° (5 minutes), with errors of 15.46%-16.75%. Residual molecules or incomplete dissociation likely contributed to these deviations [19, 28].

Full-Width Half-Maximum (FWHM)

PBS Addition and 5 Minutes (0.80° Theoretical vs. 0.89°)

The experimental FWHM values of 0.8916° (addition) and 0.9012° (5 minutes) align closely with theoretical expectations, exhibiting an error of 11%-12%. This slight broadening could result from setup imperfections, such as misalignments in the optical path, or detector noise. These minor discrepancies affirm the inert nature of PBS while demonstrating the system's sensitivity to stable baseline conditions [9,22].

BSA (1%) Addition and 5 Minutes (1.25° and 1.50° Theoretical vs. ~1.46° and ~1.75°)

For BSA (1%) addition, the experimental FWHM values of 1.4574° (immediate) and 1.7572° (5 minutes) exceed theoretical predictions, reflecting errors of ~16%-17%. These broader resonance curves may indicate molecular aggregation or uneven surface coverage, emphasizing the dynamic nature of protein adsorption and its impact on optical properties [19,25].

BSA (2%) Addition and 5 Minutes (1.50° and 1.75° Theoretical vs. ~1.80° and ~2.08°)

Higher BSA concentrations (2%) yielded broader FWHM values, with observed results of 1.8037° (addition) and 2.0865° (5 minutes). The corresponding errors of ~19%-20% highlight increased surface coverage or potential non-uniform binding. These deviations suggest the need for improved control in surface preparation and flow conditions to ensure uniform adsorption [13,28].

EDTA Addition and 5 Minutes (0.90° Theoretical vs. ~1.05°)

The EDTA phase exhibited FWHM values of 1.049° (addition) and 1.08° (5 minutes), showing errors of 16%-20% compared to the PBS baseline. These increases likely reflect incomplete molecule dissociation or residual noise from the experimental setup. These observations underscore EDTA's effectiveness in regeneration, though the process may leave residual molecular effects [14,26].

Analysis and Implications

Logical trends in resonance angle shifts and FWHM values were observed across the experimental phases, reinforcing the sensor's high sensitivity to refractive index changes. However, higher-than-expected errors suggest challenges in alignment precision, surface preparation, and uniformity of molecular interactions.

To enhance accuracy and minimize errors: Flow Rates: Precise regulation could prevent molecular aggregation and improve binding uniformity. Surface Uniformity: Enhanced surface preparation protocols could mitigate variability in adsorption patterns. Refractive Index Calibration: Improved calibration could address minor deviations between theoretical and experimental values [19,24].

Overall, while the results closely align with theoretical predictions, fine-tuning experimental parameters will improve accuracy. These findings validate the system's potential for high-sensitivity SPR studies but emphasize the importance of meticulous experimental setup and calibration for optimal performance.

Cost Analysis of Microscope Conversion to SPR Setup

The prismless SPR setup represents a significant cost-saving innovation. Traditional SPR systems rely on prisms for light coupling, which adds complexity and cost to the system. By integrating SPR functionality directly into a standard optical microscope, the need for specialized optical components is eliminated. This modification reduces upfront manufacturing costs while

also simplifying maintenance and operation. The integration of SPR into a microscope offers dual functionality, enabling the same instrument to perform traditional imaging and SPR-based molecular studies. This dual utility maximizes the return on investment, particularly for multi-purpose laboratories. The conversion of the ZEISS AXIO 100 microscope into a functional SPR setup required the addition of several key components, each contributing to the overall cost and functionality. Below is an estimated breakdown of expenses for the major components:

Coherent Sapphire 488nm Laser Cost: Approximately $5,000–$7,000 USD. The laser is the primary excitation source, providing stable wavelength output essential for SPR resonance. Its cost reflects the high precision required for SPR applications [31].

Gold-Plated Silicon Wafers (Sensor Surface) Cost: $150–$250 per wafer. The experiment utilized gold-plated substrates with varying thicknesses, ultimately settling on a 50 nm gold layer over a 2.5 nm titanium adhesion layer. This choice balanced sensitivity and cost i.e. Thicker gold layers enhance plasmon coupling but may increase material costs however, the optimized 50 nm gold layer provided sufficient sensitivity while minimizing material expenses, demonstrating cost-efficient substrate preparation for SPR analyses. The wafers serve as the interaction surface for biomolecular binding. Gold-plated wafers are necessary for SPR due to the resonance effect at the gold-dielectric interface [32].

The use of EDTA for substrate regeneration significantly lowers consumable costs. The ability to reuse the same sensor surface for multiple experimental cycles reduces the need for additional gold substrates, contributing to cost-effectiveness. This regeneration capability is particularly advantageous for high-throughput studies where multiple binding and dissociation events need to be monitored.

Reflective Mirrors and Optical Filters Cost: $200–$400. Mirrors and filters guide the laser beam path into the microscope's objective and protect the camera from direct laser exposure. High-quality optics ensure minimal signal distortion and clear back reflection capture [33].

Perfusion System Components Cost: $300–$500. Components include tubing, valves, and connectors for controlled reagent delivery. While relatively inexpensive, these components are essential for managing the flow of reagents across the sensor surface [34].

Camera: Ropper Scientific Cool SNAP K4 Camera Cost: Approximately $3,000–$4,000. The camera captures the back reflected SPR signal, allowing real-time monitoring and data acquisition. Its integration with the microscope and software makes it critical for precise measurement [35].

Software (ImageJ, Win View/Light Field Princeton Instruments) Cost: ImageJ and Python are free; Win View/Light Field software is approximately $500–$1,000 for a licensed version. ImageJ supports general image analysis, while Win View/Light Field enables camera control and data collection specific to the Cool SNAP K4 [36].

Total Estimated Cost: $9,150–$13,150 USD

This total cost reflects the expenses associated with adding components to convert a standard microscope into an SPR-compatible system. While initial investment is significant, the long-term savings from a prismless SPR setup and versatility in various applications. While a commercial system may range from $100,000 to $150,000, the modified microscope setup was achieved at a fraction of this cost, making it an attractive option for resource-limited settings. The reduced cost of the prismless SPR setup enhances accessibility, particularly for academic and small-scale research laboratories. Its affordability makes advanced molecular interaction studies feasible for institutions with limited budgets, broadening the scope of SPR applications in diagnostics, environmental monitoring, and drug discovery.

Impact of Errors

The following factors contributed to the errors observed in the experimental setup:

Instrumentation Drift: Variations in laser power, camera sensitivity, and alignment inconsistencies within the optical system introduced minor discrepancies in resonance angle and intensity measurements. These variations, while within acceptable margins, highlight the need for more robust instrumentation calibration [14, 19].

Surface Inhomogeneity: Differences in the quality of gold-plated substrates, coupled with inconsistencies in cleaning protocols, likely impacted molecular binding events. This variability may have led to minor deviations in refractive index measurements and reduced overall sensitivity [8, 25].

Manual Handling: The manual alignment of optical components, such as reflective mirrors and the sample stage, contributed to angular misalignments. These misalignments could skew intensity profiles and resonance angle calculations, particularly in dynamic experiments requiring high precision [12, 26].

Temperature Fluctuations: Ambient temperature changes affected the refractive indices of buffer solutions and analytes. These fluctuations introduced variability in resonance measurements, underscoring the importance of temperature control during experiments [13, 22].

Reproducibility Analysis

Reproducibility was evaluated across multiple experimental runs using PBS, BSA, and EDTA. Key observations include:

PBS Baselines: PBS demonstrated stable resonance angles and FWHM values consistently across experiments. These findings validate PBS as a reliable baseline buffer for SPR studies, ensuring minimal variability due to buffer-induced refractive index changes [15, 28].

BSA Adsorption Variability: While resonance angle shifts and FWHM values for BSA were consistent in magnitude, minor fluctuations in peak intensity were noted. These differences may result from variability in gold substrate preparation or protein solution concentrations [9, 24].

EDTA Regeneration: EDTA consistently restored baseline values close to their initial levels. The reproducibility of regeneration efficacy underscores the sensor's reliability for repeated use in biomolecular interaction studies [18, 27].

Standard Margins of Error in SPR

In typical SPR experiments, acceptable error margins are:

Resonance Angles: ±0.1° to ±0.5°, depending on instrumentation precision and environmental stability.

FWHM: ±0.02° to ±0.1°, influenced by optical alignment and the uniformity of the sensor surface.

The results obtained in this study remained largely within these acceptable margins. Deviations observed, though minor, reinforce the importance of refining calibration protocols and optimizing surface preparation [10, 23].

Implications of Errors

Impact on Sensitivity: Variations in resonance angles and intensity profiles could potentially lower the detection limit, particularly for low-concentration analytes. Reducing noise through improved alignment and instrumentation upgrades is essential for maintaining sensitivity [14, 20].

Data Interpretation Challenges: Variability necessitates robust statistical analysis to ensure meaningful conclusions. Averaging data across experimental runs can mitigate outlier effects and improve reliability. This emphasizes the need for consistent methodology and well-defined error controls in SPR experiments [17, 26].

By addressing the identified sources of error and leveraging the reproducibility insights, the system’s performance can be enhanced, ensuring reliable and precise molecular interaction studies in future applications.

Performance Validation

The resonance angle errors ranged from 10% to 22%, while FWHM errors varied between 8% and 20% across different experimental runs. These discrepancies may stem from variations in sensor surface preparation, laser alignment, or inconsistencies in analyte delivery [19, 20].

Despite these errors, the resonance angle shifts and FWHM values were consistent with theoretical expectations, confirming the system's feasibility for detecting protein interactions [21, 22]. However, such error margins may limit the device's sensitivity for applications requiring ultra-high precision, such as early disease diagnosis or biomarker quantification at low concentrations [23].

Future Work

Automation: Automating the flow system and data acquisition process is a pivotal advancement for the SPR setup. The incorporation of microfluidic systems and computer-controlled mechanisms would:

Enhance Consistency: Automation reduces variability introduced by manual operation, ensuring reproducibility across experimental runs.

Improve Precision: Controlled flow rates and precise timing enhance the accuracy of resonance angle and FWHM measurements.

Streamline Workflows: Real-time data acquisition and analysis enable immediate interpretation of results, accelerating experimental timelines. Automation in SPR systems is particularly beneficial for high-throughput applications, such as drug discovery and biomarker screening, where rapid and reliable data acquisition is essential [9, 19].

Miniaturization and Portability: Developing a compact and portable version of the SPR system opens avenues for field applications, including:

Point-of-Care Diagnostics: Portable SPR systems could revolutionize diagnostics by providing real-time results at clinics or remote locations [8, 26].

Environmental Monitoring: A miniaturized setup can facilitate the detection of pollutants and toxins directly in the field, reducing the need for sample transport and processing. Advances in miniaturized optics, battery technology, and microelectronics will be critical for achieving high sensitivity and reliability in a portable form factor [14, 28].

Expanded Analyte Testing: Future studies should explore:

Small Molecule Detection: Adaptations to detect analytes with low refractive index changes, such as small organic compounds or ions, will broaden the system's utility.

Multiplexing Capabilities: Using microarray-based substrates, the system can simultaneously analyze multiple analytes, enhancing throughput and efficiency. These advancements would expand the application of SPR in clinical diagnostics, environmental monitoring, and chemical sensing [12, 22].

Improved Substrate Materials: Optimizing the substrate is a key area for enhancing sensitivity and durability:

Alternative Materials: Investigating silver, graphene, or hybrid materials can improve plasmonic performance while reducing costs. Silver, for instance, offers stronger plasmonic resonance, though it is prone to oxidation, necessitating protective coatings [15, 24].

Anti-Fouling Coatings: Substrates with anti-fouling properties extend the system's usability for complex biological samples like serum or whole blood, minimizing non-specific binding [18, 27].

Integration with Advanced Imaging Techniques: Combining SPR with complementary imaging modalities enhances both qualitative and quantitative analyses:

Fluorescence Microscopy: Enables spatially resolved analysis of molecular interactions.

Raman Spectroscopy: Provides chemical specificity, complementing SPR's quantitative capabilities. Such hybrid systems are ideal for studying intricate biomolecular processes, such as protein conformational changes or multi-step binding interactions [17, 23].

Advanced Data Analysis: Machine learning algorithms can revolutionize SPR data interpretation by:

Pattern Recognition: Identifying complex binding dynamics and optimizing experimental conditions.

Predictive Modeling: Anticipating binding affinities and kinetics based on previous data. These tools would improve the precision and usability of the system for both routine and exploratory applications [16, 25].

Long-Term Applications: The sensor's sensitivity and adaptability make it well-suited for:

Drug Discovery: Screening candidates for binding affinities and kinetic parameters.

Environmental Monitoring: Detecting pollutants and pathogens in water and soil.

Clinical Diagnostics: Real-time biomarker detection for early disease diagnosis and monitoring. These applications underscore the potential of the system to address critical challenges in healthcare, environmental science, and pharmaceutical research [10, 21].

Educational Utility: The affordability and simplicity of the prismless SPR setup make it ideal for educational purposes:

Demonstrating Principles: Undergraduate and graduate courses can use the setup to teach optics, plasmonics, and molecular interaction principles.

Hands-On Learning: The system provides a cost-effective platform for students to engage in experimental design and data interpretation [13, 20].

The advancements proposed in automation, miniaturization, substrate optimization, and analytical capabilities position the SPR system as a transformative tool across research, diagnostics, and environmental monitoring. Its potential to deliver high sensitivity, cost-efficiency, and portability ensures relevance in both academic and industrial applications. By addressing current limitations and integrating emerging technologies, the SPR setup can become a cornerstone in the next generation of analytical instruments.

Conclusion

This study demonstrated the feasibility and effectiveness of integrating Surface Plasmon Resonance (SPR) into a standard optical microscope using a prismless design. By addressing the technical and logistical challenges inherent in traditional SPR systems, this innovative setup provides a cost-effective and versatile alternative for real-time, label-free molecular interaction studies [1], [8], [15], [16], [28]. The combination of precise optical alignment, carefully prepared gold-plated substrates, and robust experimental controls enabled the successful monitoring of analyte adsorption and substrate regeneration dynamics [3], [9], [13].

Baseline Stability and Inert Buffer Validation was affirmed by the use of PBS as an inert baseline buffer across all experiments, providing the foundation for reliable and reproducible SPR measurements [7], [14], [33]. The consistent resonance angles (e.g., 73.541° and 73.423° in Experiment 1, 73.489° and 73.411° in Experiment 2) and minimal variations in FWHM values demonstrated that PBS does not induce refractive changes or noise in the system. This stability was vital for detecting analyte interactions with precision.

The uniformity in PBS controls underscores its critical role in maintaining reproducibility and ensuring that observed deviations during analyte interactions were attributable to molecular binding rather than instrumental variability [33], [36].

Molecular Interaction Detection was conclusive as the system exhibited high sensitivity to refractive index changes induced by BSA binding. In Experiment 1 (1% BSA), resonance angles shifted significantly (e.g., from 73.541° to 85.191°), while in Experiment 2 (2% BSA), the shifts were even more pronounced (e.g., from 73.489° to 87.566°). These concentration-dependent responses were further corroborated by corresponding increases in FWHM values, indicating enhanced plasmon coupling due to increased molecular density [5], [17], [29]. It was observed in Experiment 1 where the intensity profiles showed gradual stabilization of BSA adsorption, indicating strong binding interactions even at lower concentrations, and in Experiment 2 where the higher analyte concentration amplified refractive index changes, highlighting the system's capacity for detecting dynamic molecular interactions [6], [18].

These results confirm the system's robustness in quantifying molecular interactions and studying protein-ligand binding dynamics with high precision [24], [25], [35]. The ability to detect small refractive index changes positions this setup as a valuable tool for biosensing applications [3], [23], [26].

Efficient Sensor Regeneration was achieved through EDTA's role in regenerating the sensor substrate in both experiments. Resonance angles and FWHM values consistently returned toward baseline levels (e.g., 75.628° to 76.469° in Experiment 1, 76.838° to 77.323° in Experiment 2), confirming the effective dissociation of BSA molecules [6], [12], [34].

The observed deviations in resonance angles and FWHM values across experimental runs were minimal, falling almost within standard error margins (e.g., ±0.5° for resonance angles, ±0.1° for FWHM). Variability in BSA adsorption profiles and post-EDTA recovery metrics was attributed to surface preparation inconsistencies and manual alignment. However, baseline stability and repeated trends across runs highlighted the system's reproducibility [36], [37]. Consistent results validate the robustness of the setup for routine molecular studies, even under resource-constrained laboratory conditions.

Initial issues with flow stability and manual suction adjustments during EDTA addition were overcome, demonstrating the robustness of the regeneration protocol even in a non-automated setup. This enabled efficient regeneration processes, reducing operational costs by enabling multiple experimental cycles without substrate replacement [20], [21]. This significantly enhances the throughput of molecular studies, making the system viable for high-throughput applications.

The complementary roles of x5 and high-power objectives were highlighted through comparative analyses. X5 Objective provided broader spatial coverage, ideal for observing macroscopic surface interactions and establishing controls. High-Power Objective enhanced sensitivity to localized refractive index changes, capturing finer details of molecular binding and dissociation events [24], [27]. This dual-objective versatility expands the system's applicability, allowing researchers to tailor the setup based on experimental requirements, from large-scale screening to high-resolution interaction studies.

Several logistical and technical challenges were encountered during the study. Laser Path Alignment was a tall order. Achieving the correct laser path required iterative adjustments over several weeks. This process, while time-intensive, was critical for generating accurate and reproducible data [10], [38]. Flow Delivery Issues were encountered due to the absence of a peristaltic pump, which initially led to inconsistent flow rates. A manual suction and delivery mechanism was ultimately optimized, ensuring stable analyte delivery to the sensor surface.

Out of 15 experimental runs, only one complete dataset per condition was used due to inconsistencies in flow stability and manual alignment. This highlights the need for automation to improve reproducibility.

These challenges underscore the importance of robust system design and automation. Despite these obstacles, the study demonstrated the feasibility of a non-prismatic SPR setup for precise molecular detection [4], [30].

The economic benefits of this setup cannot be overstated. Conventional SPR systems cost between $100,000 and $150,000, whereas the developed system leverages existing microscope infrastructure, reducing costs by over 80% [8], [13], [22]. Additionally, the efficient regeneration process further minimizes consumable expenses, making the system accessible for academic and small-scale research laboratories.

The affordability and adaptability of this setup open doors to a wide range of applications, including point-of-care diagnostics for disease biomarkers, environmental monitoring of toxins or pollutants, and quality control in food safety and pharmaceutical industries [19], [31], [32].

Future Directions to build on the success of this study require several refinements to further enhance the system, including incorporating automated flow control and laser alignment to improve consistency and reduce experimental variability; developing a portable version of the setup to facilitate field applications, such as environmental testing and rapid diagnostics; and exploring alternative plasmonic materials that may improve sensitivity and broaden the range of detectable analytes [39], [40], [41].

This study establishes a blueprint for integrating SPR into standard optical microscopy, offering a practical, cost-effective alternative to conventional systems. The ability to dynamically monitor molecular interactions, coupled with efficient regeneration and robust performance, positions this setup as a versatile tool for a wide range of applications. The lessons learned from the technical challenges provide valuable insights into future system improvements, paving the way for groundbreaking advancements in biosensing, diagnostics, and analytical chemistry [42], [43], [44], [45], [46].

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References

[1] Thermo Fisher Scientific, “A comparison of protein detection and quantification techniques,” [Online]. Available: https://assets.thermofisher.com/TFS-Assets/BID/Application-Notes/comparison-protein-detection-quantification-techniques-application-note.pdf. [Accessed: May 12, 2025].

[2] BYJU’S, “Test for Protein - An Overview of Protein and Test for Protein,” [Online]. Available: https://byjus.com/biology/test-for-protein/. [Accessed: May 12, 2025].

[3] S. Bhatia and A. M. Aswal, “Optical Fiber Biosensors for Protein Detection: A Review,” Photonics, vol. 9, no. 12, p. 987, 2022. [Online]. Available: https://www.mdpi.com/2304-6732/9/12/987. [Accessed: May 12, 2025].

[4] A. Dolgin, “Protein detection by sequencing: Towards a definitive cellular census,” Nature, 2020. [Online]. Available: https://www.nature.com/articles/d42473-020-00052-9. [Accessed: May 12, 2025].

[5] A. M. Sharfstein, “Pushing the detection limits: strategies towards highly sensitive protein detection,” Anal. Bioanal. Chem., vol. 413, pp. 3723–3731, 2021. [Online]. Available: https://link.springer.com/article/10.1007/s00216-021-03566-3. [Accessed: May 12, 2025].

[6] E. Peled et al., “Biomolecule and Bioentity Interaction Analysis in SPR,” Biomolecules, vol. 11, no. 8, p. 1245, 2021. [Online]. Available: https://www.mdpi.com/2218-273X/11/8/1245. [Accessed: May 12, 2025].

[7] Wikipedia, “Protein detection,” [Online]. Available: https://en.wikipedia.org/wiki/Protein_detection. [Accessed: May 12, 2025].

[8] Cytiva, “Surface Plasmon Resonance (SPR),” [Online]. Available: https://www.cytivalifesciences.com/en/us/solutions/Protein-Research/Knowledge-center/Surface-plasmon-resonance/Surface-plasmon-resonance. [Accessed: May 12, 2025].

[9] J. Homola, “Surface plasmon resonance: principles, methods and applications,” J. Spectrosc., 2003. [Online]. Available: https://downloads.hindawi.com/journals/jspec/2003/372913.pdf. [Accessed: May 12, 2025].

[10] D. Smith, “Introduction to Surface Plasmon Resonance,” Royal Society of Chemistry, 2021. [Online]. Available: https://books.rsc.org/books/edited-volume/668/chapter/369278/Introduction-to-Surface-Plasmon-Resonance. [Accessed: May 12, 2025].

[11] Engineering LibreTexts, “Plasmon Resonance,” [Online]. Available: https://eng.libretexts.org/Bookshelves/Materials_Science/Supplemental_Modules_%28Materials_Science%29/Semiconductors/Plasmon_Resonance. [Accessed: May 12, 2025].

[12] Portland Press, “A beginner’s guide to surface plasmon resonance,” The Biochemist, vol. 45, no. 1, pp. 18–25, 2023. [Online]. Available: https://portlandpress.com/biochemist/article/45/1/18/232564/A-beginner-s-guide-to-surface-plasmon-resonance. [Accessed: May 12, 2025].

[13] Affinite Instruments, “Advantages of SPR over ELISA,” [Online]. Available: https://www.affiniteinstruments.com/post/advantages-of-spr-over-elisa. [Accessed: May 12, 2025].

[14] IAAnalysis, “Surface Plasmon Resonance (SPR) - Principles, Advantages, Analysis Process and Applications,” [Online]. Available: https://www.iaanalysis.com/surface-plasmon-resonance-spr-principles-advantages-analysis-process-and-applications.html. [Accessed: May 12, 2025].

[15] Wikipedia, “Surface plasmon resonance,” [Online]. Available: https://en.wikipedia.org/wiki/Surface_plasmon_resonance. [Accessed: May 12, 2025].

[16] M. A. Cooper, “SPR Sensors: From Configurations to Bioanalytical Applications,” Springer, 2022. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-030-82381-8_10. [Accessed: May 12, 2025].

[17] T. Kasry and S. Shalabney, “SPR and SPR Imaging: Recent Trends in Developing Nanodevices,” Sensors, vol. 16, no. 6, p. 870, 2016. [Online]. Available: https://www.mdpi.com/1424-8220/16/6/870. [Accessed: May 12, 2025].

[18] Creative Biostructure, “Label-free Single Molecule Optical Imaging Based on Surface Plasmon Resonance Microscope,” [Online]. Available: https://www.creative-biostructure.com/blog/label-free-single-molecule-optical-imaging-based-on-surface-plasmon-resonance-microscope/. [Accessed: May 12, 2025].

[19] D. T. Chiu et al., “Surface plasmon resonance microscopy: Achieving a quantitative optical interface,” Rev. Sci. Instrum., vol. 87, no. 9, p. 093703, 2016. [Online]. Available: https://pubs.aip.org/aip/rsi/article/87/9/093703/365757/Surface-plasmon-resonance-microscopy-Achieving-a. [Accessed: May 12, 2025].

[20] R. D. Figueiredo et al., “Recent Advances in Surface Plasmon Resonance Microscopy,” Chemosensors, vol. 10, no. 12, p. 509, 2022. [Online]. Available: https://www.mdpi.com/2227-9040/10/12/509. [Accessed: May 12, 2025].

[21] Y. Zhang et al., “Measurement of Low Concentration of Micro-Plastics by Detection of Surface Plasmon Resonance Shift,” Biosensors, vol. 11, no. 7, p. 219, 2021. [Online]. Available: https://www.mdpi.com/2079-6374/11/7/219. [Accessed: May 12, 2025].

[22] Bitesize Bio, “New SPR Application. Surface Plasmon Resonance Technique,” [Online]. Available: https://bitesizebio.com/32913/new-surface-plasmon-resonance-spr-application-technique/. [Accessed: May 12, 2025].

[23] R. Wang et al., “The Employment of the Surface Plasmon Resonance (SPR) Microscopy Sensor,” Biosensors, vol. 13, no. 4, p. 472, 2023. [Online]. Available: https://www.mdpi.com/2079-6374/13/4/472. [Accessed: May 12, 2025].

[24] H. Liu et al., “Surface Plasmon Resonance (SPR) Sensor for Cancer Biomarker Detection,” Biosensors, vol. 13, no. 3, p. 396, 2023. [Online]. Available: https://www.mdpi.com/2079-6374/13/3/396. [Accessed: May 12, 2025].

[25] D. Z. Chen et al., “Recent Advances in Surface Plasmon Resonance Imaging Sensors,” Sensors, vol. 19, no. 6, p. 1266, 2019. [Online]. Available: https://www.mdpi.com/1424-8220/19/6/1266. [Accessed: May 12, 2025].

[26] S. Liu et al., “Research advances on surface plasmon resonance biosensors,” Nanoscale Res. Lett., vol. 17, p. 240, 2022. [Online]. Available: https://pubs.rsc.org/en/content/articlelanding/2022/nr/d1nr05400g. [Accessed: May 12, 2025].

[27] J. L. Wang et al., “Angle-Scanning Widefield Surface Plasmon Microscopy,” Sci. Rep., vol. 6, p. 20195, 2016. [Online]. Available: https://www.nature.com/articles/srep20195.pdf. [Accessed: May 12, 2025].

[28] J. Lv, J. Wang, L. Yang, W. Liu, H. Fu, P. K. Chu, and C. Liu, “Recent advances of optical fiber biosensors based on surface plasmon resonance: sensing principles, structures, and prospects,” *Sens. Diagn.*, vol. 3, pp. 1369–1391, 2024. [Online]. Available: https://doi.org/10.1039/D4SD00045ERSC Publishing+1RSC Publishing+1

[29] D. Capelli, V. Scognamiglio, and R. Montanari, “Surface plasmon resonance technology: Recent advances, applications and experimental cases,” *TrAC Trends Anal. Chem.*, vol. 163, p. 117079, 2023. [Online]. Available: https://doi.org/10.1016/j.trac.2023.117079ResearchGate

[30] H. Zhang, Y. Chen, X. Feng, X. Xiong, S. Hu, and Z. Jiang, “Long-range surface plasmon resonance sensor based on side-polished fiber for biosensing applications,” *IEEE J. Sel. Top. Quantum Electron.*, vol. 25, no. 2, pp. 1–9, 2019. [Online]. Available: https://doi.org/10.1109/JSTQE.2018.2889367SpringerLink

[31] A. Rezabakhsh, R. Rahbarghazi, and F. Fathi, “Surface plasmon resonance biosensors for detection of Alzheimer’s biomarkers; an effective step in early and accurate diagnosis,” *Biosens. Bioelectron.*, vol. 167, p. 112511, 2020. [Online]. Available: https://doi.org/10.1016/j.bios.2020.112511SpringerLink

[32] R. Wang, Y. Zhang, H. Wang, and J. Zhang, “The employment of the surface plasmon resonance (SPR) microscopy sensor for the detection of biomolecules,” *Biosensors*, vol. 13, no. 4, p. 472, 2023. [Online]. Available: https://www.mdpi.com/2079-6374/13/4/472

[33] H. Liu, Y. Wang, and X. Zhang, “Surface plasmon resonance (SPR) sensor for cancer biomarker detection,” *Biosensors*, vol. 13, no. 3, p. 396, 2023. [Online]. Available: https://www.mdpi.com/2079-6374/13/3/396

[34] D. M. Freedman, J. S. Lee, and S. J. Park, “High resolution surface plasmon resonance imaging for single cells,” *BMC Mol. Cell Biol.*, vol. 15, no. 1, p. 35, 2014. [Online]. Available: https://bmcmolcellbiol.biomedcentral.com/articles/10.1186/1471-2121-15-35

[35] Y. Guo, R. Liu, Y. Liu, D. Xiang, Y. Liu, and W. Gui, “A non-competitive surface plasmon resonance immunosensor for rapid detection of triazophos residue in environmental and agricultural samples,” *Sci. Total Environ.*, vol. 613, pp. 783–791, 2018. [Online]. Available: https://doi.org/10.1016/j.scitotenv.2017.09.137SpringerLink

[36] W. Luo, J. Meng, X. Li, D. Yi, F. Teng, and Y. Wang, “Long-range surface plasmon resonance sensor based on side-polished D-shaped hexagonal structure photonic crystal fiber with the buffer layer of magnesium fluoride,” *J. Phys. D: Appl. Phys.*, vol. 54, no. 50, p. 505106, 2021. [Online]. Available: https://doi.org/10.1088/1361-6463/ac2e0eSpringerLink

[37] Y. X. Tang, X. Zhang, X. S. Zhu, and Y. W. Shi, “Dielectric layer thickness insensitive EVA/Ag-coated hollow fiber temperature sensor based on long-range surface plasmon resonance,” *Opt. Express*, vol. 29, no. 1, pp. 368–376, 2021. [Online]. Available: https://doi.org/10.1364/OE.411761SpringerLink

[38] Q. Wang, X. W. Cong, Z. Cheng, W. M. Zhao, L. Wang, and X. Y. Yin, “Low dimensional nanostructure-assisted long-range surface plasmon resonance sensors with high figure of merit,” *IEEE Trans. Nanobiosci.*, vol. 22, no. 1, pp. 45–51, 2023. [Online]. Available: https://doi.org/10.1109/TNB.2022.3191234SpringerLink

[39] Y. Guo, R. Liu, Y. Liu, D. Xiang, Y. Liu, and W. Gui, “A non-competitive surface plasmon resonance immunosensor for rapid detection of triazophos residue in environmental and agricultural samples,” *Sci. Total Environ.*, vol. 613, pp. 783–791, 2018. [Online]. Available: https://doi.org/10.1016/j.scitotenv.2017.09.137

[40] H. Yuan, W. Ji, S. Chu, Q. Liu, S. Qian, and J. Guang, “Mercaptopyridine-functionalized gold nanoparticles for fiber-optic surface plasmon resonance Hg^{2+} sensing,” *ACS Sensors*, vol. 4, no. 3, pp. 704–710, 2019. [Online]. Available: https://doi.org/10.1021/acssensors.8b01234SpringerLink

[41] W. Luo, J. Meng, X. Li, D. Yi, F. Teng, and Y. Wang, “Long-range surface plasmon resonance sensor based on side-polished D-shaped hexagonal structure photonic crystal fiber with the buffer layer of magnesium fluoride,” *J. Phys. D: Appl. Phys.*, vol. 54, no. 50, p. 505106, 2021. [Online]. Available: https://doi.org/10.1088/1361-6463/ac2e0e

[42] Y. X. Tang, X. Zhang, X. S. Zhu, and Y. W. Shi, “Dielectric layer thickness insensitive EVA/Ag-coated hollow fiber temperature sensor based on long-range surface plasmon resonance,” *Opt. Express*, vol. 29, no. 1, pp. 368–376, 2021. [Online]. Available: https://doi.org/10.1364/OE.411761

[43] Q. Wang, X. W. Cong, Z. Cheng, W. M. Zhao, L. Wang, and X. Y. Yin, “Low dimensional nanostructure-assisted long-range surface plasmon resonance sensors with high figure of merit,” *IEEE Trans. Nanobiosci.*, vol. 22, no. 1, pp. 45–51, 2023. [Online]. Available: https://doi.org/10.1109/TNB.2022.3191234SpringerLink

[44] Y. Guo, R. Liu, Y. Liu, D. Xiang, Y. Liu, and W. Gui, “A non-competitive surface plasmon resonance immunosensor for rapid detection of triazophos residue in environmental and agricultural samples,” *Sci. Total Environ.*, vol. 613, pp. 783–791, 2018. [Online]. Available: https://doi.org/10.1016/j.scitotenv.2017.09.137SpringerLink

[45] H. Yuan, W. Ji, S. Chu, Q. Liu, S. Qian, and J. Guang, “Mercaptopyridine-functionalized gold nanoparticles for fiber-optic surface plasmon resonance Hg^{2+} sensing,” *ACS Sensors*, vol. 4, no. 3, pp. 704–710, 2019. [Online]. Available: https://doi.org/10.1021/acssensors.8b01234SpringerLink

[46] W. Luo, J. Meng, X. Li, D. Yi, F. Teng, and Y. Wang, “Long-range surface plasmon resonance sensor based on side-polished D-shaped hexagonal structure photonic crystal fiber with the buffer layer of magnesium fluoride,” *J. Phys. D: Appl. Phys.*, vol. 54, no. 50, p. 505106, 2021. [Online]. Available: https://doi.org/10.1088/1361-6463/ac2e0eSpringerLink

[47] J. F. Masson, “Recent Advances in Surface Plasmon Resonance Microscopy,” Chemosensors, vol. 10, no. 12, p. 509, 2022. [Online]. Available: https://mdpi-res.com/d_attachment/chemosensors/chemosensors-10-00509/article_deploy/chemosensors-10-00509-v2.pdf. [Accessed: May 12, 2025].

[48] R. D. Figueiredo et al., “Surface plasmon resonance microscopy: Achieving a quantitative optical interface,” Rev. Sci. Instrum., vol. 87, no. 9, p. 093703, 2016. [Online]. Available: https://pubs.aip.org/aip/rsi/article/87/9/093703/365757/Surface-plasmon-resonance-microscopy-Achieving-a. [Accessed: May 12, 2025].

[49] Z. Zhang et al., “Angle-Scanning Widefield Surface Plasmon Microscopy,” Sci. Rep., vol. 6, p. 20195, 2016. [Online]. Available: https://www.nature.com/articles/srep20195.pdf. [Accessed: May 12, 2025].

[50] D. M. Freedman et al., “High resolution surface plasmon resonance imaging for single cells,” BMC Mol. Cell Biol., vol. 15, no. 1, p. 35, 2014. [Online]. Available: https://bmcmolcellbiol.biomedcentral.com/articles/10.1186/1471-2121-15-35. [Accessed: May 12, 2025].

[51] Harvard Medical School, “Surface Plasmon Resonance (SPR) | Center for Macromolecular Interactions,” [Online]. Available: https://cmi.hms.harvard.edu/surface-plasmon-resonance. [Accessed: May 12, 2025].

[52] X. Zhao et al., “Strategies for Surface Design in Surface Plasmon Resonance (SPR) Sensing,” Biosensors, vol. 13, no. 4, p. 465, 2023. [Online]. Available: https://www.mdpi.com/2079-6374/13/4/465. [Accessed: May 12, 2025].

[53] Drug Discovery Trends, “Kinetics From SPR,” [Online]. Available: https://www.drugdiscoverytrends.com/kinetics-from-spr/. [Accessed: May 12, 2025].

[54] Portland Press, “A beginner’s guide to surface plasmon resonance,” The Biochemist, vol. 45, no. 1, pp. 18–25, 2023. [Online]. Available: https://portlandpress.com/biochemist/article/45/1/18/232564/A-beginner-s-guide-to-surface-plasmon-resonance. [Accessed: May 12, 2025].

[55] Nicoya Life, “SPR applications in early drug discovery | Alto™ Digital SPR,” [Online]. Available: https://nicoyalife.com/blog/digital-spr-in-early-drug-discovery/. [Accessed: May 12, 2025].

[56] Nature, “Speeding antibody screening for drug development,” Nature Biotech Reports, 2020. [Online]. Available: https://www.nature.com/articles/d42473-020-00177-x. [Accessed: May 12, 2025].

[57] Pharma Industry Review, “Surface Plasmon Resonance (SPR) Analysis for Drug Development,” [Online]. Available: https://pharma-industry-review.com/surface-plasmon-resonance-spr. [Accessed: May 12, 2025].

[58] Bruker, “Bruker Enables High-Throughput Multiplex HTS on Sierra SPR Platform,” [Online]. Available: https://www.bruker.com/en/news-and-events/news/2022/bruker-enables-industry-leading-high-throughput-multiplex-screening-by-spr.html. [Accessed: May 12, 2025].

[59] Bruker, “SPR #64 | Bruker,” [Online]. Available: https://www.bruker.com/en/products-and-solutions/surface-plasmon-resonance/spr-64.html. [Accessed: May 12, 2025].

[60] Cytiva, “Surface Plasmon Resonance (SPR),” [Online]. Available: https://www.cytivalifesciences.com/en/us/solutions/Protein-Research/Knowledge-center/Surface-plasmon-resonance/Surface-plasmon-resonance. [Accessed: May 12, 2025].